

Energy and Commerce

California's 16th Congressional District
Student Advisory Board

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Opening Remarks

Karthi Gottipati – Chair

Hello Everyone,

My name is Karthi Gottipati and I have had the privilege of serving as the Chair of the 31st and final iteration of the Student Advisory Board. It has been such an honor and I am eternally grateful to my peers for giving me this opportunity. This board has been incredibly well run this year and for that, we have to thank Nicholas, Justin, and Ariana - our amazing staff this year. Your mentorship has been so valuable to me and I am continually impressed by your dedication and commitment to ensuring this program succeeds. This report would be nowhere near the quality it is without your guidance. I also wanted to thank Sarah, the Vice Chair, for her support throughout this work. She has done an amazing job and has gone above and beyond in her responsibilities. Thank you to each of the subcommittee chairs who have all done an incredible job organizing their subcommittees and ensuring the work got done on time. Finally, and perhaps most importantly, I want to extend my gratitude to the reason we are all here today, Congresswoman Anna Eshoo - thank you for making this all possible and giving over a hundred high schoolers an unforgettable experience.

With that said, here's the logistics for today. We modeled our research organization off of the Energy and Commerce Committee on which Representative Eshoo serves and we divided up into six Subcommittees. Each Subcommittee is further divided into separate subgroups with their topic to research. We have covered a broad range of issues, so every group will give you the necessary context. In the broadest possible terms, we sought to address how our regulations can adapt to address an evolving world as we move into the future. The presentations you will see here today are the culmination of months of research and writing, and we are so proud to share our work with you. Thank you, everyone, for taking the time out of your day to be here with us.

Sarah Wong – Vice Chair

Hello Everyone,

I hope you all are doing well. Thank you so much for joining us today to celebrate the diligent and thoughtful work of the 2024 Student Advisory Board. My name is Sarah Wong, and I had the privilege of serving as the Vice Chair of this year's Student Advisory Board and working alongside our dedicated members from across the district.

This year, our SAB structure mirrored that of the Energy and Commerce Committee in Congress which Representative Eshoo serves. We organized into six Subcommittees: Environment, Manufacturing and Critical Materials; Energy, Climate and Grid Security; Innovation, Data, and Commerce; Communications and Technology; Health; and finally Oversight and Investigations. Within these Subcommittees, each and every one of our members delved into legislative research and analysis, working to identify gaps in existing policies. We then crafted policy proposals addressing specific topics within our focus areas. Working in small subgroups, we developed proposals ranging from amendments to existing bills to the creation of new programs. Today, you will hear presentations of these proposals, which we as the youth of the 16th Congressional District, believe to be crucial for our nation's overall well-being.

I'd like to first extend my sincerest gratitude to each one of our members for their wholehearted dedication to democratic principles and their efforts towards fostering a more equitable and inclusive society. Working with the board and witnessing students formulate their policy proposals has truly been a remarkable experience. However, none of this would have been possible without the support of Congresswoman Eshoo's staff. To Nicholas, Justin, and Ariana, your unwavering dedication to ensuring the seamless operation of the board is deeply appreciated. Thank you for your tireless efforts.

Finally, I'd like to display my profound gratitude for the opportunity to share our voices and recommendations with Congresswoman Eshoo. Thank you for considering our proposals. As this is the last year Representative Eshoo will be serving, we cannot thank you enough for your hard work and dedication to furthering your policy goals and to this program.

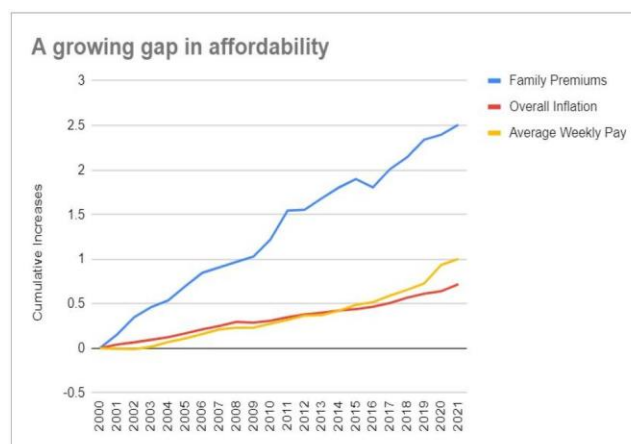
This meeting marks the culmination of months of hard work, and as your Vice Chair it has been an honor to serve on Congresswoman Eshoo's Student Advisory Board.

Insurance Policy Proposal: Offsetting Rising Costs for Medical Providers and Policy Holders Via Quarterly Audits

Proposed by: Nava Riahi, Aarohe Shah, Shepherd Cremer, Johanah Brandt

Introduction:

Healthcare in California is notable for its increasing insurance deductibles and inflated service costs. From 2000 to 2021, the cost of health insurance premiums increased by 251%¹. One practice hospitals have notoriously engaged in is “cost-shifting.” Cost shifting is a money-saving practice in which service providers seek higher reimbursement from some sources to offset the deficit created when providing charity care. However this has also been partnered with the fraudulent practice of inflating charges for individual items – saline, gauze, and bandaids – to cover the deficit caused by low reimbursement rates from insurance firms.



Current legislative attempts to confront this issue have been plagued with loopholes. Currently, hospitals are required to disclose the cost of individual items used in the treatment process. This requirement is frequently bypassed or complicated by hospitals, making it difficult for individuals to understand what they are being charged for in addition to a procedure or appointment.

Policy Proposal:

This policy aims to introduce greater transparency and accountability in billing practices within the healthcare sector. This policy intends to do so by requiring hospitals and healthcare providers to justify charges for individual items and services, as well as establish a standard for minimum reimbursement by insurance providers, to guarantee healthcare providers receive fair compensation for their services. This policy will be enforced via quarterly audits of healthcare and insurance firms; these audits would assess whether these firms are overcharging individuals, and would verify that insurance firms fairly compensate individuals and healthcare institutions. If firms and institutions do not abide by these updated standards, they will then receive an appropriate penalty that is proportional to the resulting rate hikes. This serves as a deterrent against unethical billing practices and incentivizes compliance with the established standards. Penalties could range from fines to corrective action plans; a variety of penalties will ensure that offenders face consequences proportional to the harm caused by their actions. With this policy,

¹ Scheffler & Shortell, 2023

patients will receive the care they need without being burdened by exorbitant costs, and healthcare providers will receive appropriate reimbursement for their services.²

Legislative analysis:

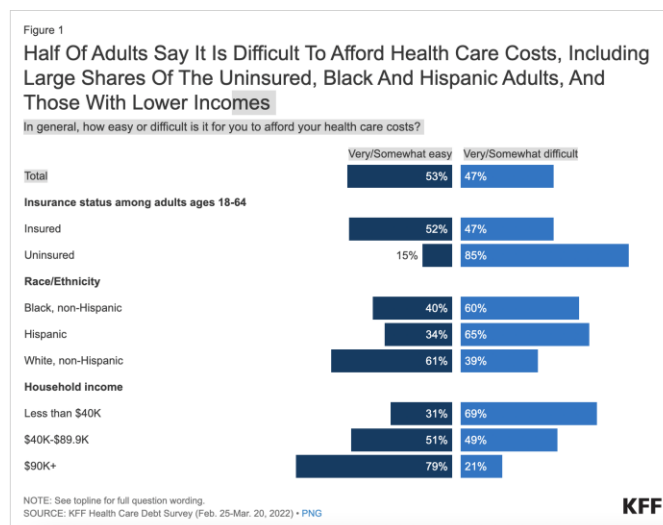
This policy will create more transparency and maintain accessibility within the healthcare system. It makes it possible to maintain a baseline standard for patients and prevents insurance firms from gaining too much influence via a checks and balances system.

Although some may argue that regulating insurance allows both state and federal government a greater influence in the medical industry, it is important to note that legislative authority is instrumental in defending common citizens and small and large medical practices from exploitation. Another argument against this policy is the potential for raised operating costs for hospitals. It is the medical provider's fiscal responsibility to purchase medical supplies for treatments, and assuming proper compensation from insurance providers, there is no need for individuals to cover operating costs. Medical providers have the opportunity to purchase in bulk, thus they pay less for both common and sophisticated medical supplies. This allows medical practices to turn a profit by charging more for individual medical supplies. For the two hospitals that received a penalty from the HHS thousands more are evasive while the HHS has not issued any further penalties. Even though 51.3% of hospitals posted negotiated prices, 49.8% failed to comply with accountability because the majority of data on the pricing was absent or unfinished.

Additionally, some may say that audits are time-consuming and hold the potential for fraud. To combat these possibilities, audits will be conducted at various overlapping periods to ensure a holistic review. The potential for fraud will be mitigated by post-audit planning that will discontinue the extensive cost to customers while avoiding harm to insurance providers. Prices are available online for x-rays and bloodwork that are provided during inpatient and outpatient admission visits. It is important to remember, that although the prices are available to pay in cash, when about one in five adults over the age of 65 ensured with Medicare the out-of-pocket expenses are high relative to their income. People with low incomes, below 200% of the FPL, had the highest rates of underinsurance, with no major difference between people in Medicare Advantage and traditional Medicare, after they accounted for the difference in income.

Conclusion:

The healthcare landscape in California is facing severe challenges due to skyrocketing insurance deductibles and inflated supply costs. The prevalence of insurance's raised pricing for service costs elucidates the necessity for greater transparency and accountability in insurance billing practices. The proposed policy seeks to rectify these shortcomings by introducing measures to catalyze transparency in insurance practices within the healthcare sector. By requiring hospitals and healthcare providers to justify charges for individual items and services



² Montero, A., Lopes, L., Presiado, M., & Hamel, L.

and establishing minimum reimbursement fee standards for insurance companies, we can ensure fair compensation for healthcare services and promote a more equitable healthcare system. Not only will there be transparency, there will be follow-through. There will be constant monitoring of large insurance practices to keep them in check and to make sure they are using the minimum reimbursement rates for healthcare practices.

This policy would lay down the foundation for a healthcare system that is transparent, equitable, and sustainable, with benefits extending to individuals, healthcare providers, and society as a whole. Increased transparency will empower consumers to make informed decisions about their healthcare, fostering a more patient-centered approach. This leads individuals to confidently navigate the healthcare system, knowing they are receiving fair treatment and value for their money.

By addressing the harsh disparities in reimbursement rates, this policy contributes to broader goals of social equity and public health by helping dismantle barriers to access and affordability, ensuring that all Californians have equal opportunities to receive quality healthcare. Furthermore, the policy supports the subcommittee's efforts to foster innovation and competition within the healthcare marketplace by preventing insurance monopolies from exerting undue influence and maintaining the checks and balances system. Therefore, this policy would combat critical issues within the healthcare system and promote equity and transparency. The policy embodies the values of the health subcommittee, ensuring a healthier future for all.

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Reproductive Policy Proposal: Regulation of Menstrual Products

Proposed by: Aanya Mittu, Aliya Koshaliev, Zisen (Samantha) Fan

Introduction

It has been almost 25 years since any new policy was created about the ingredients and chemicals allowed in menstrual products. Coupled with the lack of comprehensive research on impacts of the chemicals present in these products, it presents a concerning situation, especially when a projected 57.43 million women will use these products in 2024.³ With the studies that have been conducted, chemicals in menstrual products have been found to cause irritation, cancer, reproductive toxicity, and endocrine disruptor, making it crucial to create policies that restrict the use of these harmful chemicals.⁴

Name of chemical	Potential health impacts of exposure
Acetone	Irritation
Benzene	Cancer, anemia
Benzophenone	Cancer, endocrine disruption
Butylphenyl methylpropional	Reproductive toxicity
Butylphenyl methylpropional	Reproductive toxicity
Carbon disulfide	Reproductive toxicity
Chlorine	Irritation, corrosion
Chloroethane	Cancer
Chloroform	Cancer, reproductive toxicity, neurotoxicity

We hope with this policy proposal there will be more comprehensive research on the chemicals within menstrual products and how they impact the health of the people using them. As more research increases our knowledge, we will be better suited to make policies regarding the chemicals and ingredients that should and shouldn't be in menstrual products. We also hope this opens the door to more policies that positively benefit reproductive healthcare.

Previous legislation on the regulation of the ingredients of menstrual products has been

limited. The FDA passed a regulation in July 2005 regarding different aspects of menstrual products, including device description, necessary clinical studies, and labeling of the product.⁵ There is currently no federal legislation that requires the ingredients of menstrual products to be listed on its packaging, only state legislation. California's governor signed a law requiring ingredients in menstrual products to be listed, but it gives manufacturers trade secret protections, so not all ingredients are required to be disclosed.⁶

³ U.S.: Usage of sanitary pads and napkins 2011-2024. Statista. (2024, February 5).

<https://www.statista.com/statistics/287048/usage-of-sanitary-pads-and-napkins-in-the-us-trend/#:~:text=According%20to%20this%20statistic%2C%2057.43,pads%20and%20napkins%20in%202020.>

⁴ What's in period products? timeline of chemical testing - WVE. Women's Voices for the Earth. (2020, October 28). <https://womensvoices.org/whats-in-period-products-timeline-of-chemical-testing/>

⁵ FDA. (2005, July). Menstrual tampons and pads: Information for premarket guidance. U.S. Food and Drug Administration. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/menstrual-tampons-and-pads-information-premarket-notification-submissions-510ks-guidance-industry>

⁶ Federal rules don't require period product ingredients on packaging labels. States are stepping in. - CBS News. (2023, April 27). [Www.cbsnews.com. https://www.cbsnews.com/news/menstrual-pads-tampons-fda-ingredients-labels/](https://www.cbsnews.com/news/menstrual-pads-tampons-fda-ingredients-labels/)

Policy Proposal

Although surprising, current menstrual products used by women across the world tend to contain toxic chemicals and pesticides that slowly enter the bloodstream, causing long-term health effects. For instance, a Columbia University study conducted in 2019 found that chlorine bleaching, a common process used in manufacturing menstrual products to disinfect, contributes to the amount of toxins found in these products. Common pesticides such as organophosphates cause oxidative stress, leading to cell and tissue damage⁷. This same study found “evidence that there may be an association between tampon use and oxidative stress, but more studies are warranted,” emphasizing the great need for more modern research to be conducted on the necessity of these chemicals in the products. The effects these chemicals and toxins have on women’s bodies is not an issue to be taken lightly, especially considering that the health of American citizens should be a top priority to our policymakers.

For instance, the House of Representatives has also previously introduced a new bill titled the Robin Danielson Menstrual Product and Intimate Care Product Safety Act of 2023, but there unfortunately was no action taken since it was introduced. It’s crucial we work on this issue as soon as possible; as a solution, we propose a continuation of this bill. We demand a sharp increase in the quality and the amount of research on ingredients that are used in menstrual products. We also propose new federal policies requiring menstrual product ingredients to be listed on the packaging. Companies are not required by law to list their ingredients on the box of menstrual products, yet if most people who were to find out that chemicals such as phthalates, dioxins, and dioxin-like compound that have chances of causing cancer when used, most people would likely *not* buy or use these menstrual products.

In 2023 the National Institute of Health was reported to have spent 48 billion dollars on medical research, with some extra funds going toward grants⁸. We propose that a fraction of this money be spent on furthering research regarding menstrual products, their ingredients, and the long-term effects of these toxins in bodies.

Legislative Analysis

The benefits of this proposal are endless. First, this proposal will improve the health and hygiene of consumers as more harmful and synthetic materials will be identified, lowering the risks of health problems. By increasing research on this topic, consumer awareness will be raised, leading to consumers making more empowered and informed choices on which menstrual products they want to put on their bodies, reducing their exposure to harmful ingredients. As a result, with more research on safe menstrual products, there will be more regulatory changes aimed at improving safety standards for menstrual products, including stricter regulations on ingredients and better labeling practices. More research can also highlight the disparities marginalized communities face when it comes to accessing safe menstrual products. By recognizing this issue, there will be more equity in public health. Furthermore, more research will lead to overall education around menstruation, promoting safe and healthy reproductive health, and reducing the stigma around menstruation. However, increasing research funding for harmful ingredients in menstrual products has potential drawbacks. Funding large-scale research is very costly, which might cause product prices to go up. By forcing these groups

⁷ Pizzino, et. al. “Oxidative Stress: Harms and Benefits for Human Health.” National Library of Medicine, 2017. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5551541/#:~:text=Abstract,to%20detoxify%20these%20reactive%20products.>

⁸ National Institute of Health, October 2023 <https://www.nih.gov/about-nih/what-we-do/budget#:~:text=The%20NIH%20invests%20most%20of,research%20for%20the%20American%20people.>

to allocate their funds towards researching the ingredients in menstrual products, other public health initiatives could suffer due to the lack of resources. Intense amounts of scrutiny may inadvertently strengthen the stigma around menstruation, undermining the positive progress and education on menstrual health.

While funding large-scale research is very costly, the money invested is going towards finding harmful ingredients in menstrual products, which is vital for protecting public health. By avoiding these concerns, the number of health problems that may arise will lead to serious health problems, outweighing the short-term budget constraints. Additionally, other departments that may be affected by the amount of money being put into research funding for menstrual products can accommodate themselves. The concern about reinforcing the stigma around menstruation through intense scrutiny overlooks the potential for research to foster positive discourse and education on menstrual health. Addressing safety concerns can contribute to normalizing discussions about menstruation.

Conclusion

It is imperative to act now and increase the amount of research being conducted on the impacts of the chemicals and ingredients in menstrual products. History has often shown that we act too late in areas regarding the health impacts of certain products, such as the lack of research that was done on cigarettes during the 1930s, with such companies even promoting their cigarettes as being medically better for your throat.⁹ We do not want to find out ten, fifteen, or twenty years down the line that the chemicals in these products negatively impact the health of millions of people. Improving the standards in one aspect of reproductive health care such as menstrual products will help increase the standard of other areas of reproductive health care, also helping improve the standard of living.

⁹ Little, B. (2018, September 13). When Cigarette Companies Used Doctors to Push Smoking. HISTORY; A&E Television Networks. <https://www.history.com/news/cigarette-ads-doctors-smoking-endorsement>

Addiction Policy Proposal: Harm Reduction

Proposed by: Gin Williams, Charlie Folk, Charlotte Church, Sophia Zhu

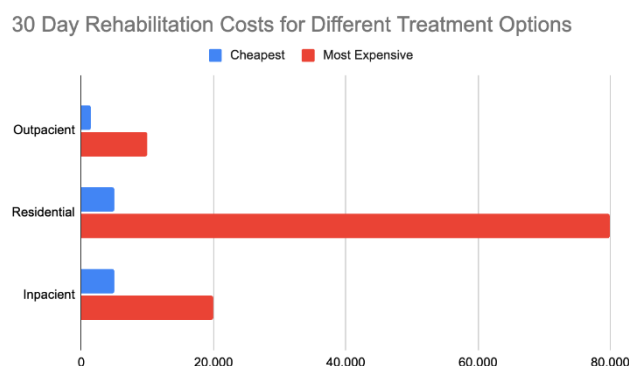
Introduction

Our policy proposal aims to address the issues of substance abuse and the lack of both access to treatment and harm reduction resources that individuals may encounter. According to a United States survey on Drug Use and Health in 2022, “an estimated 54.6 million people aged 12 and older needed treatment for a substance use disorder in the past year. Only 26% of those received any help for substance use disorder,” demonstrating the severity of the lack of treatment resources that exist.¹⁰

To help a wider range of individuals needing treatment facilities and from different socioeconomic backgrounds, our policy focuses on allowing more people to find community support and harm reduction options near them. This policy will tackle the need for increased tolerance and accessibility of substance abuse resources in economically disadvantaged locations by providing support to those who cannot afford expensive addiction treatments.

In the past, the United States’ war on drugs stance on the addiction crisis has been based on removing the source of addiction, which is good in theory, but has ultimately led to high incarceration rates for drug possession, namely in marginalized populations, and has also created a massive stigma around people who struggle with substance abuse.¹¹ The Biden-Harris administration has begun to work on changing this through increased accessibility to resources, but these programs need to be expanded and work to better support and target communities that need it the most.

An example of past legislation is the Providing Access to Treatment and Housing Act of 2023. However, this specifically extends substance abuse treatment and access to mental health services for unhoused individuals, failing to reach low-income individuals as well.¹² To address these gaps, our policy aims to expand these programs to help individuals in different socio-economic situations. Additionally, the Providing Access to Treatment and Housing Act does not work to address the stigma those struggling with addiction face, another issue we aim to address with our policy.



¹⁰ Alcohol and drug abuse statistics (facts about addiction). (2024, April 4). American Addiction Centers. Retrieved April 16, 2024, from <https://americanaddictioncenters.org/addiction-statistics>

¹¹ War on drugs. (2024, March 5). *Britannica*. Retrieved April 16, 2024, from <https://www.britannica.com/topic/war-on-drugs>

¹² Schiff, A. B., Rep. (2023, July 28). Providing access to treatment and housing act of 2023. Congress.gov. Retrieved April 16, 2024, from <https://www.congress.gov/bill/118th-congress/house-bill/4941/text>

In 2023, the Harm Reduction Through Community Engagement Act of 2023 was introduced in the Energy and Commerce Committee, which encouraged community engagement opportunities for those struggling with opioid treatment.¹³ While this legislation would improve community involvement and potentially reduce the stigma associated with addiction, we believe a separate piece of legislation targeting low-income and marginalized communities would increase accessibility for necessary addiction resources. We would also want to expand this legislation past only opioid treatment programs to help a wider range of those struggling with addiction, supporting them with safer use options and medication for a broader range of drug addictions.

Policy Proposal

Our proposed solution is to expand harm reduction facilities towards low-income communities to allow for a wider range of access. Substance abuse is prevalent in low-income households as individuals struggling with income often turn towards unhealthy coping mechanisms such as substance abuse.¹⁴ By increasing the availability of harm reduction facilities, a wider range of support can be offered to individuals in low-income areas. These facilities would offer syringe access, safer drug use and medication treatment to ensure that all drug use is done safely, and to ensure no overdoses occur, with trained personal oversight to ensure this. Having these resources would also make sure that drug use occurs more safely and would help lower the risk of harmful health effects (cross contaminated needles, overdoses) from drug use.

The War on Drugs policies, such as the “Just Say No” policy from the Nixon administration, have created a stigma around those who are struggling with drug abuse, leading to criminalization of drug possessions, and mass incarcerations of predominantly marginalized populations, as the stigma around drug use leads people to not getting health and ultimately being arrested and criminalized in many cases. This war on drugs mentality has created tight restrictions which can make it difficult for people struggling with drug abuse to get treatment without fear of police intervention and shame directed at them. Our policy means to address this by implementing a harm reduction system, which has been enacted in other countries such as the Netherlands and Germany.¹⁵ Their harm reduction policies aim to lessen the stigma around drug abuse so that those struggling can access treatment in a more community-involved approach that allows individuals to get support from a community and access to care. Instead of creating humiliation and guilt, harm reduction aims to inspire hope and create communities with resources for treatment. The Biden-Harris administration has made improvements to the harm reduction system, increasing capacity for harm reduction, but reaching this improvement into lower socioeconomic statuses would ensure that access is more broadly accessible to communities that need support the most.¹⁶ It is important to note that harm reduction facilities do

¹³Espallat, A. (2023, April 24). *Harm reduction through community engagement act of 2023*. Congress.gov. Retrieved April 17, 2024, from <https://www.congress.gov/bill/118th-congress/house-bill/2804?q=%7B%22search%22%3A%22harm+reduction%22%7D&s=1&r=2>

¹⁴ Correa, G. (2024, April 15). *Addiction and low-income americans*. Retrieved April 16, 2024, from <https://www.addictioncenter.com/addiction/low-income-americans/>

¹⁵ Mehrolhasani, M. H., Yazdi-Feyzabadi, V., Hajebi, A., & Mirzaei, S. (2019). Cross-country comparison of treatment policies facing the drug abuse in five selected countries. *Addiction and Health*, 11(2). <https://doi.org/10.22122/ahj.v11i2.233>

¹⁶*Biden-Harris administration marks two years of advancements in HHS' overdose prevention strategy with new actions to treat addiction and save lives*. (2024, February 1). Substance Abuse and Mental Health Services

not take the place of rehabilitation centers, but would offer a more supportive environment to get access to rehab and other resources, lowering overall stigma. Substance abuse treatment is

Treatment Type	Cost per Person	Cost per Person Adjusted for Inflation (2022)
Detoxification	\$121,381	\$141,001
Residential	\$43,446	\$50,469
Outpatient	\$7,219	\$8,386
Outpatient (Methadone)	\$6,414	\$7,451

expensive, with adult residential treatment ranging from \$4,008 to \$15,524 on average, and the average cost of drug rehabilitation per person being \$13,475.¹⁷ Harm

reduction facilities would offer a way for those who cannot afford these treatments to mitigate the negative consequences of drug usage and unsafe practices without an immense financial investment that they may not be able to afford.

Funding our proposal would cost significantly less than opening more treatment facilities in general, as it is more community and resource-based than actual treatments. According to a study in the *Harm Reduction Journal*, the cost per person for harm reduction treatment was estimated at \$729.72, with most of the cost going to personnel salary and overhead funds, as opposed to thousands for outpatient and residential programs, making it a more affordable to operate and therefore more affordable to open in communities which need it most.¹⁸ We would need to fund facilities to train staff and have materials for harm reduction such as overdose reversal resources, drug test kits, written educational materials, and distribution mechanisms to the community that needs it.¹⁹ To fund this policy, we would need to delve into existing budgets, potentially tacking this program onto the existing goals of the Biden-Harris administration's Advancements in HHS' Overdose Prevention Strategy with New Actions to Treat Addiction and Save Lives. Our funding may be drawn from President Biden's existing budget request of \$10 million for the 2025 fiscal year to launch more services aimed towards reducing substance abuse.²⁰ The budget could potentially allow for the expansion of harm-reduction treatment facilities in low-income communities. The funding could have a significant impact, especially considering a report from the National Library of Medicine revealing that individuals with an

Administration. Retrieved April 16, 2024, from <https://www.samhsa.gov/newsroom/press-announcements/20240201/biden-harris-administration-marks-two-years-advancements-hhs-overdose-prevention-strategy-new-actions-treat-addiction-save-lives>

¹⁷ *Average cost of drug rehab.* (n.d.). National Center for Drug Abuse Statistics. Retrieved April 16, 2024, from <https://drugabusestatistics.org/cost-of-rehab/>

¹⁸ Bartholomew, T. S., Patel, H., McCollister, K., Feaster, D. J., & Tookes, H. E. (2021). Implementation and first-year operating costs of an academic medical center-based syringe services program. *Harm Reduction Journal*, 18(1). <https://doi.org/10.1186/s12954-021-00563-8>

¹⁹ *Harm reduction.* (2023, April 24). Substance Abuse and Mental Health Services Administration. Retrieved April 16, 2024, from <https://www.samhsa.gov/find-help/harm-reduction>

²⁰ *Fiscal year 2025: Justification of estimates for appropriations committees.* (n.d.). Substance Abuse and Mental Health Services Administration. Retrieved May 1, 2024, from <https://www.samhsa.gov/sites/default/files/samhsa-fy-2025-cj.pdf>

income less than \$20,000 were 36% more likely to face issues with substance abuse compared to those with an income greater than \$75,000.²¹ The investment would help in targeting harm reduction facilities toward lower-income communities which is crucial given that the rates of substance abuse are shown to be higher.

Legislative Analysis

This proposal stands out in offering a more community- and education-based alternative, to treatment centers, for those who may need specific support from peers and realistically cannot afford high-cost treatments or have the capacity to undertake months-long recovery programs with a treatment center. We then center our proposal by targeting disadvantaged communities: working with these groups of people allows for an allocation of resources to those who need it most, as the people who cannot afford treatment may simultaneously not have the time to take off work and other responsibilities to get treated. By working to eliminate the stigma often associated with substance use, intertwining recovery efforts with counseling, and utilizing a network of referrals and services, harm reduction programs integrate individuals back into society and provoke them to let go of addiction practices.

Currently, policies have already been created in the sector of harm reduction, but the movement remains in its preliminary stages, and we strive to use the existing legislation as a framework to expand the reach of these projects to lower-income communities. Since the Biden administration has taken a harm reduction and overdose prevention stance on drug abuse rather than emphasizing abstinence, we can be in line with the existing goals of the administration. One of our largest concerns then becomes how to acquire the resources, people, and locations needed to put harm reduction in the United States on the map. Furthermore, staff would have to be trained through some sort of curriculum. The solution to that is the collaborative aspect of harm reduction. In many communities we can draw support from peers to serve as staff. Many basic treatments and services can also already be allowed under SAMHSA funds, shifting the budget discussion to instead focus on acquiring locations for distribution. The easiest locations would be large pharmaceuticals like Walgreens or Walmart, and we can partner with smaller local pharmacies also, who may be more likely to support the cause; all of these stores appear plentiful in the bay area and across the country. In terms of securing major retailers, we understand the legal concerns related to incentivizing illicit drug use, so a part of our proposal has been to also provide more information about harm reduction services, and we hope that more awareness about harm reduction can address the stigma around addiction and make it easier to overcome potential legal issues. The specific locations we have listed would also already have the instruments in place to preserve the treatment resources and keep them safe, preventing any foul play and lowering the possibility of harm reduction actually increasing rates of drug abuse. Overall, the pros of harm reduction stand out more than the cons, backed by studies that have shown harm prevention to be a more sustaining method than treatment centers at supporting those who are struggling and making an attempt to alleviate the issue at hand with less stake cost-wise than treatment facility operations.

²¹ Baptiste-Roberts, K., & Hossain, M. (2018). Socioeconomic disparities and self-reported substance abuse-related problems. *Addiction and Health*, 10(2). <https://doi.org/10.22122/ahj.v10i2.561>

Conclusion

Each member of our community deserves equal access to health care, and it is more often than not that individuals who are struggling with addiction are not properly supported, especially when they cannot afford expensive treatment. It's been proven that harm reduction and support group meetings have worked to lower the casualty rate of addiction, and also lower the amount of addiction in total. Our group hopes this policy can set up more harm reduction facilities across the United States, especially in lower social-economic communities. Through outreach and volunteer work, we hope to get trained staff to work in these harm-reduction facilities. These volunteers will not only provide clean supplies and testing strips for residents, but they will also help to educate the public on the importance of harm reduction, and provide further resources, such as treatment facilities and anonymous meetings for the disadvantaged.

Mental Health Policy Proposal: Health and Safety on Social Media

Proposed by: Ruchi Mangtani, Leo Terman, Arjun Singh, and Naiya Daswani

1 Introduction

Social media has become an aspect of most teenagers' daily lives. According to a 2022 Survey by the Pew Research Center, 95% of U.S. teens ages 13 to 17 use YouTube, 67% use TikTok, and 62% use Instagram.²² Though these platforms can be a source of enjoyment and connection, they can also be severely detrimental to teen mental health.

One of the major risks of teen social media usage is cyberbullying. Over 73% of cyberbullying occurs on social media platforms.²³ Furthermore, a survey showed that 46% of teenagers in the U.S. have been cyberbullied.²⁴ The most targeted demographic is 15 to 17-year-old girls, who are often bullied because of their physical appearance. This type of cyberbullying has been directly linked to eating disorders and comorbid weight gain.²⁵ Furthermore, adolescent perpetrators of cyberbullying often have high social media addiction scores.²⁶ Due to the prevailing cyberbullying issue, 75% of teens have said that they feel the government and social media platforms are not doing enough to solve this problem.²⁷

Even if one is not specifically cyberbullied, constant use of social media is frequently linked with mental health issues. One study found that those who used social media presented a 70% increase in depressive symptoms.²⁸ A study additionally found that social media addiction is linked to psychological stress and poor sleep quality.²⁹ Both consequences can impact teenagers' ability to pay attention in school and enjoy life outside of their screens. For these reasons, we see a need to limit social media usage among adolescents.

To limit unhealthy social media use among adolescents, we propose a guardian/parent consent requirement for ages 16 and under before being able to use social media apps. Parental/guardian involvement in social media consent could help protect vulnerable children from the negative implications of unhealthy social media usage. Such involvement could address comorbid mental health problems, inform digital consent to terms and conditions, and promote

²² Vogels, E., Gelles-Watnick, R., & Massarat, N. (2022, August 10). *Teens, Social Media and Technology 2022*. Pew Research Center. <https://www.pewresearch.org/internet/2022/08/10/teens-social-media-and-technology-2022/>

²³ *Effects of Cyberbullying*. (2024, February 26). Social Media Victims Law Center. <https://socialmediavictims.org/cyberbullying/effects/#:~:text=A%20Cyberbullying%20Research%20Center%20study,by%20many%20people%20very%20quickly>

²⁴ Vogels, E. (2022, December 15). *Teens and Cyberbullying 2022*. Pew Research Center. <https://www.pewresearch.org/internet/2022/12/15/teens-and-cyberbullying-2022/>

²⁵ Vogels, E. (2022, December 15). *Teens and Cyberbullying 2022*. Pew Research Center. <https://www.pewresearch.org/internet/2022/12/15/teens-and-cyberbullying-2022/>

²⁶ Kao, K. (2021, March 30). Social media addiction linked to cyberbullying. *UGA Today*. <https://news.uga.edu/social-media-addiction-linked-to-cyberbullying/>

²⁷ Vogels, E. (2022, December 15). *Teens and Cyberbullying 2022*. Pew Research Center. <https://www.pewresearch.org/internet/2022/12/15/teens-and-cyberbullying-2022/>

²⁸ Karim, F., Oyewande, A., Abdalla, L. F., Chaudhry Ehsanullah, R., & Khan, S. (2020). Social media use and its connection to mental health: A systematic review. *Cureus*. <https://doi.org/10.7759/cureus.8627>

²⁹ Karim, F., Oyewande, A., Abdalla, L. F., Chaudhry Ehsanullah, R., & Khan, S. (2020). Social media use and its connection to mental health: A systematic review. *Cureus*. <https://doi.org/10.7759/cureus.8627>

online self-efficacy.³⁰ With over 81% of adults supporting parental involvement in minor social media usage, such a proposal is feasible to have a significant impact.³¹

There are already many ways that children are restricted on social media. For example, according to Title 16 of the COPPA Act, children under 13 are protected from personal and targeted data collection by social media companies.³² The COPPA Act has directly prompted many social media companies, such as TikTok, Snapchat, Instagram, and Discord, to set 13 or older age limits to use their social media platforms. The Communications Decency Act (CDA) helps regulate harmful content online that children may be exposed to, such as indecency or obscenity.³³

However, Section 230 of the CDA states that social media companies are not liable for user-generated content, meaning that there is often a lack of incentive for social media companies to take down content that may be prompting harm to adolescents.

As such, addressing adolescent users being harmed/harming themselves on social media is difficult. With multiple amendments and attempts to repeal the section failing, a solution that aims to attack the liability of social media companies to more stringently regulate content may not be the most effective short-term solution.

Recently, Utah passed bills SB 194 and HB 464 which lock harmful and addictive features on social media as well as require social media companies to provide security measures in the form of tools and resources for parental/guardian regulation. A federal-level implementation of a similar framework may be an effective solution.³⁴

In order to promote health and safety on social media platforms, we propose requiring parental/guardian consent for individuals under the age of 16 to use social media platforms. Parents/guardians could help facilitate healthier social media usage and become a part of ensuring their children's digital safety.

2 Policy Proposal

To solve the aforementioned issues, the health subcommittee proposes the federal government to require social media companies to provide parental/guardian consent options for individuals under the age of 16 prior to opening social media accounts. With our policy, we recommend that there is increased federal regulation requiring social media companies to follow a framework that includes parental or guardian verification and a default curfew that runs from 10:30 pm to 7 am will be implemented by social media companies as a part of such framework, with an option for parental/guardian adjustment of the timing. Because the restrictions will be implemented by social media platforms with parent/guardian consent, the federal government can promote adolescent safety on social media without restricting First Amendment rights.

³⁰ Hammer, M., Scheiter, K., & Stürmer, K. (2021). New technology, new role of parents: How parents' beliefs and behavior affect students' digital media self-efficacy. *Computers in Human Behavior*, 116, 106642. <https://doi.org/10.1016/j.chb.2020.106642>

³¹ Anderson, M., & Faverio, M. (2023, October 31). *81% of U.S. adults – versus 46% of teens – favor parental consent for minors to use social media*. Pew Research Center. <https://www.pewresearch.org/short-reads/2023/10/31/81-of-us-adults-versus-46-of-teens-favor-parental-consent-for-minors-to-use-social-media/>

³² Part 312—Children's Online Privacy Protection Rule. (2013, January 17). Code of Federal Regulations. <https://www.ecfr.gov/current/title-16/chapter-I/subchapter-C/part-312>

³³ Brannon, V., & Holmes, E. (2024, January 4). *Section 230: An Overview*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/R/R46751#:~:text=Section%20230%20of%20the%20Communications,users%20of%20interactive%20computer%20services.>

³⁴ Home. socialmedia.utah.gov. (2024, April 25). <https://socialmedia.utah.gov/>

Additionally, until a teen turns 16, parents/guardians will have access to their child's account with the ability to view and monitor messaging, search history, and content displayed on their feed. Lastly, messaging regulations will block unapproved direct messages ensuring user privacy. Through these measures, adolescents can have a safer digital experience while prioritizing their mental health and well-being.

3 Legislative analysis

The proposed policy of implementing parental/guardian consent for individuals aged 16 and under, coupled with a curfew restricting app usage like after 10 pm, presents several notable advantages. Firstly, it prioritizes the promotion of healthy sleep habits among minors. Extensive research from the American Academy of Child and Adolescent Psychiatry (AACAP) indicated that excessive screen time before bedtime disrupts sleep patterns, potentially leading to detrimental effects such as difficulty concentrating and mood disturbances. By establishing a curfew, the policy seeks to ensure that minors obtain adequate rest, mitigating the risks associated with sleep deprivation. Studies have shown a correlation between nighttime screen time and sleep disturbances in adolescents, impacting mood, learning, and overall mental health.³⁵ By setting a curfew better sleep habits will be encouraged and therefore the developing adolescents will be able to avoid the aforementioned issues. Additionally, the proposal aims to address the growing concern of screen time addiction among adolescents. Furthermore, studies have shown a correlation between excessive social media usage and mental issues like anxiety and depression.³⁶ By mandating a screen time limit, the policy endeavors to reduce addictive behaviors and encourage diversification of activities. Moreover, the implementation of parental/guardian consent for app usage strengthens parental/guardian oversight and control, aligning with recommendations from the Society for Research on Adolescence. Granting parents/guardians the authority to regulate their children's online interactions empowers them to safeguard against inappropriate content and potentially harmful influences on their children. This policy proposal addresses pressing concerns surrounding adolescent well-being and digital consumption and underscores the importance of parental involvement in fostering a safe and balanced online environment for minors.

Limitations: One significant concern is the issue of privacy. The requirement for parental/guardian verification inherently involved the collection and possibly the storage of sensitive information. This raises questions about the security and privacy of the data of both children and their parents/guardians but also about who has access to this information and how it is protected.³⁷ Furthermore, the effectiveness of implementing a curve to restrict app usage from 10:30 pm to 6:00 am could be quite limited. Minors might circumvent these restrictions using alternative devices that are not monitored by their parents/guardians, or by altering their account details to appear older. Additionally, such a curfew does not address the nature of the content accessed but merely the time of access, which may not effectively mitigate the risks or exposure to harmful content. More comprehensive content moderation strategies might offer a robust

³⁵ Hale, L., & Guan, S. (2015). Screen time and sleep among school-aged children and adolescents: A systematic literature review. *Sleep Medicine Reviews*, 21, 50-58. <https://doi.org/10.1016/j.smrv.2014.07.007>

³⁶ Schønning, V., Hjetland, G. J., Aarø, L. E., & Skogen, J. C. (2020). Social Media Use and Mental Health and Well-Being Among Adolescents – A Scoping Review. *Frontiers in Psychology*. <https://doi.org/10.3389/fpsyg.2020.01949>

³⁷ Wong, M. (2017, August 2). *Pizza over privacy? A paradox of the digital age*. Institute for Economic Policy Research. <https://siepr.stanford.edu/news/pizza-over-privacy-paradox-digital-age>

solution by focusing directly on the appropriateness of content rather than solely limiting the time of access. While well-intended, the policy may not only impinge on privacy rights but also fall short of achieving its intended protective measures against the unsupervised digital engagement of minors due to various loopholes.

4 Conclusion

Teenagers are exposed to the entire world on social media, where they can face cyberbullying, compare themselves to others, and spend hours a day scrolling. All of these factors can cause disastrous, long-term consequences on their mental health and well-being. Rather than prohibiting teenagers from social media entirely, we hope to create a more guided and informed experience on these platforms by involving their parents/guardians and minimizing their time online. In doing so, we intend to set teenagers up with healthy social media habits so that as they transition into adulthood where they will have free reign over these platforms, they may be more cognizant of how to maintain a balanced relationship with social media in order to create an overall healthy lifestyle.

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Integrating AI in Education

Brandon Ha, Abigail Fukuhara, Kate Lee, Hayden Leung, Asheeka Noronha

1. Introduction

With the rise in AI comes new ways to utilize artificial intelligence, both as a beneficial tool for good or for more negative impact. One debated and important facet of this is the use of AI in education in the US. Although AI is often seen as a cause of harm in the education system, there should be increased emphasis on the unique and important ways it can benefit students and teachers. However, there currently is no funding specifically allocated towards supporting the implementation and long-term use of AI in education.

The implementation of AI should increase efficiency in education and prepare students to utilize AI tools when they graduate and enter the real world. Our policy aims to aid teachers in decreasing their busy work and provide more personalized learning plans to students. We do not want the education field to fall behind in technology as the outside world advances at a rapid rate.

The Universal Service Program for Schools and Libraries, also known as E-rate, is part of the Telecommunications Act of 1996. It provides subsidies of up to 90% to help ensure that qualifying schools and libraries have access to high-speed internet and telecommunications at affordable rates.³⁸ However, this program addresses increasing internet access, while our policy will include grants that increase specifically AI resources. Also, our policy will apply to a wider range of educational institutions, not just those in rural and economically disadvantaged areas.

The Every Student Succeeds Act (P.L. 114-95) of 2015, which reauthorized the Elementary and Secondary Education Act of 1965, authorized the use of computer adaptive testing in state student academic assessments mandated under the act. This was the first time Congress explicitly approved an AI testing technique for widespread use in schools.³⁹ However, this Act merely approved AI testing methods, while our policy will fund more AI mechanisms.

The Elementary and Secondary Education Act (ESEA) (P.L. 89-10, as amended) was enacted in 1965 and was last reauthorized in 2015 by the Every Student Succeeds Act (ESSA; P.L. 114-95). Title I-A, the largest ESEA program, provides compensatory grants to local educational agencies (LEAs) to fund educational and related services for low-achieving and other students who attend elementary and secondary schools with relatively high concentrations of students from low-income families.⁴⁰ However, this Act is more general in what it'll fund, while our policy will specifically fund AI resources to not just low-income but all students.

³⁸ Lu, J. J., & Harris, L. (2018). *Artificial Intelligence (AI) and Education - CRS Reports*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/IF/IF10937>

³⁹ Lu, J. J., & Harris, L. (2018). *Artificial Intelligence (AI) and Education - CRS Reports*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/IF/IF10937>

⁴⁰ Stoll, A., Skinner, R. R., & Smole, D. (2022). *A Summary of Federal Education Laws Administered by the U.S. Department of Education - CRS Reports*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/IF/IF10551>

2. The Policy Proposal

We propose allocating funds from the Higher Education Act of 1965 towards obtaining Artificial Intelligence resources for educational institutions to increase classroom efficiency and teach students how Artificial Intelligence can be utilized to improve their lives. Funds previously allocated to section 402C of the Higher Education Act--which provided funds to generate higher interest in education beyond secondary school--will be redirected to this new AI program. As of 2021, 91% of U.S. citizens hold a high school diploma; therefore, the precedent for pursuing higher education is well established, reducing the funds required to continue promoting higher education. The funds for programs authorized by section 402C are no less than \$190,000, which is enough to fund services, such as [AI for Education](#)⁴¹ and Gruvy Education⁴², that:

- a. [Train](#) teachers on how to utilize AI
- b. Help schools [adopt](#) a strategy for AI implementation
- c. [Develop](#) AI policies that can be used at school

Our policy addresses concerns over the rapid growth of AI, especially regarding education and how it can effectively be used to help all students. Particularly since the release of OpenAI's ChatGPT program, there have been worries about academic integrity. Students know that AI technologies like ChatGPT exist, and many are already experimenting with them, but they need guidance about using them responsibly. Thus, educators and educational programs should not ignore or pretend AI does not exist, as doing so only further allows potential irresponsible use.⁴³ Our policy will not only encourage schools to confront AI, but it will also provide AI resources so that AI can be used as a supportive and adaptive tool that's not taken advantage of.

We will fund our proposed policy by redirecting funds from the Higher Education Act of 1965 (HEA; P.L. 89-329, as amended) that authorizes a broad range of federal student aid programs including those that provide federal support to postsecondary institutions of higher education (IHEs) in improving their capacity and ability to offer postsecondary education programs. The HEA is organized into eight titles including Title II, Teacher Quality Enhancement, that authorizes grants for improving teacher education programs. Part B of Title II authorizes competitive grants for teacher training programs such as preparing graduate teacher candidates to use technology-rich teaching methods.⁴⁴ Our policy of providing AI resources can be considered a technology-rich teaching method.

The Department of Education (ED) administers the programs authorized by the HEA. In FY2022, ED provided approximately \$3.6 billion in federal support to IHEs under the HEA.⁴⁵

⁴¹ AI for Education. (n.d.). <https://www.aiforeducation.io/>

⁴² Gruvy Education (n.d.). <https://gruvyeducation.com/traininginfo>

⁴³ Ross, E. M. (2023). Embracing Artificial Intelligence in the classroom. Harvard Graduate School of Education. <https://www.gse.harvard.edu/ideas/usable-knowledge/23/07/embracing-artificial-intelligence-classroom>

⁴⁴ Shohfi, K. & Zota, R. (2023). *The Higher Education Act (HEA): A Primer - CRS Reports*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/R/R43351>

⁴⁵ Shohfi, K. & Zota, R. (2023). *The Higher Education Act (HEA): A Primer - CRS Reports*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/R/R43351>

Additionally, in FY2018, ED's Teacher Quality Partnership (TQP) competition plans to award approximately \$14 million in grants to preparation programs for teachers.⁴⁶ According to the Congressional Research Service, "If Congress decides to support funding teacher preparation for AI, options could include redirecting funds toward teacher technology training and directing ED to develop the best practices for teacher technology competency."⁴⁷

3. Legislative Analysis

Our policy contains various benefits for both school systems, teachers, and students.

Personalized Learning, which allows each student's personalized needs and learning styles to create additional support systems and a less intimidating curriculum, ensures that students will receive instruction for their level and pace. Personalized learning may be particularly helpful for students with disabilities for the best chance of academic success. **Immediate feedback** creates quick progress and updates regarding a student's performance that do not depend on other students. **Accessibility** ensures that every student will have access to their learning that is quick and efficient, without conflicting with other students and being easy to pull up. The issue of **scalability** will no longer be a problem, meaning that large numbers of students can be taught without compromising quality. **A more knowledgeable basis in school**, with aid from both teachers and AI, provides many more resources for students when they are struggling.

Our policy contains various benefits for both school systems, teachers, and students. However, possible concerns regarding the implementation of this technology may arise. **Lack of Human Interaction**; Personalized AI tutoring may lack the human touch and interpersonal connection, with some students preferring face-to-face interaction for emotional support. However, teachers should still be present in classrooms and Human interaction will not be completely cut off as the AI system is just a tool. **Technological Barriers**; Students need access to technology or digital literacy to participate, requiring initiatives to ensure equal access. However, the cost of technology should be addressed by the funds allocated by the money available. **Privacy Concerns**; Privacy concerns arise from extensive data collection. This means that data protection measures are necessary. Terms and policies can be added to fit users' needs. **Training and Jobs**; Adapting to AI-based learning methods may require extensive teacher training. However, as a part of implementing AI technology, teachers and educators would be provided with the necessary training and instruction to prepare and use AI systems. Additionally, there's the potential for a decrease in the demand for teaching positions with increased AI adoption.

4. Conclusion

Our proposal will instigate a new era in education marked by introducing AI into the school curriculum. According to [Bloomberg](#)⁴⁸, the generative AI market is predicted to grow to

⁴⁶ Lu, J. J., & Harris, L. (2018). *Artificial Intelligence (AI) and Education - CRS Reports*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/IF/IF10937>

⁴⁷ Lu, J. J., & Harris, L. (2018). *Artificial Intelligence (AI) and Education - CRS Reports*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/IF/IF10937>

⁴⁸Bloomberg. (2023, June 1). *Generative AI to become a \$1.3 trillion market by 2032, research finds*. Bloomberg.com. <https://www.bloomberg.com/company/press/generative-ai-to-become-a-1-3-trillion-market-by-2032-research-finds/>

\$1.3 trillion over the next 10 years. With AI becoming an integral part of people's lives, it will become essential that all individuals are educated about the topic. Additionally, with AI implementation, ethical implications also arise. Our proposal will help schools develop policies that make implementing AI smoother.

With the introduction of technology into education a few years ago, there were many observed improvements including better access to information, more interactive and engaging learning, higher efficiency, and more⁴⁹. Similarly, introducing AI will cause even bigger changes in efficiency and flexibility at school.

⁴⁹Ascione, L. (2023, December 11). *The Impact of Technology on Education*. ESchool News.
<https://www.eschoolnews.com/it-leadership/2023/12/11/the-impact-of-technology-on-education/>

The AI Accountability Act

Hayden Fu, Claire Luo, Lakshya Agarwal, Zijing Zhang

1. Introduction

- a. Established in 1996, Section 230 of the Communications Decency Act⁵⁰ was an influential policy that provided immunity to providers of online media services from third-party content created by users. While Section 230 successfully catalyzed innovation on the early internet by protecting platforms from liability for user-generated content, it unintentionally reduced the incentives of online platforms to actively address illegal activity on their services and, at the same time, left them free to sabotage transparency or accountability.

Today, as the rise of AI becomes an ever-growing trend, the implications of AI's ability to directly generate content of all sorts without limitation raises a huge concern. The lack of regulation around AI-generated content creates uncertainty around liability, copyright, transparency, and oversight obligations. Without new guidelines, generative AI could be exploited to produce harmful and factually flawed content like misinformation, hate speech, impersonations, and copyright infringement backed by believable media evidence at unprecedented scales with limited accountability.

- b. The proposed **AI Accountability Act** aims to address the existing gap in the law by generating accountability for malicious or unwanted AI usage. It establishes clear responsibilities for AI developers to ensure their systems are safe and transparent, obligations for operators to use AI responsibly, and liability standards when AI outputs cause harm. The Act mandates disclosure of AI-generated content, prohibits high-risk use cases, requires human oversight for sensitive applications, and enables auditing of AI models.
- c. Section 230 of the Communications Decency Act has been in place for almost half a century, yet it is not adequately able to keep up with the pace of rapidly developing technological advancements to regulate them. The once-designed policy to protect internet service providers should not be carried over to the emerging field of generative Artificial Intelligence. This is because the internet is focused on communication and information sharing, but AI can be expanded on automation and decision-making, according to Dr. Oscar W Mannbro⁵¹. AI has the potential to automate many tasks that are currently performed by humans, and left alone, AI can independently study and simultaneously improve itself at an exponential speed every second through unsupervised learning techniques like

⁵⁰ Benson, P. J., & Brannon, V. C. (n.d.). Section 230 immunity and Generative Artificial Intelligence. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/LSB/LSB11097>.

⁵¹ Oscar W Mannbro, "Ai and the Internet: A Comparison of Their Impact on Society," March 2, 2023, <https://www.linkedin.com/pulse/ai-internet-comparison-impact-society-oscar-w-mannbro#:~:text=Similarities%20and%20Differences%20between%20AI%20and%20the%20Internet&text=While%20the%20Internet%20is%20focused,data%20entry%20and%20customer%20service.>

reinforcement learning or GAN (generative adversarial networks). Therefore, taking into account the unprecedented power and influence of AI, new, carefully thought-out, and stricter limitations desperately need to be implemented.

2. The Policy Proposal

- a. The AI Accountability Act is aimed to hold AI developers and companies of generative AI models liable to an extent if the AI they developed commits harm or damage to third party individuals or groups.
- b. Even though the developers may not necessarily have control over what the users will request the AI models to do, our policy is only trying to form an incentive among the developers that they have to be wary of how they train the AI model and how to prohibit certain types of response from possibly generating.
- c. We do realize that most of these generated contents will be used on social media platforms such as Facebook, YouTube, or X. However, these platforms, as long as they are not or not related to the developers of an AI model, will not be held responsible for the scope of our policy.
- d. The AI Accountability Act will only hold developer companies that programmed and trained any given AI model to be responsible. This is crucial to preventing and minimizing occurrences of AI models that generate harmful and abusive content in the first place.
 - i. *Ethical implications:* AI-generated content can easily be manipulated by the operator (user). With such a wide range of not only easily accessible but also near-instantaneous and high quality generative AI models such as ChatGPT and Midjourney, users can easily instruct and guide the AI model to generate hateful, harmful, abusive, false (or any combination of the above) content without proper regulation on the technology.
 - ii. *Legal accountability:* If AI contents cause harm, there needs to be accountability. While the most common voice will be to penalize the operator who demanded those types of content, we think it is much more effective and straight to the root of the problem by requiring the service providers to make corresponding regulations to prevent any creation of these types of contents in the first place.
 - iii. *Responsible development:* Thus, when developer companies know that they will be held primarily responsible for the consequences of their AI models, they are more likely to be careful during the training and programming of the AI model by adding more safety measures and detection methods. Some of the methods that should be taken to test the immunity to abuse of these AI models may include human testing, sample validations, and ongoing monitoring.
- e. There exist dozens of instances of abuse and potential threats that AI-generated content can bring; some specific real-world cases are voice cloning scams, fake

video evidence, political manipulation, and disinformation, and medical diagnosis counterfeit.

- i. *Voice cloning scams*⁵²: An individual receives a phone call from their family member or an intimate person and after picking it up, a familiar voice sounds from the other side of the line. They subconsciously let their guard down and will likely agree to follow what the scammer directs them to do (e.g. transferring money to an account for a “medical emergency.”) This is not irrational behavior given the often high-stakes situation that the scammer utilizes to pressure prompt action, and this type of scam is difficult to prevent given that AI technology nowadays can clone anyone’s voice with near-perfect accuracy with only an input of a short snippet or sample of an individual’s real voice. Fake voices can then generate audio that is indistinguishable from the real person and be used in phone calls to trick victims into providing sensitive information or transferring money.
- ii. *Fake video evidence*⁵³: Deepfake technology that has been used to create fake videos is no longer much of a stock today, but what if a completely forged and fake video were to be created as evidence for legal cases? Such examples could involve altering surveillance footage or tampering with testimonial videos. This completely false and manipulated video evidence can damage an individual’s reputation and severely undermine the integrity of the legal system.
- iii. *Political manipulation and disinformation*⁵⁴: Equipped with the unbounded possibilities of AI, extreme political organizations may employ AI technology to conduct disinformation campaigns aimed at influencing elections, undermining the democratic process. This strategic technique is not new, yet AI mockups and generated content heavily complicate the process of debunking or disproving such campaigns, something which already requires significant time and effort on the voters’ end. The AI Accountability Act can be implemented to completely block the possibility of such self-generated content by prohibiting AI models from being able to do so.

⁵² Charles Bethea, “The Terrifying A.I. Scam That Uses Your Loved One’s Voice,” *The New Yorker*, March 7, 2024, <https://www.newyorker.com/science/annals-of-artificial-intelligence/the-terrifying-ai-scam-that-uses-your-loved-ones-voice#:~:text=Last%20year%2C%20the%20F.T.C.,use%20of%20voice%2Dcloning%20technology>.

⁵³ Writingsofrach, “AI Deepfakes Are Poised to Enter Court Proceedings at Time of Low Trust in Legal System,” *CNBC*, March 6, 2024, <https://www.cnbc.com/2024/03/06/ai-deepfakes-are-headed-to-court-at-time-of-low-trust-in-legal-system.html>.

⁵⁴ Ali Swenson and Kelvin Chan, “Election Disinformation Takes a Big Leap with AI Being Used to Deceive Worldwide,” *AP News*, March 21, 2024, <https://apnews.com/article/artificial-intelligence-elections-disinformation-chatgpt-bc283e7426402f0b4baa7df280a4c3fd>.

- iv. *Medical diagnosis counterfeit*⁵⁵: AI algorithms used in medical diagnosis and healthcare systems can be manipulated to produce false results or recommendations. For example, attackers can command AI to misdiagnose patients by changing/forging first-hand data in the internet system first hand, which would be virtually impossible to find out.
- v. *Other cases*: Other abuses of AI can be defined as a serious violation of legal rights, the destruction of social image, and/or a lack of responsible development.
- f. We have carefully considered the potential issues in funding such an AI Accountability Act may raise, and luckily, only passing a policy into law won't require as much budget and funding as compared to other more direct regulations. The only downside that may come is on the company's side, as currently existing Artificial Intelligence giants would need to update and enhance their current systems and policies, which can be tedious and require significant resources. However, once changes following the Accountability Act have been implemented on existing models and platforms, all future AI models should become easier to follow the predecessors' paths.

3. Legislative analysis:

- a. *Pros*: All major abuses of AI-generated content have been addressed and further violations have been outlined clearly, so no loopholes can be exploited in terms of a designation of an unmentioned abuse. Additionally, this act can be implemented without major costs or funding needed, and the creation of an act rather than a regulatory agency or a more major legislative option facilitates this economic possibility. This policy will set a precedent for future legislation involving people or companies that create AI so that people will receive meaningful protection from AI no matter how much AI has developed. Section 230 of the Communications Decency Act did not foresee the creation of AI to this degree, particularly in regards to generative AI, nor did it predict the advent and rapid adoption of social media in the last two decades. With AI developed as it is, this AI Accountability Act will remedy the issue of legislation being outdated.
- b. *Cons and Refutations*: Upon the passing of the AI Accountability Act, developer companies may need to make drastic changes to pivot and adapt to the new policies proposed. This initial process may be seen as tedious and resource-draining; however, once these changes have been made, they can serve as a model for future AI and as a baseline as to what is expected of future developer companies involved with AI. Furthermore, this proposal could be taken by some as an attack on AI-generated content; however, this is not the case - the proposal

⁵⁵ Jessica Davis, "Study: Medical AI Diagnosis Models Can Be Manipulated to Output Wrong Diagnosis," SC Media, June 23, 2022, <https://www.scmagazine.com/analysis/study-medical-ai-diagnosis-models-can-be-manipulated-to-output-wrong-diagnosis>.

merely seeks to assign accountability for misuse and misconduct regarding AI-generated content. Also, this proposal could be taken as similar to a previous bill that has been introduced in the House recently, H.R 7766 - Protecting Consumers from Deceptive AI Act. However, this act only serves to make “the identification of content created by generative artificial intelligence”⁵⁶ clear in the context of protecting consumers, and not to discern accountability for AI-generated content. Finally, while Section 230 provides the basis for our argument as to what needs to be changed, we do not seek to change Section 230 to accommodate AI. Instead, our proposal specifically considers AI and who gets accountability for models that generate unwanted content; similar to what Section 230 does, but a different argument entirely and completely separate from the already-passed law. For example, in LSB11097, Benson discusses the possibility of Section 230’s implementation on AI, stating that “Section 230’s application could therefore vary across different generative AI products, different applications of a single product, and different legal claims about an application or product”⁵⁷. Because Section 230 has already been defined as a separate entity that did not predict the rise of AI, the ambiguity present in the possibility of Section 230 being modified to accommodate AI is exactly what the AI Accountability Act seeks to avoid, with clear definitions and goals specifically geared towards change in AI.

4. Conclusion:

- a. Our proposal will hold AI developer companies primarily responsible when their AI is misused to create malicious content. We believe that at this time, this legislation is crucial because AI is a rapidly evolving and advancing technology, and the freedoms granted by Section 230 of the Communications Decency Act should not be so readily applied to the products of AI. We also believe that AI still has plenty of room to grow—it may remain the most influential technology in society for several decades given its rate of advancement in the past few years. Thus, no matter what form AI takes years from now, it is necessary that legislation be established to hold developer companies accountable and most efficiently handle the root of the problem.
- b. Our subcommittee is primarily concerned with the larger issue of people and corporations not being held liable for dangerous content that is spread online. The issue of malicious content being generated is not unique; anyone can be impacted by AI content being generated which can create fake voice clones, generate fake videos, affect politics through misinformation, and provide false medical

⁵⁶ S.2691 - 118th Congress (2023-2024): Ai Labeling Act of 2023 | congress.gov | library of Congress. (n.d.). <https://www.congress.gov/bill/118th-congress/senate-bill/2691?s=1&r=55>

⁵⁷ Benson, P. J., & Brannon, V. C. (n.d.). *Section 230 immunity and Generative Artificial Intelligence*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/LSB/LSB11097>

diagnoses. All of these issues of misinformation should be taken into consideration when limiting the freedoms that AI-generated content should be allocated.

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Misinformation

Names: Alex X., Emma Yang, Aadit Noronha, Vayun Krishna, Rishan Patel, and Jonah Frenkel

Introduction

Over 50% of Americans obtain news from social media, and the nature of social media makes it easier for misinformation to spread. The misinformation on social media can be disproportionately impacted by a relatively small amount of people - 15% of users were able to contribute to 40% of misinformation on Facebook, for example.⁵⁸ Support for greater regulation and moderation is present in America. This proposal exists in order to make information spread online more reliable and reduce the spread of misinformation that has been facilitated by social media. This proposal's goal is to hold social media companies accountable for the disinformation they spread and encourage them to disincentivise disinformation on their platforms. Section 230 has run counter to the proposal's goal to reduce disinformation by not holding companies liable for information posted by someone else on their platforms. The problem is that many social media companies do not make an effort to combat disinformation, or implicitly encourage misinformation's spread. Our policy would prevent disinformation from getting a foothold on social media platforms, encouraging them to eliminate it early so they are not held accountable for the content on the sites.

Reason for policy proposal

Over 50% of Americans get at least some of their news through social media.¹ Disinformation has been spread more often and more easily due to social media and its own algorithms - 15% of users on Facebook contributed 40% of the platform's misinformation due to how the platform's algorithm functioned.¹ In fact, 65% of Americans believe that tech companies should be moderating false information on their platforms, with 55% agreeing that the federal government should be in charge.⁵⁹

Goal

The goal of this policy is to hold social media companies accountable for disinformation spread on their platforms, and encourage social media companies to improve their moderation effectiveness.

Previous Legislation

Section 230 prevents computer-based services from being held liable for information posted by someone else. However, these services often implicitly encourage disinformation or false

⁵⁸ Micich, A. (2023). *How misinformation on social media has changed news*. U.S. PIRG Education Fund. <https://pirg.org/edfund/articles/misinformation-on-social-media/>

⁵⁹ Aubin, C. St., & Liedke, J. (2023, July 20). *Most Americans favor restrictions on false information, violent content online*. Pew Research Center. <https://www.pewresearch.org/short-reads/2023/07/20/most-americans-favor-restrictions-on-false-information-violent-content-online/>

information for engagement metrics or do not make enough of a systematic effort to prevent it. Section 230 acts as a legal shield for these companies, allowing them to promote malicious content and disinformation with no consequence. Our policy fills in these gaps by holding companies accountable for malicious disinformation present and implicitly promoted by the site.

The policy proposal

Our proposed solution to this problem of misinformation spread without any consequences for the platform is to amend Section 230 of the Communications Decency Act to prevent disinformation spread with malicious intent, by removing Section 230 immunity when social media companies knowingly utilize algorithms that spread and promote harmful/false news. Nevertheless, it will likely face resistance by social media companies and by Congress across party lines, as it would involve the changing of an entrenched policy that has existed since the 1990s.

Proposed solution

Amending Section 230 of the Communications Decency Act to prevent disinformation spread with malicious intent, by removing Section 230 immunity when social media companies knowingly utilize algorithms that spread and promote harmful/false news. The problem addressed is the question of whether or not social media companies are responsible for false information spread by users of their respective platforms.

Funding

Given that this is a proposed amendment to a law, there is no necessity for funding.

Difficulties

Resistance from social media companies, Congress, both across party lines and due to the fact that this policy would involve changing a piece of legislature from the 1990s. However, amendments *have* been made. See:

- “Fifth, after passage of the Allow States and Victims to Fight Online Sex Trafficking Act of 2017 (FOSTA), Section 230 immunity does not extend to claims under certain sex trafficking provisions.”

This sets precedent for follow-up acts to create exceptions to Section 230. In our case, this means misinformation, as opposed to sex trafficking.

However

Removing section 230 does *not* inherently render social media companies liable, it simply incentivizes them to stop promoting disinformation. There has to be another law prohibiting the activity as well.

The last time this happened, it was not a simple amendment, it came with its own act. Presumably, a “Fight Misinformation” act would have to be created, which is difficult due to its broad, abstract nature and subjectivity.⁶⁰

Legislative analysis

Our proposal would improve the quality of information accessed by people online. Misinformation currently has a great deal of influence on US citizens⁶¹ - 64% of adults believe it is causing “a great deal of confusion” and 23% admitting to have shared misinformation, whether knowingly or not. The rapid spread of misinformation allowed by the Internet could be curbed by our proposal, and would lead to greater trust in shared information in today’s heated political climate. However, defining misinformation has historically been difficult. A key distinction to keep in mind when proposing a way to counter misinformation is to focus on *disinformation* - misinformation created with malicious intent, thus being more dangerous than simple misinformation in general. Spreaders of disinformation generally get the benefit of the doubt when considering “malicious intent,” as it is difficult to determine intent and under-censorship is preferred over over-censorship. For practical purposes, it is often defined as misinformation with a cost to the recipient, more specifically, misinformation that benefits the sender in some way.

Benefits & Drawbacks

Curbing the spread of disinformation by holding social media companies liable will improve the quality of information on social media platforms. Additionally, an increase in the quality of information will increase the trustworthiness and reliability of social media sites. By holding companies accountable, we will see an increase in effective moderation.

The internet has long avoided censorship, and this lack of censorship has been the backbone of innovation for it. With the possibility of over censorship, innovation on social media could plummet, potentially stifling small companies. Furthermore, the question of what is and isn’t considered false are quite difficult to resolve in certain intents, as well as how one can determine malicious intent.

It would additionally require a limitation of user-generated content eg. comments under a video/blog. This could be seen as overly restrictive to certain posters.

When it’s hard to determine whether or not something is disinformation, a human team could be used, even if it does bring in subjectivity, in order to fairly determine what is and is not disinformation and to firmly define what is and is not malicious intent.

⁶⁰ Allow States and Victims to Fight Online Sex Trafficking Act of 2017, H.R.1865, 115th Cong. (2017) <https://www.congress.gov/bills/115/congress/house-bill/1865>

⁶¹ Barthel, M., Mitchell, A., & Holcomb, J. (2016, December 15). *Many Americans believe fake news is sowing confusion*. Pew Research Center’s Journalism Project. <https://www.pewresearch.org/journalism/2016/12/15/many-americans-believe-fake-news-is-sowing-confusion/>

Conclusion:

By curtailing disinformation and holding social media companies accountable for maliciously misinformative content on their platform, our proposal will lead to greater trust in information on social media and will prevent users from being exposed to constant falsehoods. Our subcommittee aims to address and reduce the spread of misinformation. By holding social media companies accountable for disinformation, we hope to ensure that the quality and spread of information will improve, while simultaneously reaffirming social media as a positive source of knowledge.

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Transparency of Social Media Companies Subgroup

Arthur Cheong, Allegra Hoddie, Anusha Jain, Naveli Singh, Christopher Yoberd

Act Name: Digital Integrity and Accountability Act

I. Introduction

- A. The content moderation on various social media platforms is often ineffective in preventing the spread of misinformation and hate speech. Oftentimes, content that is unproblematic is flagged and other times content that is problematic is not flagged. Currently, social media companies—organizations that provide online platforms and services facilitating the creation, sharing, and exchange of user generated content, as well as enabling social interaction and communication among individuals or groups⁶²—are entirely reliable for regulating content on their platforms. In order to ensure that there is no political bias among other biases when content is being regulated, we believe there is a need for a political agency that holds digital platforms accountable for what content they remove from their platforms.² This will keep users aware of why their content is being taken down and in general what content is taken down in addition to ensure that social media companies take proactive steps to safeguard online discourse and promote digital integrity.⁶³
- B. Section 230 of the Communications Decency Act (CDA) does find that companies may not be held liable for the content on their websites, but unlike policies on direct content moderation, we are aiming to create more transparency between social media companies and their users as well as the federal government.⁶⁴ Additionally, the Digital Millennium Copyright Act (DMCA) fails to address the rapidly evolving landscape of social media, particularly in combating misinformation and hate speech.⁶⁵ We find that most policies pertaining to social media companies are in need of updated information and with our policy proposal, we hope to give the federal government more transparent access to what social media companies are regulating, in order to create more effective legislation in the future regarding social media companies.⁶⁶

⁶² *Social media*. (2021, December 31). Wikipedia. https://en.wikipedia.org/wiki/Social_media#cite_note-Kietzmann-1

⁶³ Draper, D. (2022, July 5). *Summarizing the Section 230 Debate: Pro-Content Moderation vs Anti-Censorship* | Bipartisan Policy Center. <https://bipartisanpolicy.org/blog/summarizing-the-section-230-debate-pro-content-moderation-vs-anti-censorship/>

⁶⁴ MacCarthy, M. (2022, November 1). *Transparency is essential for effective social media regulation*. Brookings. <https://www.brookings.edu/articles/transparency-is-essential-for-effective-social-media-regulation/>

⁶⁵ Gallo, J., & Cho, C. (2021). *Social Media: Misinformation and Content Moderation Issues for Congress*. <https://crsreports.congress.gov/product/pdf/R/R46662/6>

⁶⁶ Huff, H. S., Dana Brusca, Bart. (2024, March 4). *New California Content Moderation Rules Highlight Legislative Focus on Social Media Companies* *New California Content Moderation Rules Highlight Legislative Focus on*

II. Proposed Policy

- A. Through our Digital Integrity and Accountability Act, we propose the creation of a new department under the Federal Communications Commission (FCC), the “The Office of Social Media Accountability,” which monitors social media companies by requiring them to report what content on their platform they flag and take down. This aims to increase transparency in the social media moderation teams; there are no requirements to what they take down or how they moderate, rather just that they inform the government on their activity.
- B. The problem addressed by this policy is the ineffective moderation of social media platforms, which leads to the widespread dissemination of misinformation and hate speech. By establishing a dedicated agency to monitor and report on content moderation activities, the policy aims to enhance transparency, accountability, and effectiveness in addressing harmful content online.
- C. Funding: Money would be pulled from government funding allocated to FCC.

III. Legislative Analysis

- A. Pros: Enhanced Transparency, Increased Accountability, centralized oversight, improved user trust. Cons: Potential for censorship, Regulatory complexity, and resistance from social media companies
- B. Counter arguments:
 - Potential for censorship:* While centralized oversight might raise concerns over calls for censorship, the main goal of this proposed agency is to combat harmful content such as misinformation and hate speech, not to restrict legitimate speech. No new speech is censored within this agency, it simply reports what is already being censored. No content is being regulated by the government specifically. With clear guidelines, this can help ensure that content moderation decisions are based on objective criteria rather than subjective biases, therefore minimizing the risk of censorship because social media companies are aware that they have to report the decisions they make.⁶⁷
 - Regulatory Complexity:* It might make the regulatory landscape more complicated but it is necessary to adapt to evolving challenges created by digital communication. By creating clear guidelines for content moderation, this proposed agency would provide certainty for both social media companies and

Social Media Companies. ZwillGen. <https://www.zwillgen.com/ftc-state-ag/california-content-moderation-legislative-focus-social-media-companies/>

⁶⁷ Justice Department. (2023, January 9). *Justice Department and Meta Platforms Inc. Reach Key Agreement as They Implement Groundbreaking Resolution to Address Discriminatory Delivery of Housing Advertisements*. Www.justice.gov. <https://www.justice.gov/opa/pr/justice-department-and-meta-platforms-inc-reach-key-agreement-they-implement-groundbreaking>

the users, which also promotes responsible online behavior.⁶⁸

Resistance From Social Media Companies: It's better to collaborate with regulators and showcase a commitment to combating harmful content. By working together with industry stakeholders, this agency would develop solutions that balance the need for content moderation with respect from platform autonomy and user rights, that ultimately benefits the entire digital ecosystem.

IV. Conclusion

- A. In the long-term, the safety of children and their experiences on social media is crucial for societal growth. This policy would enhance transparency around the type of content that children are being exposed to on social media platforms through providing clear parameters for what is being regulated. This increased transparency can lead to more effective moderation efforts to remove harmful content such as explicit material, cyberbullying, and predatory behavior. This office would work in collaboration with the National Center for Missing & Exploited Children (NCMEC) as well as other agencies within the FCC along with the Department of Justice and Federal Trade Commission.
- B. Over time, this implementation of this policy can contribute to a cultural shift toward greater awareness and prioritization of children's safety in the online environment. By establishing clear standards for content moderation and holding social media companies accountable, the policy fosters a culture of responsibility and collective action to create a safer digital space for children to learn, explore, and connect.
- C. The proposal would ensure safety against harmful misinformation by holding companies accountable for negligence; misinformation can have serious real-world consequences. By analyzing the reported data on flagged or removed content, the regulatory agency can identify patterns and trends related to misinformation. When objectively incorrect information is allowed to continue being shared on platforms, it becomes a threat to democracy. Through identifying what content is taken down, observers can gauge the range of content that companies consider in violation of their policy. This enables proactive measures to be taken to address emerging forms of misinformation before they become widespread. Continuous monitoring and adaptation to evolving challenges and technologies ensure that content moderation practices remain effective in combating misinformation. This adaptability allows for the development of new

⁶⁸ Ahn, A. (2022, December 23). Facebook parent Meta will pay \$725M to settle a privacy suit over Cambridge Analytica. *NPR*. <https://www.npr.org/2022/12/23/1145303268/facebook-meta-cambridge-analytica-privacy-settlement>

strategies and tools to address emerging forms of misinformation as they arise.⁶⁹

- D. Our proposal would greatly aid in the tackling of misinformation spread. After the arrival of social media, misinformation now travels faster than ever before. Americans increasingly receive their news through social media, and fake news found on social media sites indeed has a detrimental impact on our society. The spread of fake news, furthermore, is further amplified by social media algorithms promoting them for views. Our proposal would combat the spread of this misinformation by establishing “The Office of Social Media Accountability,” that would monitor social media companies by requiring them to report what content is flagged and taken down.

⁶⁹ Siripurapu, A., & Merrow, W. (2021). *Social Media and Online Speech:: How Should Countries Regulate Tech Giants?* JSTOR. <https://www.jstor.org/stable/resrep31160>

The Updated Reclamation Performance Bond (RPB)

Rebecca Bossow, Sydney Li, Arjun Krishna

I. Introduction

During an age of increasing coal bankruptcies due to its decreasing share of US power generation as the country shifts towards renewable and cleaner energy sources, an alarming number of coal mines are being left abandoned⁷⁰. For many coal mines, declaring bankruptcy helps evade any responsibility for reclamation required by the Bureau of Land Management⁷¹. Environmental liabilities take a back seat, while creditors and banks receive priority. In a study done by the Natural Resources Defense Council, over 500 mountaintop removal sites were surveyed with only 4% having any post-mining economic activity and 89.3% having no form of verifiable post-mining economic reclamation⁷².

In order to ensure reclamation of permit land, the BLM requires a performance bond, which is outlined in the Surface Mining Control and Reclamation Act of 1977 and managed by the US Department of Interior's Office of Surface Mining Reclamation and Enforcement⁷³. This bond would ensure the cost of reclamation in the "worst case scenario." The SMRCA allows for three types of performance bonds: surety bonds, collateral bonds, and self-bonds. The most problematic of these bonds is a self-bond, in which mining operators demonstrate sufficient corporate assets⁷⁴. These bonds do not restrict the use of assets meant to insure, and this lack of accountability often leads to bankruptcy or a lack of sufficient reclamation resources.

This policy proposal aims to amend the Surface Mining Control and Reclamation Act by removing self-bonds as a credible type of performance bond. The SMRCA currently fails to account for the efficacy of performance bonds, specifically self-bonds. Currently, the Abandoned Mine Land Reclamation Program, created by SMRCA, attempts to address the issue by enacting a fee per ton on coal mined⁷⁵. The fee goes towards a grant fund that can be rewarded only to land reclaimed before 1997 and prior to the SMRCA being enacted. However, it is important to address the root of the issue and prevent mine abandonment by ensuring that permit miners follow up on their duty to reclaim mined lands by standardizing bond procedure. Failure to reclaim lands can pose serious economic and health repercussions, endangering the welfare of the public.

II. The Policy Proposal

The use of self-bonds, in which mining companies act as their own sureties, has raised concerns regarding their adequacy and effectiveness in fulfilling reclamation responsibilities. This policy proposal seeks to amend SMCRA to eliminate self-bonds as an acceptable form of performance bond and establish stringent criteria for bonds to ensure effective reclamation.

⁷⁰ <https://www.sierraclub.org/sierra/2016-3-may-june/feature/coal-industry-bankrupt>

⁷¹ <https://www.propublica.org/article/west-virginia-coal-blackjewel-bankruptcy-pollution>

⁷² <https://www.nrdc.org/bio/rob-perks/mountaintop-removal-reclamation-fail>

⁷³ <https://www.osmre.gov/laws-and-regulations>

⁷⁴ <https://crsreports.congress.gov/product/pdf/R/R46610>

⁷⁵ <https://www.osmre.gov/programs/reclaiming-abandoned-mine-lands>

Elimination of Self-Bonds:

- Amend SMCRA to prohibit the use of self-bonds as a form of performance bond for surface coal mining operations
- Existing self-bonds should be phased out within a reasonable timeframe to ensure continuity of reclamation efforts.

Surety Bonds:

- Require all surface mining operators to secure performance bonds from third-party surety companies or financial institutions.
- Third-party surety bonds should meet minimum requirements to ensure adequate coverage for reclamation costs.

Alternative Financial Measures:

- Explore the feasibility of alternative financial instruments, such as collateral requirements to supplement traditional surety bonds.

Regulatory Enforcement:

- Strengthen regulatory enforcement mechanisms to ensure compliance with bonding requirements and timely reclamation activities.
- Implement regular monitoring and auditing procedures to assess the effectiveness of performance bonds and address any deficiencies promptly.

III. Legislative Analysis

The benefits of this proposal would help to guarantee a baseline of accountability to ensure that reclamation efforts are completed in accordance with the original grant framework. Moreover, by creating a concrete standard of entry, the costs of reclamation would be equitably created by an independent agency to decrease the likelihood of incomplete or improper reclamation within a predetermined cost. An NRDC survey⁷⁶ of over 500 mountain-top removal sites were analyzed to determine that: “410 supposedly reclaimed mine sites, for which we found that: 366 (89.3%) had no form of verifiable post-mining economic reclamation excluding forestry and pasture 26 (6.3% of total) yield some form of verifiable post-mining economic development.” This evidence clearly substantiates the notion that, left to their own devices, mining companies do not effectively follow through with reclamation measures due to a lack of accountability. By securing deposits before ground is broken and excluding guaranteed self-bonds, the chance of successful reclamation is high.

Some might counter these assertions by suggesting that this might monopolize mining and ensure exclusivity of government mining rights only to companies who have sufficient capital to put down for reclamation. This argument fails to consider the economic and environmental responsibility of mining companies to the BLM established in individual contracts. Such responsibilities include both environmental and economic reclamation. Moreover, under the status quo, even mining giants regularly evade reclamation costs with subsidiaries and guaranteed self-bonds. Opponents of the policy might also suggest that these new regulations might strain the already scant resources of regulatory bodies. This proposal

⁷⁶<https://www.nrdc.org/bio/rob-perks/mountaintop-removal-reclamation-fail#:~:text=Of%20the%20500%20mountaintop%20removal,reclamation%20excluding%20forestry%20and%20pasture>

focuses on creating guarantees of reclamation prior to breaking ground. Therefore, instead of expending resources litigating to reinstate reclamation, the funds and resources to do so would already be in place before the start of the operation.

IV. Conclusion

This policy proposal aims to minimize the long-term environmental impact of mines and allow abandoned mine land (AML) to return to its previous natural state, while preventing the dangers that come with AML. Prominent dangers associated with AMLs include cave-ins, undetonated explosives, deadly gasses, and toxic chemicals, all of which pose a threat to both humans and wildlife.⁷⁷ In the past, mine operators had no legal obligations to remove on site hazards post mining, but the creation of future hazards can be prevented by forcing companies to comply with current regulations. Some examples of reclaimed mines include a Mississippi Bar gravel extraction site being converted into a riparian wildlife habitat and mined lands being used for crops, such as alfalfa and strawberry fields.⁷⁸

The Environment, Manufacturing, and Critical Materials Subcommittee is responsible for “all matters related to soil, air, and water contamination. [...] The subcommittee focuses on the regulation of solid, hazardous, and nuclear wastes, including mining.”⁷⁹ By preventing environmental destruction and the leaching of toxic chemicals into groundwater and soil, this policy proposal falls in line with the mission of the subcommittee.

⁷⁷https://www.blm.gov/sites/default/files/uploads/AML_PUB_DangersAtAbandonedMines.pdf

⁷⁸<https://www.conservation.ca.gov/dmr/SMARA%20Mines/reclamation/Pages/Index.aspx#:~:text=%E2%80%8BReclamation%20is%20the%20combined,or%20residential%20and%20commercial%20development.>

⁷⁹<https://energycommerce.house.gov/committees/subcommittee/environment>

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The Manure Methane Reduction Program (MMRP)

Karina Pallikala, Julia Luckow, Claire Bauschlicher, Anoushka Lahiri

1. Introduction

Colorless, odorless, and flammable: methane could be described as an invisible climate killer. Methane traps solar energy in the Earth's atmosphere and, like other greenhouse gasses, is a major driver of global warming and the current climate crisis. Due to the fact that methane traps more heat into the atmosphere per molecule than carbon dioxide does, around 30% of current global warming is attributed solely to methane emissions.⁸⁰ This means that methane is 80 times more detrimental than carbon dioxide over a 20 year period.⁸¹ Additionally, the agriculture sector is the largest source of methane emissions in the United States.⁸² California, specifically, is the most productive agricultural state⁴, and dairy is its number one product.⁹ Methane is produced through the natural digestive processes of various livestock, like cattle, which is called enteric fermentation; it is also emitted from animal manure that is stored in liquid lagoons or holding tanks.⁵ In terms of cattle farming, one cow produces anywhere from 154 to 264 pounds of methane gas yearly.⁶ As seen in the chart below, 25% of US methane emissions come from enteric fermentation, while another 9% stems from manure management. With both sources classified under the agricultural sector, it is clear that in order to reduce methane emissions, there needs to be a focus on change in agriculture.

In 2022, the California Department of Food and Agriculture (CDFA) formed the Alternative Manure Management Program (AMMP), a grant program to encourage agriculture producers to use technology and specific management practices that will lower long-term methane emissions and environmental damage. Through the Budget Act of 2021, \$32 million dollars from the California state budget was allocated to the CDFA. Of this amount, \$12.2 million was used for AMMP and funded grants of up to \$750,000. Additionally, in December of 2023 the White House released their plan to accelerate the reduction of methane emissions in all parts of the economy, including oil, landfills, and agriculture. It states that the US Department of Agriculture (USDA), specifically through the Natural Resources Conservation Service (NRCS), programs will deliver \$19.5 billion dollars to fund climate-friendly conservation practices. Additionally, the USDA's Rural Energy for America Program offered over \$43 million in grants for anaerobic digester programs as well as over \$115M towards loans for renewable fuel that is produced by animal waste, called biogas. Both of these initiatives, while not being perfect, do

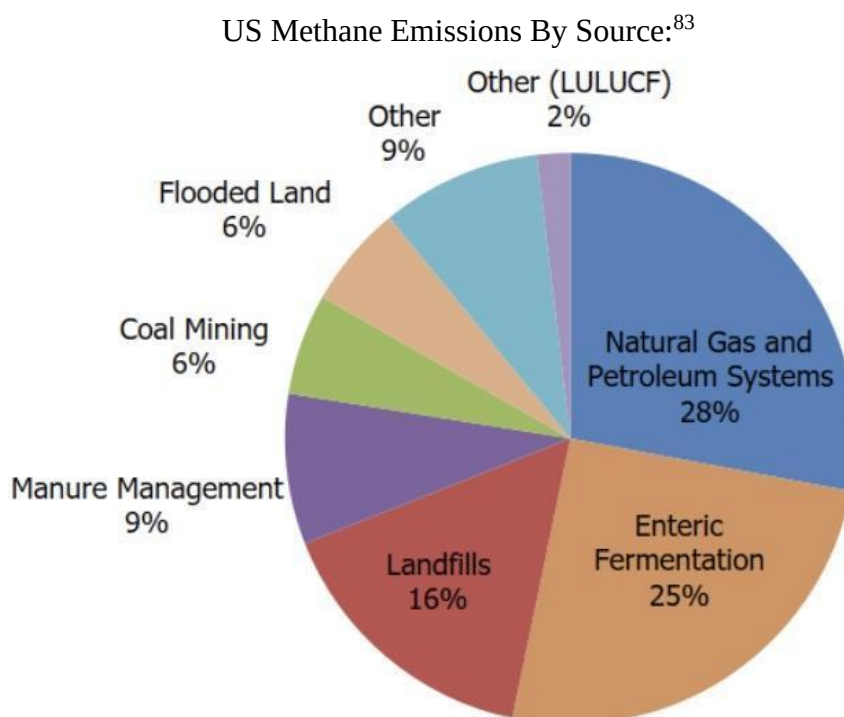
⁸⁰ *Methane and climate change*. IEA. (2022). Retrieved April 17, 2024, <https://www.iea.org/reports/global-methane-tracker-2022/methane-and-climate-change#>

⁸¹ *What's the deal with methane?* UNEP. (2022, October 18). <https://www.unep.org/news-and-stories/video/whats-deal-methane#:~:text=It%20is%20responsible%20for%20more,years%20after%20it%20is%20released.>

⁸² *What's the deal with methane?* UNEP. (2022, October 18). <https://www.unep.org/news-and-stories/video/whats-deal-methane#:~:text=It%20is%20responsible%20for%20more,years%20after%20it%20is%20released.>

⁴ *USDA FSA California Fact Sheet What's the deal with methane?* UNEP. (2022, October 18). <https://www.unep.org/news-and-stories/video/whats-deal-methane#:~:text=It%20is%20responsible%20for%20more,years%20after%20it%20is%20released.>

address pertinent problems in the American agriculture sector, and are promoting practices of eco-friendly farming.



Currently, dairy lagoons of cattle manure and the storage of liquid manure in tanks in the US are a major source of agricultural greenhouse gas emissions.⁸⁴ We propose the Manure Methane Reduction Program (MMRP) which is a financial incentive for farmers that provides grants for the implementation of more sustainable manure management systems that would reduce overall methane emissions. We hope to see a decrease in methane emissions from manure management from 9% to around 5% since practices such as well-covered outdoor storages with manure slurry at rest can reduce more than 50% of its methane emissions.⁸⁵ Thus, the goal of almost halving the amount of methane emissions from manure management through the MMRP is a viable expectation.

⁸³ Overview of Greenhouse Gases. Environmental Protection Agency. (2024, April 11).. <https://www.epa.gov/ghgemissions/overview-greenhouse-gases#methane>

⁸⁴ Leytem, A. B., Bjorneberg, D. L., Koehn, A. C., Moraes, L. E., Kebreab, E., & Dungan, R. S. *Methane emissions from dairy lagoons in the Western United States - Sciencedirect*. Science Direct. (2017, April 9). <https://www.sciencedirect.com/science/article/pii/S0022030217305799>

California Department of Food and Agriculture. (2024). *California Agricultural Production Statistics*. CDFA. <https://www.cdfa.ca.gov/Statistics/>

⁸⁵ Hilhorst, M.A., Melse, R.W., Willers, H.C., Groenestein, C.M., Monteny, G.J. *Reduction of Methane Emissions from Manure*. IMAG. <https://edepot.wur.nl/303610>

2. The Policy Proposal

As stated earlier, certain manure management systems, such as storage in lagoons and liquid manure in tanks in the US, are a major source of agricultural greenhouse gas emissions.⁸⁶ In order to lower methane emissions in Agriculture and manure management, we propose the Manure Methane Reduction Program (MMRP) which provides grants to farmers and agricultural companies for the implementation of more sustainable manure management systems. This would follow the model of California's Alternative Manure Management Program (AMMP) where farms and organizations can submit applications describing their plans to rework their manure management systems that will result in methane reduction.⁸⁷ Farms will also submit the amount of money they are requesting, and detail how they will utilize the funds. This program would essentially take the model of the California Alternative Manure Management Program (AMMP), but simply make the grants available nationwide. Our Manure Methane Reduction Program would be administered directly through the EPA and different amounts of funds will be appropriated to each state's local government to carry out the MMRP based on the number of livestock farms located in the state.

In terms of funding, approximately \$12.2 million is made available for the California AMMP, and since California is one of the states with the largest amount of livestock farms, we estimate this to be the maximum amount of funding necessary for each state's MMRP.⁸⁸ Naturally, each state's funding for their MMRP will differ based on the amount of livestock farms in the state, but the maximum ask for the program entirely would be around \$445.3 million assuming the top 25 states with the most livestock farms would need \$12.2 million each and the bottom 25 with the least amount of livestock farms would need half of that or \$6.1 million each. This number is likely an overestimate of the real cost for the program as each state's MMRP funding amount would be more specific and tailored to the number of the livestock farms in the state, while our stated amount is more of a general estimate based on the funding of California's AMMP. The funding for this program will come from and be compiled in the EPA's budget.

3. Legislative Analysis

Atmospheric methane has proliferated at an alarming rate, and far more than any other environmental pollutant in history, due to its molecular structure trapping heat at a higher rate⁸⁹. This is due to the ever increasing rate of human consumption and production. Since the 1980's,

⁸⁶ Methane emissions from dairy lagoons in the Western United States - Sciencedirect. (n.d.).
<https://www.sciencedirect.com/science/article/pii/S0022030217305799>
<https://www.sciencedirect.com/science/article/pii/S0022030217305799>

⁸⁷ 2022 alternative Manure Management Program. California Grants Portal. (2024, February 5).
<https://www.grants.ca.gov/grants/2022-alternative-manure-management-program/>
<https://www.grants.ca.gov/grants/2022-alternative-manure-management-program/>

⁸⁸ 2022 alternative Manure Management Program. California Grants Portal. (2024, February 5).
<https://www.grants.ca.gov/grants/2022-alternative-manure-management-program/>
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⁸⁹ What's the deal with methane? UNEP. (n.d.).
<https://www.unep.org/news-and-stories/video/whats-deal-methane#:~:text=Due%20to%20its%20structure%2C%20methane,years%20after%20it%20is%20released.>

there has been a worrying increase in methane production⁹⁰, and there have been no successful attempts at curbing the ever increasing rates. This is especially prevalent in the agricultural sector in which methane emissions rank number one among annual emissions. This is why we propose an incentivization program, entailing grants for farmers, which will bring methane emissions down through the usage of renewed manure storage systems.

While it is not the purpose of farmers, agriculture exhibits the highest rate of methane emissions, with an average expenditure of 40- 46%⁹¹ of global methane emissions, and is only set to increase, with the rising trends of food production. Due to this alarming trend of increasing methane emissions, the agricultural industry must be incentivized to mitigate their expenditure.

Alternate manure management systems are imperative because current manure management systems account for 10% of methane emissions.⁹² Current manure management systems consist of settling bins, or storage lagoons, in which water flushing systems remove the excrement, however while the manure is left in anaerobic environments, it releases methane. This is increasingly evident when breaking down the prominent components of methane emissions, in which faulty manure management systems is one of the highest emitters. Due to this circumstance, farmers must be met with government grants that will aid in the installation of renewed systems. Emission percentages can easily be curbed with the addition of compostable pack barns, and solids separation, which are manure management upgrades that have already proven to be effective in lowering methane emission rates.⁹³ With the aid of manure management planning, and funds via government grants, farmers have the possibility to implement entirely new systems of manure management to benefit both themselves, and those around them.

Despite the numerous benefits, and funds provided by the government, the renewed manure management system could run into some issues, including a lack of incentives for farmers, due to the fact that it is not profitable, as well as the fact that methane reductions will only be incentivized to be reduced to a regulated level. However, this can be combated with mandatory information disclosure from the agricultural sector's side, and voluntary action for firms, which creates a competitive edge via social appeal.

4. Conclusion

The Manure Methane Reduction Program is significant to our world's environment and climate as greenhouse gas emissions pose a substantial threat. The mission of the MMRP is to permanently reduce methane emissions by aiding with the establishment of more sustainable

⁹⁰ Increase in atmospheric methane set another record during 2021. National Oceanic and Atmospheric Administration. (n.d.). <https://www.noaa.gov/news-release/increase-in-atmospheric-methane-set-another-record-during-2021>

⁹¹ Princeton University and Cornell University. (n.d.-a). *Princeton*. Opportunities to Reduce Methane Emissions from Global Agriculture. https://searchinger.princeton.edu/sites/g/files/toruqf4701/files/methane_discussion_paper_nov_2021.pdf#:~:text=Agriculture%20contributes%20around%2040%2D46,7%25%20are%20from%20managed%20manure

⁹² White House Office of Domestic Climate Policy . (n.d.). *U.S. Methane Emissions Reduction Action Plan*. U.S. METHANE EMISSIONS REDUCTION ACTION PLAN. <https://www.whitehouse.gov/wp-content/uploads/2021/11/US-Methane-Emissions-Reduction-Action-Plan-1.pdf>

⁹³ 2017 alternative Manure Management Program - california ... 2017 Alternative Manure Management Program. (2018, March 27). <https://www.cdfa.ca.gov/oefi/AMMP/docs/2017AMMP-GrantAwardees.pdf>

manure systems nationwide. Since there currently is a lack of a widespread federal system that provides this assistance in manure management, our proposal will make a positive impact across the nation and provide an avenue for climate-friendly practices. With the goal of reducing methane emissions in the large industry of agriculture, the MMRP would also foster a more environmentally aware mindset in this industry and its success could pave the way for other environmentally focused agricultural legislation. The long-term impact of the proposal would be the development and implementation of permanent sustainable manure management systems across all 50 states as well as a permanent reduction in methane gas emissions. As the climate crisis continues to afflict our planet, the MMRP is an important action to help decrease global warming.

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Regulating Air Pollution Summary Memo

By: Sarah Wong, Shay Brandes, Rafferty Villar Perez, Devansh Malik

Intro

Being the driving contributor to our climate crisis, carbon emissions have been exponentially rising without proper regulation and it is now time to take action. The United States has annually emitted 5.13 billion metric tons of energy-related carbon dioxide.⁹⁴ Companies that utilize fossil fuels, specifically the automobile industry, threaten our environmental security as there are approximately 275 million registered vehicles in the United States. The sole manufacturing of a car battery can emit approximately 2400-16,000kg of carbon into our atmosphere.⁹⁵ As a result, our solution aims to gradually reduce carbon emissions produced by these companies to play our part in combating climate change. Although previous cap-and-trade legislation has been passed such as the Clean Air Act Amendments of 1990, we seek to specifically impose a cap on greenhouse gasses emitted during the manufacturing process of motor vehicles due to their significant contribution to climate change.

Policy Proposal

Our proposed solution to the problem of excessive carbon emissions during the production of motor vehicles involves implementing an emissions trading system (cap-and-trade) for automotive companies in order to provide an incentive against producing more carbon. This policy explicitly affects the emissions created during factory production, and nothing else related to the emissions gas cars produce when driving. Under this cap-and-trade system, the government would set a cap on the amount of emissions a company could produce.

Ideally, companies would try to stay under this level of carbon production. But, if they wished to exceed that level, they would have to get a permit, which is an allowance for making more carbon above that level. These permits would be created by the government. If a company wants to exceed the cap, they must get a permit either by purchasing some directly from the government, or by trading with another company. However, there would be a limited amount of permits available at any given time, as determined by the cap. Additionally, the set amount of permits would decrease over time and the annual cap would lower⁹⁶. With the amount of firms that want permits fluctuating with their decisions regarding their carbon emissions levels, the permit price will work with supply and demand, with a variety of prices in between because of the trading system.

Cap-and-trade systems have been proven to work in the past. For example, New England's Regional Greenhouse Gas Initiative (RGGI) and the California Air Resources Board's Cap-and-Trade Program have been fairly effective in offsetting state or region-wide carbon emissions and boosting their clean energy economies. In this specific proposal to regulate

⁹⁴ "How Much Carbon Dioxide Does the United States and the World Emit Each Year From Energy Sources? | U.S. Geological Survey," March 22, 2019.

⁹⁵ "How Much CO2 Is Emitted by Manufacturing Batteries? | MIT Climate Portal."

⁹⁶ "Cap-and-Trade Program | California Air Resources Board," n.d. ww2.arb.ca.gov/our-work/programs/cap-and-trade-program/about.

emissions created during the production of automobiles, these permits and caps would only apply to auto manufacturers. Essentially, if cap-and-trade or emissions trading systems are implemented, it will help address the problem of air pollution caused by vehicle production. The bearers of the cost would be automakers, who may incur significant costs when updating their manufacturing practices to be more carbon efficient, or when paying the fines for exceeding limits before this transfer. However, the only cost to the government is the cost for regulating the permits. If these permits are initially sold at auction, that can fund the program. Additionally, the punishment for going above the carbon emissions limit could be a fine that is used for funding the management of the program.

Legislative Analysis

Implementing emissions trading systems, more commonly known as “cap and trade”, presents several advantages and disadvantages. On the positive side, such systems work to reduce greenhouse gas emissions in a cost-effective manner.⁹⁷ By setting a cap on maximum emission levels and creating permits for each unit of emissions allowed, our government can incentivize firms to reduce their emissions. In turn, this encourages businesses to adopt cleaner technologies and practices, leading to a significant decrease in overall emissions. By adopting renewable energy sources and energy-efficient methods, emissions trading can play a crucial role in achieving climate change mitigation targets and reducing emissions for the covered sectors.

Nonetheless, there are drawbacks that must be considered. One concern raised is that emissions trading may lead to economic disparities between participating firms, also exemplified through worldwide examples and corporations, are falling under the scope of the emissions trading policy and those outside of it.⁹⁸ Firms subject to emissions regulations risk facing increased costs that may potentially impact their competitiveness compared to non-regulated entities. While emissions trading systems have proven to be successful in certain regions and industries across the globe, their effectiveness in reducing greenhouse gas emissions in a reasonable manner may vary in regards to US companies.⁹⁹ Therefore, careful consideration and regular evaluation are necessary to guarantee that emissions trading policies effectively balance environmental protection with economic considerations.

However, such drawbacks previously mentioned can be avoided. Quotas for a lower carbon price could be considered for smaller companies in the industry to reduce emission costs. Additionally, governments can utilize funds raised from emission trading schemes in order to further redistribute income, including funding programs such as the cap-and-trade program, simultaneously reducing greenhouse gas emissions¹⁰⁰. Furthermore, actions to combat present emission issues are currently being taken by the current presidential administration that outlines the multitude of benefits with emissions regulations, including \$62 billion annual decrease in

⁹⁷ “How Do Emissions Trading Systems Work?” Grantham Research Institute on Climate Change and the Environment. December 11, 2023. www.lse.ac.uk/granthaminstitute/explainers/how-do-emissions-trading-systems-work/. [accessed April 2, 2024].

⁹⁸ Patel, Urjit R. “The Challenges of Emissions Trading.” Brookings. November 16, 2007. www.brookings.edu/articles/the-challenges-of-emissions-trading/. [accessed April 2, 2024].

⁹⁹ Ibid.

¹⁰⁰ Zhang, J., & Zhang, Y. (2023). How emissions trading affects income inequality *Climate Policy*, 23(5), 593–608. doi.org/10.1080/14693062.2022.2099789 [accessed April 17, 2024].

fuel costs and \$13 billion annual public health benefits, accumulating to approximately a \$100 billion annual benefit to society¹⁰¹. In the United States, the National Oceanic and Atmospheric Administration's analysis shows a large decrease in greenhouse gas emissions nationwide in 10 years due to the efforts in reducing such emissions alongside emission trading systems¹⁰². Although drawbacks must be considered, significant progress has been shown already addressing such concerns and can be exemplified on a much larger scale in the future.

Conclusion

Once implemented, our cap-and-trade policy seeks to decrease carbon emissions caused by the car manufacturing industry. By regulating and enforcing permits, we incentivize car companies to not go beyond a specific limit thereby reducing carbon emissions caused by excessive manufacturing over time. Essentially, our policy aims to address this world's climate crisis and acknowledge our nation's role in mitigating the over emission of carbon by regulating the excessive production of automobiles.

¹⁰¹ Biden Administration. "Biden-Harris Administration Finalizes Strongest Ever Pollution Standards for Cars." Environmental Protection Agency, March 20, 2024, www.epa.gov/newsreleases/biden-harris-administration-finalizes-strongest-ever-pollution-standards-cars-position#:~:text=These%20standards%20will%20avoid%20more,and%20repair%20costs%20for%20drivers. Accessed April 17, 2024.

¹⁰² Vergoth, Karin. "Potent Greenhouse Gas Declines in the U.S. Confirming Success of Control Efforts." NOAA Research, March 7, 2023, <https://research.noaa.gov/2023/03/07/potent-greenhouse-gas-declines-in-the-us-confirming-success-of-control-efforts/#:~:text=A%20new%20NOAA%20analysis%20shows,and%20the%20electric%20power%20industry>. Accessed April 17, 2024.

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Members: Thanisha Kapur, Lara Dumanli, Max Tan, Robert Fields, Aaron Bao

IDC AI Innovation and Security Final Summary Memo

I. Introduction

In our growing democracy, with AI development increasing, it's integral that we must implement mandatory assessments to ensure that AI systems are developed and deployed responsibly, fostering trust among users and stakeholders. By requiring more stringent assessments for larger companies, the policy promotes equity in AI development and deployment, ensuring that entities with greater resources and access to data bear a proportionate responsibility for mitigating risks and safeguarding societal interests. Furthermore, mandating algorithmic impact assessments for developers of artificial intelligence algorithms, in alignment with the National Institute of Standards and Technology's AI Risk Management Framework¹⁰³, aims to manage risks associated with AI deployment, including potential negative impacts on individuals, organizations, and society at large. In addition, algorithmic impact assessments and design evaluations help safeguard individual rights and privacy by scrutinizing the design, uses, training data, and outputs of AI algorithms. Public disclosure of these evaluations ensures transparency and accountability, empowering individuals to make informed decisions about their interactions with AI systems.

A previous legislation that followed risk assessment standards within Artificial Intelligence use is the *American Data Privacy and Protection Act*.¹⁰⁴ Although this act has only been introduced, if enacted, would develop proper standards and strategies to address the collection of private data, either "traditionally" sensitive data such as health and geolocation or government identifiers such as Social Security numbers. Following these measures, they would require companies to conduct an algorithm risk impact assessment, highlighted in Section 207, so that companies are held accountable. However, as detailed in the act, the impact assessment is unclear on how companies are required to submit the assessment. However, this act is unlikely to be introduced again and there is currently a new discussion draft of a bill called the "American Privacy Rights Act". This act would be the successor of the *American Data Privacy and Protection Act* as it would set clear national data privacy rights and protections. Further, previous legislation such as the *Federal Artificial Intelligence Act*¹⁰⁵ would require federal agencies to use the National Institute of Standards and Technology (NIST) framework to specify appropriate cybersecurity tactics and protect civil and online liberties. This act, if enacted, would

¹⁰³ AI Risk Management Framework. NIST. (2024, January 5). <https://www.nist.gov/itl/ai-risk-management-framework>

¹⁰⁴ The American Data Privacy and Protection Act: Is federal regulation of AI finally on the horizon?: Insights: Mayer Brown. Insights | Mayer Brown. (2024, January 10). <https://www.mayerbrown.com/en/insights/publications/2022/10/the-american-data-privacy-and-protection-act-is-federal-regulation-of-ai-finally-on-the-horizon>

¹⁰⁵ S.3205 - Federal Artificial Intelligence Risk Management ... (n.d.-a). <https://www.congress.gov/bill/118th-congress/senate-bill/3205>

only require federal agencies to follow the NIST framework, limiting the scope of its impact. Our policy, more specifically, would spread across all agencies, regardless of their federal standing, ensuring that these efforts are being met with widespread implementation. Finally, previous legislation, the H.R. 4704 - *Artificial Intelligence and Biosecurity Risk Assessment Act*, introduced on July 17, 2023, requires ASPR to conduct regular risk assessments to understand the biological risk from advancements in AI overall. Compared to these various legislations, our policy would be efficient in ensuring that the outcomes would be effectively met within society.

II. Our Policy Proposal

We propose to mandate algorithmic impact assessments for developers of artificial intelligence algorithms and incorporate the National Institute of Standards and Technology's AI Risk Management Framework into these evaluations. Entities that have annual gross revenues over \$250 million and possess data from more than five million individuals or devices would need to carry out in-depth "impact assessments." These assessments, which would be conducted annually, must examine the design, uses, training data, and outputs, and list specific risks. Secondly, all other entities falling under this regulation are required to undertake a broader "design evaluation." This evaluation focuses on the design, structure, and inputs of algorithms while addressing the same risks identified for the impact assessments by large data holders. Both the impact assessments and design evaluations must be submitted to the Federal Trade Commission and also disclosed to the public.

Our policy aims to address the increasing concerns around the ethical use, transparency, and accountability of artificial intelligence algorithms. By mandating annual algorithmic impact assessments for developers, especially for "large data holders" with significant revenue and data, we seek to ensure that AI algorithms are developed and utilized responsibly. Incorporating the National Institute of Standards and Technology's AI Risk Management Framework into these assessments further standardizes the evaluation of AI risks, promoting a uniform approach to identifying and mitigating potential harms of these algorithms. This ensures that AI systems do not inadvertently perpetuate bias, violate privacy, or pose other societal risks, fostering a safer and more ethical AI ecosystem. The policy would be overseen by the Federal Trade Commission. Recognizing the financial burden on companies, especially smaller entities, our approach would include government support if necessary for those required to conduct evaluations.

III. Legislative Analysis

The implementation of such a mandate has both potential benefits and drawbacks. On one hand, implementing enhanced security measures and rigorous testing for AI prior to public deployment can significantly improve overall security standards. Additionally, instituting mechanisms for constant feedback and thorough testing can mitigate bias within AI algorithms, reducing its impact on specific demographics. Lastly, regular updates and strict adherence to security protocols can effectively manage risks associated with AI technologies. However, some potential drawbacks include resistance from critics who fear that increased regulation may hinder AI innovation, particularly within large tech corporations. Furthermore, smaller companies may lack resources for comprehensive evaluations, which is why applying a uniform

regulatory framework to diverse AI companies may pose challenges.

But these concerns are insignificant in the bigger picture. For example, recent legislative attention to AI policies and heightened public awareness of associated risks further support the need for regulatory measures. Unlike past instances where big tech lobbyists impeded progress, current circumstances suggest a more favorable environment for advancing regulatory initiatives. Additionally, for long run sustainability of the field, AI regulation is necessary so the programs do not get out of hand. We saw this with cryptocurrency, where a lack of regulation caused it to go out of control, which now makes public perceptions of the field much less optimistic. Furthermore, if programs get out of control, it will possibly cause policymakers to create extremely harsh regulations in the future, which will further stifle innovation¹⁰⁶. We acknowledge the fact that smaller companies might not have the capability to follow such frameworks, however, these resource problems are why there are different policies for large and small tech companies. For tech companies that may not be able to afford a design evaluation, grants may be afforded if a company is classified as a small business by the U.S. Small Business Administration (SBA). The SBA will determine a company's eligibility for a grant based on standards such as the scale of the company (number of employees and net worth) with respect to its NAICS industry, the company's affiliation with parent companies, and the company's outside sources of funding, like investments.

Lastly, even though all AI companies are different and might require diverse standards, NIST works towards a general safety protocol that can mitigate risks for all these corporations¹⁰⁷. For example, both Tesla's self-driving cars and a hospital's diagnostic machines would benefit from guidelines that mandate increased testing and scrutiny of algorithms before launch. Plus, these companies can work with the FTC to overcome specific obstacles or challenges that arise.

IV. Conclusion

Deploying a formal framework and assessment ensures public trust as well as security and satisfaction within the companies. Furthermore, standardization ensures that stakeholders of the proposal are ensured across every domain, not just regional corporations. By ensuring long-term trust within AI systems and corporations, public confidence and trust can slowly be built up. This ensures that the integration of AI into systems of public domains is smooth and not rejected. Overall, transparency and regular checks ensure not just safety, but also acts as a form of reassurance for the general public, who might have misgivings or concerns about the direction and expanded use of AI. Our proposal for mandatory algorithmic impact assessments aligns with the Subcommittee on Innovation, Data, and Commerce's objectives of overseeing technological innovation, data privacy, and commercial activities responsibly. By setting standards for the deployment of artificial intelligence systems, this policy directly addresses the subcommittee's

¹⁰⁶ Scharre, P., & Chilukuri, V. (2024, March 5). *What an American Approach to AI Regulation Should Look Like*. Time.com. <https://time.com/6848922/ai-regulation/>

¹⁰⁷ Encryption Consulting. (2024, April 4). What is the NIST? What is the purpose of the NIST?. Encryption Consulting. <https://www.encryptionconsulting.com/education-center/nist/#:~:text=The%20goal%20of%20releasing%20NIST,for%20exploitation%20by%20threat%20actors.>

mandate to balance technological advancement with the protection of individual rights and societal well-being. The public disclosure of these evaluations supports transparency and accountability, key principles in maintaining trust in the digital marketplace. This policy approach reflects the subcommittee's broader aim to guide the digital economy toward a future where innovation does not come at the expense of security, privacy, and ethical standards, ensuring that AI technologies benefit society as a whole.

Introducing Annual Reviews for Sanctions

By Aleksandra Rombakh and Griffin Sturino

California 16th Congressional District Student Advisory Board for Congresswoman Eshoo

I. Introduction

Surprisingly, the most widely utilized form of foreign policy, sanctions, has astonishingly little oversight. At any given time, twenty to thirty or more countries are under sanctions by the United States government, yet there is a notable lack of a formal, legislated requirement for an annual review of sanctions. As trade networks and economic landscapes rapidly evolve, regularly unassessed sanctions inadvertently harm international markets and domestic economic interests. An annual review process would refine our geopolitical strategy and protect vital economic interests that sustain global trade and foster domestic economic growth.

The US has a precedent of passing acts concerning the review of sanctions, though only for particular cases on a small scale. For example, as part of the Countering America's Adversaries Through Sanctions Act (CAATSA)¹⁰⁸, provisions enacted by Congress ensured congressional review of the President's actions regarding sanctions made on Iran, North Korea, and Russia. Similarly, the Comprehensive Iran Sanctions, Accountability, and Divestment Act of 2010¹⁰⁹ which expanded U.S. sanctions against Iran's energy sector and financial institutions, required regular reporting to Congress on enforcing the sanctions and analyzing their effectiveness. Further, the North Korea Sanctions and Policy Enhancement Act of 2016¹¹⁰ also requires review of enacted sanctions. However, overall, there is a significant lack of a continuous annual review and report from the Office of Foreign Assets Control (OFAC). While the OFAC did publish the Treasury 2021 Sanctions Review¹¹¹, it "was neither an assessment of each of the 37 existing sanctions programs administered ... nor a full examination of all economic statecraft tools" (OFAC 2021). Case in point: the Department of the Treasury, which the OFAC is a part of, rarely publishes reviews.

II. Policy Proposal

We propose that the US Congress legislate the requirement for the OFAC to undertake an annual review of all sanctions to evaluate their benefits and their impact on the US domestic economy and international trade. This review should include a comprehensive analysis of the sanctions' impact on the U.S. domestic economy and international trade relations, ensuring that they align with the current geopolitical landscape and economic objectives. With a broad range

¹⁰⁸ Countering America's Adversaries Through Sanctions Act, H.R.3364 (2017).
<https://www.congress.gov/bill/115th-congress/house-bill/3364/text>.

¹⁰⁹ Comprehensive Iran Sanctions, Accountability, and Divestment Act of 2010, H.R.2194 (2010),
<https://www.congress.gov/bill/111th-congress/house-bill/2194>.

¹¹⁰ North Korea Sanctions and Policy Enhancement Act of 2016, H.R.757 (2016),
<https://www.congress.gov/bill/114th-congress/house-bill/757>.

¹¹¹ The Office of Foreign Assets Control (2021, October) *The Treasury 2021 Sanctions Review*.
<https://home.treasury.gov/system/files/136/Treasury-2021-sanctions-review.pdf>.

of U.S. sanctions affecting numerous countries, the absence of a structured, routine evaluation can lead to misalignments between evolving economic conditions and international political dynamics. Therefore, this oversight by the OFAC is necessary for sanctions to serve their political purpose while avoiding harming US economic interests.

To fund these reviews, we will allocate a relatively small portion of the Department of Treasury's annual budget. We estimate that the change will be low-cost as these are annual meetings and will not take very long. The budget would allocate funds to operational aspects of the review process, such as manpower, technology updates, data analysis, and reporting. This targeted funding approach to sanctions review ensures that the necessary resources are available to effectively enhance US policy without requiring a significant increase in the overall budget, maintaining fiscal efficiency.

III. Legislative Analysis

Pros	Cons
Increase trading ability: Annual reviews allow for changes in export and import sanctions, leading to more accurate sanctions for individual countries. Strengthen economic relations: Creating a framework for sanctions review allows for more accuracy and less strain on trade relations due to failure in changing and examining current policies. Increasing oversight: A review process allows lawmakers to examine the purpose of past implemented sanctions to ensure continuous efficacy.	Resources and Time: We would need to allocate additional resources and personnel to ensure the successful completion of this process. 106 Counterargument: However, this newly implemented oversight increases economic stability, government efficiency, and response flexibility.

IV. Conclusion

In the long term, these annual reviews will serve to strengthen global trade as they consider the waning effectiveness of sanctions and their impacts on the US economy and trade relations. Our proposal for legislating an annual review of all sanctions by the Office of Foreign Assets Control (OFAC) is designed to significantly enhance the longevity and relevance of U.S. economic sanctions to prevent them from becoming obsolete. By ensuring that sanctions are regularly assessed and aligned with the latest economic and geopolitical developments, we can prevent outdated measures that may inadvertently hinder international trade and economic growth while furthering our political goals. **However, this proposition may infringe on the jurisdiction of the Foreign Affairs Committee. We acknowledge this discrepancy, despite the affect on sanctions on both affairs and international trade.** Nevertheless, such an approach supports the goals of the Innovation, Data, and Commerce Congressional Subcommittee by fostering a dynamic and responsive economic policy framework that can adapt to global changes and challenges.

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4. The Office of Foreign Assets Control (2021, October) The Treasury 2021 Sanctions Review. <https://home.treasury.gov/system/files/136/Treasury-2021-sanctions-review.pdf>.

Innovation, Data and Commerce Subcommittee: Proposal to Reduce and Hinder Agricultural and Food Monopolization

Members: Kayan Shah, Leah Bei, Yoonseo Lee, Ruth Moss, Derek Qian

I. Introduction

Monopolization and business tactics often associated with this behavior have encompassed rebates, pools, price fixing, and, most prominently, trusts. The prevalent topic of regulating monopolies lingers despite historical efforts to address these concerns. With monopolies commonplace in almost every industry, the subcommittee has focused on agriculture. With recovery since the Pandemic, the food industry has been described with the “graph theory.”¹¹² At its core, the theory reveals how a few significant markets hold a firm grasp of several small businesses, ruining a balanced amount of economic competitors.

First and foremost, it is critical to address the difficulties in solving monopolistic trends within our economy. Even with current policies that oversee and regulate unruly business practices, extracting evidence of price gouging takes a lot of work. One clear indicator lies in the record corporate profits. As rising food costs simply don’t reflect higher production costs, it is necessary to analyze net earnings at a historical level.

This proposal aims to produce legislation that addresses the capitalist threats agricultural monopolies have posed on our society. These corporate entities damage more than business entities, from the farmers and agricultural laborers to the consumers. For example, in the seed sector, the leading four companies dominate 80% of the corn market and 70% of the soybean market.¹¹³ Surprisingly, merely 13.5% of employees in the food sector earn a sufficient income, with farms and meat processing facilities identified as among the riskiest and most unfair workplaces across the country.¹¹⁴ Essentially, only 15 cents of every dollar spent in supermarkets goes to farmers, while the rest is distributed towards processing and marketing our food items.¹¹⁵ Consumers are also significantly impacted by these corporations; grocery prices are at all-time highs, and food processors are reporting record-breaking profits.¹¹⁶ According to

¹¹²“Graph Theory on Food Monopolies : Networks Course Blog for INFO 2040/CS 2850/Econ 2040/SOC 2090.” Accessed April 17, 2024. <https://blogs.cornell.edu/info2040/2021/09/17/graph-theory-on-food-monopolies/>.

¹¹³ “Inside the Seed Industry | AgWeb.” Accessed April 1, 2024. <https://www.agweb.com/news/crops/corn/inside-seed-industry>.

¹¹⁴ Finz, Stacy. “Low Wages of Food Industry Workers Documented.” SFGATE, June 6, 2012. <https://www.sfgate.com/business/article/low-wages-of-food-industry-workers-documented-3612372.php>.

¹¹⁵The Guardian. “Revealed: The True Extent of America’s Food Monopolies, and Who Pays the Price,” February 14, 2019.,<http://www.theguardian.com/environment/ng-interactive/2021/jul/14/food-monopoly-meals-profits-data-investigation>.

¹¹⁶Schweizer, Errol. “Why Your Groceries Are Still So Expensive.” Forbes. Accessed April 17, 2024. <https://www.forbes.com/sites/errolschweizer/2024/02/07/why-your-groceries-are-still-so-expensive/>.

an Axios Study, Walmart controls 25% of the grocery industry.¹¹⁷ Large grocers like Walmart can do this by undercutting smaller grocery operations and putting them out of business. Once they do this, they can pump up grocery prices and make it harder for Americans to access food. This is why we see millions of low-income Americans living with higher health risks; because of sky-high pricing, they are forced to resort to more unhealthy options. This results in double the obesity rates for low-income families.¹¹⁸ The agricultural industry is highly corporate-concentrated, and this issue needs to be solved as it directly harms millions of Americans.

Our goal with this legislation is to fix the monopolization in the agricultural industry and help millions of Americans.

Historically, key statutes such as the Sherman Antitrust Act of 1890 (15 U.S. Code 2) assisted in establishing monopolizing trade as a felony. The legislation eventually supported the curbing of labor unions, further damaging a government interest in improving working conditions. In following years, the Clayton Act of 1914 amended and strengthened the Sherman Antitrust Act, preventing anyone from acquiring assets when “the effect of such acquisition, of such stock or assets...may substantially lessen competition, or tend to create a monopoly.” (15 U.S. Code 18). The Antitrust Division at the Department of Justice has investigated and brought enforcement actions against firms under sections 1 and 2 of the Sherman Act, as well as section 7 of the Clayton Act.¹¹⁹ However, some important exceptions to standard antitrust law have been permitted for agriculture. For example, the Capper-Volstead Act allowed the creation of “agricultural cooperatives” that could “jointly set prices and market their products.”¹²⁰ Also, the Packers and Stockyards Act of 1921 “was designed to prevent meatpackers and processors from using unfair or deceptive practices against farmers and ranchers who sell them livestock” and regulated by the USDA’s Grain Inspection, Packers and Stockyards Administration (GIPSA). Despite this existing legislation, concentration in the agricultural sector continues to be a problem. Several recent bills have been introduced to address the continued concentration in the farm sector, including the Food and Agribusiness Merger Moratorium and Antitrust Review Act and the Farm System Reform Act (FSRA).¹²¹ Both have been introduced but have not proceeded further.¹²² The Biden Administration has also passed new rules to increase the effectiveness of

¹¹⁷ Fitzpatrick, Alex, and Erin Davis. “The Most Popular Grocery Stores in the U.S.” Accessed April 1, 2024. <https://www.axios.com/2023/04/20/most-popular-grocery-stores>.

¹¹⁸ “Rising Food Prices Will Make Obesity Rates Worse, Not Better | WIRED.” Accessed April 17, 2024. <https://www.wired.com/story/rising-food-prices-worse-obesity-rates/>.

¹¹⁹ Antitrust Enforcement and Agriculture. (n.d.). <https://www.justice.gov/atr/file/519371/download>

¹²⁰ Fiore, Katherine. “ACLU v. Miami-Dade County School Board: Reading Pico Imprecisely, Writing Undue Restrictions on Public School Library Books, and Adding to the Collection of Students’ First Amendment Right Violations.” *Villanova Law Review* 56, no. 1 (January 1, 2011): 97.

¹²¹ Don, Stephanie. “What’s The Beef With Tax Credits? Feeding California’s Animal Production Industry.” *UC Law Business Journal* 20, no. 1 (January 1, 2024): 121.

¹²² Sen. Booker, Cory A. [D-NJ. “S.4245 - 117th Congress (2021-2022): Food and Agribusiness Merger Moratorium and Antitrust Review Act of 2022.” Legislation, May 18, 2022. 2022-05-18. <https://www.congress.gov/bill/117th-congress/senate-bill/4245>.

the Packers and Stockyards Act and increase competition.¹²³ Nonetheless, “extremely high levels of concentration in virtually all sectors of agriculture” persist and require more effective legislation.

II. Policy Proposal

We propose reform to demonopolize the agricultural industry.

First - Federal antitrust regulators should block future mergers and retroactively liquidate past mergers that created companies that control over 20% of the market share of any industry. In addition, businesses that exceed that market share should be ordered to divest assets. Antitrust regulators should adopt similar maximum thresholds for cooperative markets to prohibit mergers that would create one cooperative that holds beyond a 5% share of the market. This maximum threshold may be revised and tailored to specific industries within the agricultural sector. This would enable cooperatives to bargain more effectively for smaller farmers, not allowing more giant corporations to gain market share. It would also foster more equitable cooperatives by creating incentives for competition between each other.

Secondly, we need to induce further FTC regulations on agricultural mergers. The FTC needs to create new merger and acquisition rules before approval for heavily impacted industries such as agriculture. Over the past two decades, the FTC has approved numerous harmful mergers that have negatively impacted farmers. We recommend updated regulations from the FTC that establish clear parameters for fair competition and prohibit detrimental practices such as exclusive dealing, which unfairly excludes small food producers and businesses from vital markets.¹²⁴ The FTC should implement measures for evaluating heavy fluctuations in net prices, ensuring price controls are not independent of economic downturns. The commission must set a standard percentage that justifies increases and decreases in net profit.

Third, Congress should shift the burden of proof regarding mergers onto companies. Given that of the 15,000 merger and acquisition deals between 2005 and 2014, only 3% were subject to antitrust scrutiny, mandating proactive justifications for such action would enable more effective FTC oversight.¹²⁵

The policy proposal aims to ensure fair competition for the agricultural sector, protect farmer interests, and promote democratic governance in agricultural cooperatives.

Implementation and funding will involve federal and state antitrust regulators to enforce bright-line rules. Funding will come from discretionary USDA and FTC budgets to oversee compliance and provide guidance on democratic governance. Enforcement of the provision regarding CAFOs would utilize discretionary EPA funding.

¹²³“USDA Announces Next Major Step in Promoting Competition in Agriculture and Advancing Economic Opportunity and Fairness for Growers | USDA.” Accessed April 17, 2024. <https://www.usda.gov/media/press-releases/2024/03/05/usda-announces-next-major-step-promoting-competition-agriculture>.

¹²⁴ Open Markets Institute. “Memorandum - Stop Agricultural Monopolies,” November 16, 2020. <https://www.openmarketsinstitute.org/publications/memorandum-stop-agricultural-monopolies>.

¹²⁵Mooney, Pat. “Too Big to Feed | FAO,” 2017. <https://www.fao.org/family-farming/detail/en/c/1637562/>.

III. Legislative analysis

There are significant upsides to regulating agricultural monopolies. First, setting maximum thresholds for market share and prohibiting mergers beyond certain limits can foster a more competitive landscape. This prevents monopolistic practices and encourages innovation and efficiency. In addition, by blocking mega-mergers and enforcing divestment of assets for companies exceeding certain market shares, the policy aims to protect more minor players in various industries, including farmers and cooperatives. This can prevent more giant corporations from dominating markets and give smaller businesses a fair chance to compete.

Potential downsides to our policy include the adverse effects of decentralizing the agriculture industry. For example, some¹²⁶ have argued that status-quo industrial agriculture has improved production efficiency and that transitioning from monopolized agriculture would diminish yields.

However, industrial agriculture is one of many solutions for producing yields. On a large enough scale, sustainable and organic farms are more productive enough to replace large-scale conventional farms. Various trials and meta-analyses have found that yields for organic fields equal or surpass conventional farms, attributed to their efficient use of water, light, and space.¹²⁷ A long-term trial demonstrated that a diversified industry would provide higher yields. In addition, industrial agriculture is already failing to provide equitable food distribution.¹²⁸ Millions are experiencing food insecurity, and unsustainable farming practices are destroying our environment.

IV. Conclusion

The previous policies provide the groundwork for promoting equitable competition within the agricultural industry. By blocking and liquidating mergers with over 20 percent of market share, the threshold evaluates monopolistic trends that have risen from the consolidation of multiple businesses. The policy aims to reduce the concentration of market power mergers in the industry. Additionally, authorizing the divestment of business assets for companies exceeding the policy threshold further decentralizes monopolistic control. By inducing further Federal Trade Commission regulations to be enforced, new standards on agricultural mergers can ensure close scrutinization of potential monopolies. To conduct continued oversight over authorized mergers, the policy aims to monitor the performance of mergers in the market even after their

¹²⁶Lusk, Jason. "Opinion | Why Industrial Farms Are Good for the Environment - The New York Times," September 25, 2016. <https://www.nytimes.com/2016/09/25/opinion/sunday/why-industrial-farms-are-good-for-the-environment.html>.

¹²⁷Morris, Katlyn S., and Gabriela Bucini. "California's Drought as Opportunity: Redesigning U.S. Agriculture for a Changing Climate." Edited by Anne R. Kapuscinski, Kim A. Locke, and Ernesto Méndez. *Elementa: Science of the Anthropocene* 4 (December 9, 2016): 000142. <https://doi.org/10.12952/journal.elementa.000142>.

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formation. The mergers mentioned above will also be observed in terms of adverse environmental effects through the EPA. By proactively increasing transparency and accountability of mergers, the policy ensures scrutinization despite failing to meet the twenty percent threshold.

This policy proposal falls under the jurisdiction of the Energy and Commerce Committee's Innovation, Data, and Commerce Subcommittee. Still, it *may* overlap with the Judiciary Committee's Competition Policy, Antitrust, and Consumer Rights Subcommittee.

In doing so, we join the Energy and Commerce Committee's unified measures. As a subgroup of that committee, we develop this proposal to follow the mission statement of the Innovation, Data, and Commerce subcommittee in the form of collective group action to regulate national trade and commerce.¹²⁹

¹²⁹House Committee on Energy and Commerce. "House Committee on Energy and Commerce." Accessed April 17, 2024. <https://energycommerce.house.gov/committees/subcommittee/energycommerce.house.gov>.

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Innovation, Data, and Commerce Subcommittee: Regulations to Decrease Bias in AI

Members: Eugenie Cha, Karthi Gottipati, Clayton Nguyen, Cathy Sheng, Irene Wang

I. Introduction

The development and use of AI have grown exponentially, and following the slew of ethical considerations, algorithmic bias cemented itself as a significant threat. Advancements in generative AI have unlocked endless information dissemination and creative expression, empowering businesses to leverage AI in guiding process automation, data analysis, and customer and employee engagement.

As a fledgling field, opportunity in innovation has drawn the attention of many companies who are turning their sights to developing and utilizing AI for their customers and employees. Unfortunately, little legislation exists for a field expanding as rapidly as this one, and urgent problems have risen, chief among them being algorithmic bias. The data sets used to train AI algorithms are at the heart of the issue. Such sets determine the outcomes generated by the algorithm, thereby perpetuating biases in the data. The ramifications of these biases have already been discovered in various sectors, from credit scoring to healthcare and criminal justice, often resulting in discriminatory practices such as racial profiling.

Our proposed policy hopes to achieve data set standardization and mitigate biases through two parts: auditing done by the NIST under the AI Risk Management Framework and creating standards that identify “trusted” training datasets. By implementing these standards, we can ensure fair and equitable outcomes for individuals, businesses, and governments.

While existing initiatives, such as President Biden’s “Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence”¹³⁰ and the Blueprint for an AI Bill of Rights¹³¹, represent crucial steps forward, further legislative action is imperative. President Biden’s Executive Order underscores the dual capability of AI to address urgent challenges and exacerbate societal harms. This document outlines what AI legislation could look like going forward; however, it currently requires Congressional action and budgeting. The AI Bill of Rights is a set of five principles for businesses to model their systems after, which are made to inform policy decisions and provide values for ethical AI development and deployment. Addressing and creating a regulatory environment for responsible innovation is vital in navigating AI governance. Collaboration between government, industry, and academia is critical, and an inclusive and ethical AI standard must be set.

¹³⁰ Exec. Order No. 14110, 3 C.F.R. (). <https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/>

¹³¹ The White House. 2022. “Blueprint for an AI Bill of Rights: Making Automated Systems Work for the American People.” <https://www.whitehouse.gov/ostp/ai-bill-of-rights/>

II. Policy Proposal

To address the present unethical use of AI, we propose legislation that will implement a two-part regulation for AI in commercial use:

- **Auditing by the National Institute of Standards and Technology (NIST):** NIST is a government agency that promotes and works with industry to help them develop standards¹³². In light of recent AI advancements, starting in 2020, NIST has held workshops and discussions with the private and public commercial sectors about using AI reliably¹³³. Therefore, they are a suitable agency to assess AI models used in the commercial sector. Moreover, we propose NIST use the AI Risk Management Framework. NIST established this framework to help organizations in the industry manage the risks involved with using AI. Previously, organizations only used this framework voluntarily to evaluate their AI. However, we propose that NIST use this framework to evaluate and audit AI models, adding a seal of approval to AI models. These seals may benefit organizations by building credibility and greater trust with the public while ensuring that their AI models are being used appropriately.
- **Increased Database Transparency and Regulation:** As mentioned in the introduction, the databases used to train AI significantly impact its ability to make ethical judgments. Therefore, assessing whether non-biased databases were used in AI training is crucial in ensuring that the AI used by organizations is ethical. We propose that NIST creates a standard for “trusted” training databases to encourage non-biased databases. By following these standards, organizations may market that they align with NIST database standards and gain further reputability and trust with the public¹³⁴.

With NIST auditing commercial AI systems under the AI Risk Management Framework and encouraging the use of standardized, non-biased data, our legislation ensures that AI systems minimize profiling bias.

Lastly, our proposed legislation does not require funding, as it is based on regulation by existing committees.

III. Pros

Implementation of the NIST auditing paired with the extra regulation of ensuring standardized/non-biased data used will increase fairness and reduce biases in AI. The NIST

¹³² *National Institute of Standards and Technology*. NIST. (2024, April 17). <http://www.nist.gov/>

¹³³ *Artificial Intelligence*. NIST. (2024a, March 26). <https://www.nist.gov/artificial-intelligence>

¹³⁴ *Artificial Intelligence: Background, selected ... - CRS reports*. (n.d.). <https://crsreports.congress.gov/product/pdf/R/R46795>

values of accountability, transparency, security, and privacy¹³⁵ are crucial in helping industries develop guidelines for safe and neutral AI, allowing impartial treatment towards marginalized people and the ethical progress of technology. The rigid and thorough checks grant greater assurance and reliability in the quality of AI that earn the NIST's audit, rekindling public trust. Furthermore, the transparency of regulation to the public will not only allow consumers to feel their safety is highly valued in using NIST-approved products but also let them know exactly how they are being protected. This increased public support will only fuel innovation and strengthen AI advancements. The clearly defined, set guidelines of the AI Risk Management Framework will significantly expedite and mainstream the vague ethical territory through which current AI innovation is floundering. Additionally, it's the introduction of the first framework for which AI is built, helping increase standardization. In the equity and social context, it will increase inclusivity and accessibility for marginalized communities, minimizing the risks and harms seen with modern AI innovations. Whether be racial profiling in face-recognition apps¹³⁶ or statistically hiring fewer women¹³⁷ in AI-interview screening, which could be attributed to the underrepresentation of minority people in training data or biases encoded in computer algorithms, our comprehensive approach pairing NIST auditing and collaboration with industries, as well as recommending NIST to create a standard for non-biased databases will combat this prevalent issue, protect vulnerable communities, and steer AI into a new future of innovation.

IV. Cons

Seals of Approval: While providing a seal of approval from the government generally avoids the dangers associated with complete and mandated standardization, the intent and the specific context for which these credentials are applicable rarely come across to consumers. For example, despite a proliferation of environmental certification initiatives seeking to give the ethical consumer the information to make environmentally conscious purchasing decisions, such programs have been marred by scandal and oversight failures, where low environmental standards gave consumers the impression of eco-friendliness in a phenomenon often called greenwashing. The danger of providing a government-sponsored seal of approval is two-fold for consumers. The first is largely ubiquitous among this kind of certification of safety, quality, or any other range of positive adjectives - we might be wrong. Artificial intelligence is inherently risky, and by providing government endorsement, the risk becomes greater that consumers may place their confidence in models that may not always be correct - sometimes with consequences.

¹³⁵ National Institute of Standards and Technology. (2024). The NIST Cybersecurity Framework (CSF) 2.0. The NIST Cybersecurity Framework (CSF) 2.0, 2.0. <https://doi.org/10.6028/nist.cswp.29>

¹³⁶ Findley, B. (2020, November 3). *Why Racial Bias is Prevalent in Facial Recognition Technology*. Harvard Journal of Law & Technology. <https://jolt.law.harvard.edu/digest/why-racial-bias-is-prevalent-in-facial-recognition-technology>

¹³⁷ Reuters. (2018, October 11). *Amazon Ditched AI Recruiting Tool That Favored Men for Technical Jobs*. The Guardian; The Guardian. <https://www.theguardian.com/technology/2018/oct/10/amazon-hiring-ai-gender-bias-recruiting-engine>

Look to the cases like ChatGPT providing incorrect legal information or medical advice - with government approval, these models may become even more trusted and unjustifiably have their results taken at face value. This ties into the second issue: Government sponsorship of AI models, especially generative ones, may lead to an uptick in consumer trust even outside of the range for where this approval may be designed. Per the Federal Trade Commission, consumers often interpret such certifications with the broadest possible scope - often outside of what the regulatory agency intended to convey.¹³⁸ For example, current recycling regulations

While ensuring that AI training databases are standardized and non-biased is crucial for ethical AI development, the proposed approach of categorizing AI systems and standardizing databases horizontally may oversimplify the complexity of AI applications. Different industries have unique data requirements, and a one-size-fits-all approach to database standardization may not be practical or effective. Additionally, achieving true database neutrality is challenging, as biases can be unintentionally introduced at various stages of data collection, labeling, and processing. Government regulation and standardization also tend to suppress innovation¹³⁹ - Currently, some companies privately develop similar systems, which this proposal may undercut. Similar to the USB-C charger mandate in the EU for mobile phones, standardizing database architecture may have the unintended consequence of suspending future progress to the point of standardization. Furthermore, datasets are currently in high demand and low supply¹⁴⁰, especially for larger models such as facial recognition systems and biometrics. Limiting access to datasets or restricting which datasets can be used in training such models by an arbitrary level of diversity or unbiasedness risks preventing the development of those same models. The consistent risk of all government regulation is the burden of red tape and bureaucratic inefficiencies. Furthermore, auditing machine learning systems can be time-consuming and fruitless - any audits need to have a highly specific purpose because even computer scientists cannot fully understand how a model may function once it is trained and released. The overall risk is that we may destroy the industry instead of direct it by overregulating it and creating burdensome requirements or compliance costs.

V. Counterarguments

Despite these risks, the tradeoffs are a net positive at both a corporate and societal level. For corporations, especially those engaged in innovative development, the standardization of datasets provides significant commercial upside. One of the largest roadblocks to expanding

¹³⁸ Fair, L. (2015, September 14). Performing seals. Federal Trade Commission. <https://www.ftc.gov/business-guidance/blog/2015/09/performing-seals>

¹³⁹ Aghion, P. (2021, January 21). The Impact of Regulation on Innovation. NBER. <https://www.nber.org/papers/w28381>

¹⁴⁰ Mori, L. (2023, January 20). Clearing data-quality roadblocks: Unlocking AI in manufacturing. McKinsey. <https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/clearing-data-quality-roadblocks-unlocking-ai-in-manufacturing>

machine learning models is a lack of access to a broad range of properly labeled data. Creating a set of industry standards while undercutting some specific commercial sectors may support more innovation in the AI industry by providing a foundational basis for building more technology. The current lack of an industry standard has led to the duplication of effort across corporations seeking to create proprietary technology, each attempting to create their walled garden. By standardizing and expanding the accessibility and diversity of datasets, the government contributes to an overall higher quality of AI infrastructure - the same way a rising tide lifts all boats; by improving the baseline for data, every innovator is better off.

VI. Conclusion

The regulation of AI algorithms and their databases is the most critical step in reducing AI biases. By giving AI a wide selection of databases rather than certain ones targeted only at a certain group of people and regulating AI data training, racial biases and other forms of biases would be reduced in most cases. These would affect many issues in the long term for the better, including more accurate results for underrepresented groups and minorities. As stated, AI databases are usually tested by data that favors a specific group. Regulations would reduce this. Alongside this, AI biases are also present in the societal context in which they are used. Biased AI algorithms can be harmful when deciding if a person is accepted to a school, eligible for a loan, or becoming a rental applicant. Additionally, current AI algorithms are inefficient in today's society: only computational and statistical biases are noted in AI training. NIST can easily use its power to consider human and systemic biases in AI training to further improve their responses in social situations. Human biases primarily involve filling in missing information based on prior experiences and knowledge. Systemic bias is when an institution operates in ways to disadvantage underrepresented groups and minorities. By taking these biases into account and recognizing that AI operates very socially, we can make efforts to solve the social aspects of the socio-technical solution. In conclusion, NIST auditing and regulation of AI training data will significantly improve AI's ethical and social uses. Eliminating biases, especially in the social context, will benefit all groups rather than just one group.

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Undergrounding Interstate Electrical Lines Proposal

1. Introduction

Wildfires pose a significant and rapidly rising threat to communities, ecosystems, and infrastructure across the western United States. The dangers of wildfires show the urgent need for proactive measures to reduce risks and protect lives and property. One aspect of wildfire prevention is protecting against downed power lines, which can start fires in remote locations. Furthermore, undergrounding these high-voltage transmission lines allows them to be less susceptible to damage, thus increasing reliability. By burying the lines underground, potential hazards are mitigated, and the overall resilience of the electrical grid is strengthened.

2. Proposal

The proposed legislation allocates \$6.65 billion in funds to fund undergrounding 1900 miles of interstate high-voltage transmission lines in high-risk wildfire regions as defined by federal agencies such as the U.S. Forest Service and the National Interagency Fire Center. This proposal is based on an estimated cost of 3 million dollars per mile by the California Public Utilities Commission to underground these lines, with some additional funds allocated within the proposal if the project exceeds its expected budget. The legislation will also allow the adoption of advanced underground transmission technologies to our aging infrastructure to further ensure the reliability and efficiency of the grid while minimizing environmental damage. This may include the use of resilient materials, smart grid technologies, and advanced monitoring systems to enhance grid performance and response capabilities. The purpose of undergrounding interstate high voltage transmission lines in high-risk wildfire areas is to make the nation's electrical grid more resilient and reduce the risk of catastrophic wildfires. This bill seeks¹⁴¹ to minimize the ignition potential of power lines during extreme weather conditions, thereby reducing the frequency and severity of wildfires.

3. Legislative analysis

The proposed policy would reduce the threat of wildfires while also increasing the reliability of the national power grid. The current power grid consists of more than two million miles of overhead power lines, most of which are between 30-40 years old, and was not built with our current climate in mind, and with the effects of climate change this system has become increasingly vulnerable to extreme weather conditions like strong winds and wildfires¹⁴². In the face of climate change, the dangers these conditions pose will continue to be exacerbated; thus it is critically important to make improvements to the grid that will allow it to withstand variable weather conditions, and an effective solution is moving more power lines underground

¹⁴¹ Scenic America 2023

¹⁴² McBride 2022

where they are not as susceptible to damage from severe weather or other security threats posed by animals, vandalism, and falling trees. Most of all, the proposal aims to remove the threat of downed overhead power lines igniting wildfires. This proposal aims to prevent such catastrophes as the 2018 Camp Fire in California - the deadliest wildfire in the state's history that caused 85 deaths and \$16.5 billion in damages and was ignited from a fallen power line - from occurring by undergrounding lines in the specified areas of high risk, thus reducing our reliance on overhead power lines that hold the risk of falling down in storms and igniting vegetation as well as causing property damage.³

Drawbacks of this proposal include the high cost to construct underground power lines, damage to the environment as a byproduct of the installation process, and the difficulty of maintaining underground lines. The largest concern surrounding the move to underground power cables is their high price to install in comparison to overhead lines; however, this perspective fails to consider that the economic and environmental cost of devastation from wildfires is much greater. For example, the Dixie Fire, the second-largest wildfire in California state history, was ignited by a tree falling onto distribution lines,⁴ and burnt 963,000 acres of land and caused a total of \$637 million in damages.⁵ Thus the cost to move wires underground is offset by money spent on wildfire damages, economic losses due to extended power outages, and the price of human suffering and ecological devastation that results from catastrophic wildfires. Another concern with undergrounding lines is with the environmental disruption caused by excavating land for underground wires; however, the damage from this process is not long-term, in comparison to overhead lines, which cause disruption to the environment post-installation due to their visual impact and the risk to birds of colliding with the lines.⁶ While the installation of underground cables may be expensive and disruptive to the environment in the short-term, once installed they have very little long-term impact on the environment, and the benefits they provide far outweigh the high cost. Finally, the reliability of underground cables is diminished by the longer restoration times in face of equipment failure due to the lines being harder to access and the difficulty of locating the source of an outage; however, studies have shown an overall positive correlation between undergrounding lines and increased reliability compared to overhead lines. This correlation is strongest for heavily forested areas where falling trees are most likely to bring down overhead power lines,⁷ meaning that underground cables are particularly more reliable in high-risk areas that are densely vegetated. Furthermore, improvements in technology, such as intelligent switches, Outage Management Systems, and Geographic Information Systems, are likely to help offset the difficulty of locating outage sources in the future, decreasing the longer outage times for underground lines and further improving reliability.⁸

³Zimmerman 2020

⁴CPUC

⁵Tutella 2022

⁶Zavolas 2013

⁷Fenrick *et al.* 2012

⁸Fenrick *et al.* 2012

4. Conclusion

The proposed legislation of \$6.65 billion to underground interstate high voltage transmission lines in high-risk wildfires is important for the future reliability of the power grid. Approving this legislation would address the immediate threat posed by wildfires saving lives and mitigating the cost used for wildfire restoration. It would also lay the groundwork for the future, with climate change making weather events more unpredictable and extreme, where the grid would be safe and reliable through almost any event. Although there is a large initial investment for the undergrounding project, the long-term benefits would far outweigh the cost. Our proposal would give many benefits such as the reduced risk of wildfires, preparedness for the future and reliability while also reducing the operation and maintenance costs, saving money in the long-term. This legislative proposal is an investment in resilience that will pay dividends for generations to come, ensuring that our electrical infrastructure can withstand the challenges of a changing climate while safeguarding lives, property, and the environment.

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Energy, Climate, and Grid Security: Fracking

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I. Introduction:

Hydraulic fracturing, commonly referred to as fracking, is a process of extracting natural gas and oil from deep inside the ground. Because of its ease and economic advantages, large corporations have set up fracking stations across the country's natural gas deposits to obtain the resources within. However, fracking can produce large amounts of pollutants in the form of greenhouse gasses and wastewater, often devastating local communities and the environment.

The principal harms of fracking are threefold:

1. **Release of Pollutants:** Fracking has severe impacts both in the United States and worldwide. It accounts for 63% of the increase in global emissions, 89% of which comes from the United States.¹⁴³ Fracking is so significant because it releases a high amount of methane, which has an outsized impact on climate change compared to other pollutants.¹⁴⁴ In the United States alone, 17.6 million people live within a mile of a fracking site which poses a grave danger to public health.¹⁴⁵ As a direct result of fracking-related air pollution, children in this region between the ages of 2 and 7 are two to three times more likely to be diagnosed with leukemia than average, while the elderly face a substantially higher risk of death.
2. **Chemical Contamination:** For fracking to occur, a mixture of chemicals is injected deep into the ground at high pressure. However, the unfortunate reality is that such chemicals can poison water supplies. In a study that analyzed 1000 chemicals used in fracking, researchers realized that the released chemicals have devastating effects on those who consume them.¹⁴⁶ Even worse, the same researchers found that there is limited information about many of these chemicals, suggesting that the full extent of the harms of fracking may be even greater than we currently believe.
3. **Overconsumption:** An often overlooked problem with fracking is the immense amount of water necessary to maintain the process. A recent report found that fracking companies have used 1.5 trillion gallons since 2011, with the rate of consumption increasing exponentially since then. In Texas alone, the state is expected to see a $\frac{1}{3}$ decrease in its groundwater supply by 2070 due to fracking alone.¹⁴⁷

¹⁴³ Howarth, R. W. (2019, August 14). *Ideas and perspectives: Is shale gas a major driver of recent increase in global atmospheric methane?*. Biogeosciences. <https://bg.copernicus.org/articles/16/3033/2019/>

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¹⁴⁵ Hurdle, J. (2022, November 17). *As Evidence Mounts, New Concerns About Fracking and Health*. Yale Environment 360. Retrieved April 17, 2024, from <https://e360.yale.edu/features/fracking-gas-chemicals-health-pennsylvania>

¹⁴⁶ Tabuchi, H., & Migliozi, B. (2023, September 25). 'Monster fracks' are getting far bigger. and far thirstier. *The New York Times*. <https://www.nytimes.com/interactive/2023/09/25/climate/fracking-oil-gas-wells-water.html>

¹⁴⁷ Elliott, E. G., Ettinger, A. S., Leaderer, B. P., Bracken, M. B., & Deziel, N. C. (2016). A systematic evaluation of chemicals in hydraulic-fracturing fluids and wastewater for reproductive and developmental toxicity. *Journal of Exposure Science & Environmental Epidemiology*, 27(1), 90-99. <https://doi.org/10.1038/jes.2015.81>

While the issue of fracking is multi-faceted, it is in large part driven by continued drilling on federal public lands and water. Around a quarter of all US emissions come from drilling on federal lands.¹⁴⁸ The problem is only growing worse today, as the previous Trump administration leased 5.4 million acres of public lands for new drilling projects.¹⁴⁹

II. Proposal:

We propose the following solution: a ban on fracking in federal lands and water. This would be done by ending new permits for drilling and allowing existing permits to phase out. The proposal represents a significant step in meeting our climate targets as demonstrated by a study that states that restricting fossil fuel extraction would reduce emissions by 37% of 280 million tons by 2030.

The proposal is additionally beneficial in encouraging the transition towards cleaner forms of energy. This occurs in three main ways:

1. The proposal opens up many federal lands for the development of renewable energy. Fracking on federal lands actively hinders renewable energy projects.¹⁵⁰ The Bureau of Land Management concluded that such projects are “incompatible” with continued oil and gas extraction. The regulatory process for renewable development is far more robust than the process for oil and gas, which has resulted in the vast majority of federal lands being used for oil and gas development. However, restricting fracking on these lands would lead to a significant portion of federal territories shifting towards renewable forms of energy production.
2. The policy has spillover effects on the private market in changing investment patterns. Research demonstrates that climate policies enacted between 2015 to 2021 accounted for a 12% reduction in investment in oil and gas because private markets respond to changes in the industry when making future investments.¹⁵¹ The study found that policy “can reduce uncertainty, encourage investment in renewable energy, and incentivize the proper amount of fossil fuel divestment”.
3. This policy will be instrumental in restoring US climate leadership. The impacts of US climate policy are often both domestic and international. This is confirmed by how, for every ton the US has pledged to reduce emissions by, other countries have followed suit by pledging to reduce 6-7 tons of international emissions.¹⁵² Furthermore, US climate

¹⁴⁸ Erickson, P., & Lazarus, M. (2018). Would constraining US fossil fuel production affect global co2 emissions? A case study of US leasing policy. *Climatic Change*, 150(1-2), 29-42. <https://doi.org/10.1007/s10584-018-2152-z>

¹⁴⁹ Holden, E., Tobias, J., & Chang, A. (2020, October 26). Revealed: the full extent of Trump's 'meat cleaver' assault on US wilderness. *The Guardian*. <https://www.theguardian.com/environment/ng-interactive/2020/oct/26/revealed-trump-public-lands-oil-drilling#:~:text=New%20research%20conducted%20by%20the,to%20oil%20and%20gas%20companies>.

¹⁵⁰ Rowland-Shea, J., & Mirza, Z. (2022, July 19). *The Oil Industry's Grip on Public Lands and Waters May Be Slowing Progress Toward Energy Independence*. Center for American Progress. Retrieved April 17, 2024, from <https://www.americanprogress.org/article/the-oil-industrys-grip-on-public-lands-and-waters-may-be-slowing-progress-toward-energy-independence/>

¹⁵¹ Bogmans, C., Pescatori, A., & Prifti, E. (2023). The Impact of Climate Policy on Oil and Gas Investment: Evidence from Firm-Level Data. *International Monetary Fund*, No. 2023/140.

¹⁵² Houser, T., & Larsen, K. (2021, January 21). *Calculating the Climate Reciprocity Ratio for the US*. Rhodium Group. Retrieved April 17, 2024, from <https://rhg.com/research/climate-reciprocity-ratio/>

action would make clean energy more globally competitive: by restricting continued fracking on federal lands, the U.S. would signal to the world that our climate commitments are credible encouraging greater action.

III. Legislative Analysis:

Naturally, any major proposal to address climate change will cause concerns. The most significant objection is the potential job losses that might result because of fracking restrictions. However, it should be noted that this proposal does not seek to eliminate fracking entirely, but rather restrict fracking on federal land, which will significantly decrease the number of lost jobs. Workers who are affected can seek out job training programs under the Infrastructure Investment and Jobs and Inflation Reduction Act, which have received hundreds of billions of dollars in investment. This would enable a relatively smooth transition as the skills used in fracking operations are often similar to those used in renewable energy production.¹⁵³ Additionally, fossil fuel jobs have been declining ever since our nation's recovery from the COVID-19 pandemic, whereas the number of jobs in the renewable sector has dramatically increased.¹⁵⁴ This is attributed to the increasingly reduced cost of production as the renewable energy industry expands. Our proposal will help accelerate the U.S.' transition to clean energy.

The next major concern is the potential impact that fracking restrictions would have on the economy. However, this argument is addressed by the negative impact fracking has on today's economy. In the industry, the level of debt has doubled over the past twenty years, while the return on investment was considered disappointing in 70% of cases.¹⁵⁵ Furthermore, fracking costs the U.S. \$600 billion annually due to the decrease in productivity and premature deaths in neighborhoods near fracking sites: 1.8 billion days of work absences and 1.13 times the number of birth defects are caused by fracking alone.¹⁵⁶ The shift to renewable energy would solve such issues as cleaner energy as a recent study found that shifting to renewable energy would save households \$500 annually. That would spur greater consumer spending which benefits the overall economy.

Finally, critics may argue that this policy may result in tremendous economic shocks due to the oil industry crumbling. However, the fall of the oil industry is inevitable due to a green transition but this is better done sooner rather than later. Furthermore, by ending the leases of federal lands now, investors will have the opportunity to slowly back out of their holdings, allowing the relevant parts of the industry to gradually dissolve rather than suddenly disappear. Ultimately, this will have a salutary effect on the economy.

¹⁵³ Frank M. R., & Lim, J. (2023, September 26). *Fossil fuel workers have the skills to succeed in green jobs, but location is a major barrier to a just transition*. The Conversation. <https://theconversation.com/fossil-fuel-workers-have-the-skills-to-succeed-in-green-jobs-but-location-is-a-major-barrier-to-a-just-transition-214125>

¹⁵⁴ Technology, E. I. P. and. (2022, November 8). *Clean energy jobs are booming, making up for rising fossil fuel unemployment*. Forbes. <https://www.forbes.com/sites/energyinnovation/2022/06/29/clean-energy-jobs-are-booming-making-up-for-rising-fossil-fuel-unemployment/?sh=d31c91b1c138>

¹⁵⁵ Taylor, B. (2021). Why are natural gas prices high? because fracking isn't really profitable. <https://www.barrons.com/articles/why-are-natural-gas-prices-high-because-fracking-isnt-really-profitable-51635439006>

¹⁵⁶ *New research: Air pollution from fossil fuels costs the world \$8 billion every day*. (2020, February 12). Green Peace. Retrieved April 17, 2024, from <https://www.greenpeace.org/usa/news/new-research-air-pollution-from-fossil-fuels-costs-the-world-8-billion-every-day/#:~:text=Researchers%20also%20estimate%20global%20economic,billion%20in%20economic%20losses%20annually.>

IV. Conclusion:

Fracking, as a process, generates a devastating amount of pollution. With the United States producing 89% of global fracking emissions, we must introduce legislation to decrease the scale of fracking to protect the 17.6 million people in the US within a quarter mile of a fracking site. Although there will be an economic and job cost for this plan to phase out fracking on national land, the benefits to our nation's health and environment heavily outweigh the negatives. We believe that this policy should be seen as an investment in our future, not as a cost. By investing in our environment, we can drastically reduce the total emissions in the United States and pave the way for a greener, more sustainable future.

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[transition#:~:text=At%20home%2C%20Just%20Transition%20means,and%20protection%20for%20disadvantaged%20communities.](https://www.dol.gov/agencies/ilab/just-transition#:~:text=At%20home%2C%20Just%20Transition%20means,and%20protection%20for%20disadvantaged%20communities)

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Nuclear waste

Written By:

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Introduction:

We have noticed the lack of initiative in the nuclear waste disposal industry and wondered if there were better ways to handle the waste produced to sustain our energy systems. With the increasing population growth, there is an increasing demand for effective clean energy sources. We noticed this problem and decided to create alternative strategies for dealing with nuclear waste. Because LLNW (low-level nuclear waste) has an abundance of management strategies, with most of them being incredibly effective whether it be underground storage or recycling, we will be focusing most of our effort on HLN (High level-nuclear waste), as we noticed a few gaps in the management of this particular type of waste.^{157 158} We hope to bring this issue to your attention. Hopefully, some of our ideas will inspire the government and spur them to consider this problem seriously, leading to increased funding and budget in nuclear waste management.

The Nuclear Waste Policy Act of 1997 (H.R. 1270) encourages the use of deep geologic repositories to store spent nuclear fuel, which is a short-term solution we desire to supplement.¹⁵⁹ However, in the long term, geologic repositories may not be a logical choice because, as they reach capacity, the environmental impact of spent nuclear fuel will increase drastically as more facilities are built. So, encouraging the exploration of long-term solutions now can help supplement our current systems and prepare for the country's future energy needs.

Policy Proposal:

Our proposal to the nation is to increase funding and budget for research initiatives in nuclear waste reprocessing technology. We recognize the risks of new technological advancements and the varying results. However, research will always be an effective use of tax dollars. The lack of budget for nuclear waste management will be problematic as our society demands more and more energy to accommodate our increasing population. This proposal relies on a persuasive article to convince the government to allocate more funds to nuclear waste management and disposal. This overall endeavor can be funded through non-profit supporters and tax revenue. Signatures for this cause can also be a point of persuasion. Still, the support and

¹⁵⁷ Backgrounder on Radioactive Waste. (n.d.). NRC Web. Retrieved April 3, 2024, from <https://www.nrc.gov/reading-rm/doc-collections/fact-sheets/radwaste.html>

¹⁵⁸ Radioactive Waste Management | Nuclear Waste Disposal—World Nuclear Association. (n.d.). Retrieved April 3, 2024, from <https://world-nuclear.org/information-library/nuclear-fuel-cycle/nuclear-wastes/radioactive-waste-management.aspx>

¹⁵⁹ Rep. Upton, Fred [R-MI-6. H.R.1270 - 105th Congress (1997-1998): Nuclear Waste Policy Act of 1997. 2 June 1998, <https://www.congress.gov/bill/105th-congress/house-bill/1270>. 1997-04-10.

funding come from the community who want to see change and don't like living near nuclear waste disposal reservoirs.

Legislative Analysis:

We acknowledge that this may not be a short-term solution, and there are many caveats that we also recognize from the backlash in public perception. Short-term solutions are key, and we realize that as a team. However, long-term sustainability is the end goal of most environmental proposals; we recognize that our current crisis needs to be resolved to reduce the acute situation. Public perception is also tricky, as nuclear waste management is controversial. That being said, we offer two solutions to the problem of nuclear waste management.

1. Although long-term sustainability is the overall goal, there are also solutions to deal with the problem in the short term that reduces the situation's urgency. During the Obama administration, a facility known as Yucca Mountain, a storage unit, was closed amid political turmoil. There was much discussion around this unit, with many questions about modes of transportation and permanent disposal once the facility reached its maximum capacity. While there are obstacles to reinstating this space, it is a realistic short-term solution while we build up long-term solutions in the background. The initial issues surrounding the repository closure should be addressed; however, as already stated, most were due to the current administration, political events, and a negative public perception of nuclear energy. We must emphasize the research benefits of this re-opening and highlight how this will buy us the necessary and valuable time to create the most sustainable solutions. Political turmoil will always exist, but the specific problems that arose at the time of the Yucca Mountain closure do not carry over. However, environmental issues and the controversy around nuclear energy and waste management will not change with the political tides. This problem needs to be addressed as swiftly as possible, and Yucca Mountain is exactly that quick, decisive solution we need. With the addition of some genuine intentional effort in funding, coordinating, and rebuilding, Yucca Mountain has the potential to be an excellent location as it has already been extensively researched for its viability as a high-level nuclear waste storage facility. The risk is minimal because of the geographic location; natural risk factors such as water contamination and seismic activity do not threaten Yucca Mountain.¹⁶⁰
2. Several countries reprocess waste from nuclear facilities to be used again as fuel. However, the United States currently does not, yet 25 to 30 percent more energy from uranium can be gained from reprocessing, potentially reducing waste to one-fifth⁴ the amount produced by direct disposal.¹⁶¹ Compared to direct disposal, the small amount of waste from spent nuclear fuel reprocessing means less storage space would be needed to manage increasing nuclear energy reliance. Uranium resources would also be further extended, as less would have to be mined, which is an additional environmental benefit

¹⁶⁰The research division September 2018 legislative counsel bureau, Yucca Mountain Project. September 2018, www.leg.state.nv.us/Division/Research/Publications/Factsheets/YuccaTimeline.pdf.

¹⁶¹ Processing of Used Nuclear Fuel—World Nuclear Association. (n.d.). Retrieved April 3, 2024, from <https://world-nuclear.org/information-library/nuclear-fuel-cycle/fuel-recycling/processing-of-used-nuclear-fuel.aspx>

to reprocessing. Although studies have demonstrated that reprocessing costs would be higher than directly storing nuclear waste, reprocessing could be economically viable in the long run.¹⁶² Increasing uranium prices on top of an ever-growing need for storage space for spent nuclear fuel points to reprocessing as a viable, cost-effective option for dealing with nuclear waste. Ongoing research continues to explore safer and more effective reprocessing techniques. Current concerns associated with reprocessing can be addressed by funding research focused on the topic.¹⁶³

The public eye is critical and may be one of the most significant obstacles if this proposal focuses on shifting storage locations. However, our primary focus is research funding. The government has proven to be interested in scientific advancement and historically financially capable of funding it. We acknowledge that a sudden change in nuclear management can be difficult; however, simply funding the research of new techniques and technology is easier for the public to get behind. Scientific advancements are generally celebrated for the potential benefits they can offer the country. By funding research that makes advancements focusing on long-term solutions, the public will be able to see the concrete data for why reprocessing is the best way to not only deal with current reactor output but also for the potential expansion of nuclear energy in the future.

By taking advantage of the Yucca Mountain facility and funding research related to nuclear fuel reprocessing, we hope to create a system where both the present and the future are secured in environmental sustainability as it relates to nuclear waste.¹⁶⁴ Managing nuclear waste is increasingly vital as an ever-growing population puts a greater demand on the energy supply. As we slowly shift from fossil fuels and “natural” productions of energy such as water, wind, solar, etc., to nuclear power, we need to be more responsible as to the consequences of these hazardous materials’ effects on the environment.¹⁶⁵ To continue progressing in society, we need to balance advancement and sustainability, something that has continuously been brought up but has recently become more of an acute and widespread issue.

Conclusion:

Our society is ever-growing and expanding, putting strain on the environment through various means. We must start to address these factors, one of which is the disposal of nuclear waste, particularly high-level nuclear waste. Recognizing and addressing this concern allows our society to continue and sustain its energy systems. Our proposed policy entails areas in our

¹⁶² Costs of Reprocessing Versus Directly Disposing of Spent Nuclear Fuel | CBO TESTIMONY by Peter R. Orszag Retrieved April 3, 2024, from <https://www.cbo.gov/sites/default/files/cbofiles/ftpdocs/88xx/doc8808/11-14-nuclearfuel.pdf>

¹⁶³ Nuclear Reprocessing: Dangerous, Dirty, and Expensive | Union of Concerned Scientists. (n.d.). Retrieved April 3, 2024, from <https://www.ucsusa.org/resources/nuclear-reprocessing-dangerous-dirty-and-expensive>

¹⁶⁴ Processing of Used Nuclear Fuel - World Nuclear Association. <https://world-nuclear.org/information-library/nuclear-fuel-cycle/fuel-recycling/processing-of-used-nuclear-fuel.aspx>. Accessed 17 Apr. 2024.

¹⁶⁵ Ewing, A. M., Rodney C. (n.d.). Nuclear Waste Is Piling Up. Does the U.S. Have a Plan? Scientific American. Retrieved April 3, 2024, from <https://www.scientificamerican.com/article/nuclear-waste-is-piling-up-does-the-u-s-have-a-plan/>

management systems that have gaps. By funding research and increasing the budget for nuclear waste management, we hope to address the environmental and societal impacts of nuclear waste disposal. Research and science can flourish under our government's funding.¹⁶⁶ In that case, we can find strategies to handle waste sustainably, effectively, and efficiently.

This policy proposal is only one small part of the greater issue. Our society lacks the initiative to create environmentally sustainable solutions and has had a historical tendency to push aside matters until they reach a tipping point. We cannot risk this with environmental issues, as consequences are often unseen and extreme. Much of our focus and funding has been allocated to fixing the symptoms, not the root of the problem. Still, if we can invite science and technology into these governmental issues, we will take one step forward in a sustainable future. We invite Congress to recognize the situation's urgency and take proactive measures to address nuclear waste management and environmental issues. Together, with different ideas from scientific minds, we pave the way towards a safer, sustainable energy future for generations to come.

¹⁶⁶Office, U. S. G. A. (2017, February 15). Nuclear Waste Disposal | U.S. GAO. <https://www.gao.gov/nuclear-waste-disposal>

Energy, Climate, and Grid Security: Renewable Energy - Renewable Energy Permitting and Sustainability Act

Members: Kabir Buch, Sahngwie Yim, Sophia Tian

I. Introduction

Developing effective, sustainable renewable energy sources is crucial to address the current climate change crisis which has had widespread effects on the economy, human health, the environment, and other important sectors of our nation. The best way to address this crisis is by transitioning to a green economy, which renewable energy is a key component of.¹⁶⁷ As energy demands rise and the need for renewable energy grows, the United States grid must adapt and expand. Renewable energy production is currently bottlenecked by the lack of grid infrastructure to distribute it. Although challenges to the grid have gained more attention and federal funding in recent pushes for reform, funding allotted to the grids has been limited in scope and thus unable to resolve many of the distribution problems renewable energy companies face.¹⁶⁸ Our legislation aims to modernize and expand the grid, enabling renewable energy distribution while simultaneously incentivizing its production, aiding the United States' transition to a zero-carbon economy.

Effective in 2022, the Inflation Reduction Act (IRA) was the largest investment in climate policy to date, putting the US at the forefront of tackling the climate crisis. The IRA supported issuing new loans, provided more funding, and dedicated resources to transition to a green economy.¹⁶⁹ While the IRA successfully provided resources and funding for new renewable energy development, it failed to address the inefficiencies in connecting these renewable energy sources to the grids. As a result, over 8,100 completed renewable energy projects have been unable to deliver their energy and replace nonrenewable energy sources. By targeting the shortcomings of current renewable energy development policies, our policy aims to expand energy access through improving America's grid systems.

Interconnection is the system through which energy projects are approved and connected to electric grids. Backlogs in this interconnection system are one of the main reasons why projects fail. There are a number of steps required to build a renewable power plant, including appointing a grid operator and power engineer. Due to a limited number of people to evaluate and approve these projects, the permitting process is often slow and inefficient. Because there are clogs in the system, trying to build one new project may require upgrading multiple other aspects of the grid, which substantially increases the amount of money required to implement renewable energy projects. Because the onus is on the companies to pay for the expansion of the capacities of their local grids, many companies are disincentivized from pursuing renewable energy projects due to the high costs of development and long waiting times for implementation.¹⁷⁰

¹⁶⁷ Allison R. Crimmins et al., "How the United States is Addressing Climate Change," U.S. Global Change Research Program, last modified 2023, <https://nca2023.globalchange.gov/>.

¹⁶⁸ Brad Plumer, "Energy Dept. Pours Billions Into Power Grids but Warns It's Not Enough," New York Times, last modified October 30, 2023, accessed April 3, 2024, <https://www.nytimes.com/2023/10/30/climate/energy-department-electric-grid.html>.

¹⁶⁹ Loan Programs Office, "INFLATION REDUCTION ACT OF 2022," Energy Department, last modified 2022, <https://www.energy.gov/lpo/inflation-reduction-act-2022>.

¹⁷⁰ Brad Plumer, "The U.S. Has Billions for Wind and Solar Projects. Good Luck Plugging Them In.," The New York Times, last modified February 23, 2023, <https://www.nytimes.com/2023/02/23/climate/renewable-energy-us-electrical-grid.html>.

II. Policy Proposal

Our proposal, the Renewable Energy Permitting and Sustainability Act (REPSA), centers on three policy planks: sustainability, resiliency, and modernization.

To strengthen grid resiliency, REPSA invests in expanding microgrids and standalone systems. Microgrids are energy distribution networks that are insulated from any central grid infrastructure, serving as backstops when the grid is adversely impacted. Specifically, H.R. 8033 funds the expansion of microgrids in accordance with a recent Department of Energy white paper, which aims to cut construction costs while expanding microgrids' share in overall energy production.¹⁷¹

To successfully modernize the grid and fully harness the power of renewable energy, removing "soft barriers" is key. Many steps in the permitting process are superfluous and delay the efficiency of deploying renewable energy. Despite the 2022 Inflation Reduction Act investing billions of dollars in renewable energy, most projects are stuck in administrative processes, waiting for approval. There is an accumulation of thousands of projects, which can take up to seven years to build.¹⁷² REPSA invests in making review and permitting processes more efficient, transparent, and collaborative. REPSA proposes establishing better means of communication with various stakeholders, such as key businesses, environmental groups, technology providers, and local policymakers to garner more support for grid policies, which can help speed up the process. REPSA would revise superfluous, overly complex, and outdated regulations and create an easier approval process. This can be done by providing grid companies with the funding to expand their grid capacities so that interconnection can proceed faster. The bill would also add expedited permit approvals for projects in areas with a lack of renewable energy to meet demand. Creating a dedicated platform to outline different land regulations and permitting processes would expedite and increase transparency for the permitting process as well.¹⁷³

Funding for REPSA will be appropriated from the Department of Energy, specifically the Federal Energy Regulatory Commission's discretionary spending. In fiscal year 2025, the Federal Energy Regulatory Commission requested \$192 million, which is sufficient to fund REPSA.¹⁷⁴

III. Legislative Analysis

REPSA is one of many necessary steps in moving the United States to a zero-carbon economy and reducing the nation's role in climate change. Climate change is anticipated to have catastrophic consequences beyond an increase of 1.5°C, so allaying emissions before the world

¹⁷¹ Lawrence Livermore National Laboratory, *Microgrids as a Building Block for Future Grids – Topic 4*, by Vaibhav Donde, et al., August 12, 2022, accessed April 17, 2024, <https://www.energy.gov/sites/default/files/2022-09/4-Microgrids%20as%20a%20Building%20Block%20for%20Future%20Grids.pdf>.

¹⁷² Madison Pobis, "Removing 'soft barriers' to the clean energy transition," Stanford Woods Institute for the Environment, last modified November 21, 2023, <https://woods.stanford.edu/news/removing-soft-barriers-clean-energy-transition>.

¹⁷³ Beret Walsh, "Overcoming Energy Permitting Barriers to Renewable Energy Development," Landgate, last modified February 12, 2024, <https://www.landgate.com/news/overcoming-energy-permitting-barriers-to-renewable-energy-development>.

¹⁷⁴ Department of Energy, *FY 2025 Congressional Justification*, by Office of Chief Financial Officer, report no. 3, 1, March 2024, accessed April 17, 2024, <https://www.energy.gov/sites/default/files/2024-03/doe-fy-2025-budget-vol-3-v2.pdf>.

reaches this critical point is essential.¹⁷⁵ REPSA will also reduce vulnerability to adverse weather events, supplying Americans with power even during emergencies. Lastly, it will make power generation more reliable even in the absence of extremities.

As with all climate policies, the most significant roadblock to REPSA is the cost. The Grid Deployment Office allocates around 51% of funds to transmission planning and permitting but has requested 31.4% more funds than in 2023; this is compounded by the fact that competition for funds—especially in climate sectors—remains high. Appropriating from existing fiscal spending for renewable energy development in favor of grid expansion will mitigate much of the Congressional hand-wringing over budgetary constraints; while renewable energy companies will temporarily suffer a loss in funds, the current issue most companies face at the present is having no way to connect their existing, finished projects to the grid to turn a profit in the first place.

There are concerns about how streamlining the process could lead to safety, fairness, and community engagement issues. However, these problems would be addressed by increasing collaboration and communication among different stakeholders, which would also aid in modernizing grid permitting. We can get different stakeholders on board by emphasizing the benefits of renewable energy in a community. For example, construction projects for renewable energy create more jobs, attract new businesses, and support community project goals, like eco-friendly spaces.¹⁷⁶

IV. Conclusion

The Renewable Energy Permitting and Sustainability Act, or REPSA, focuses on grid sustainability and permitting efficiency. It appropriates funds to deploy resilient renewable energy systems—such as microgrids—which are not dependent on each other, allowing for a faster and more sustainable transition to a green economy. Our bill also streamlines the renewable energy project approval process by gaining input from different committees, making the process for building projects more transparent, and establishing more resources to process and shorten applications. Making the approval process efficient is key to ensuring the implementation of renewable energy projects so renewable energy companies around the country can fully utilize federal funding to support a green transition.

¹⁷⁵ IPCC, 2023: Summary for Policymakers. In: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, Switzerland, pp. 1-34, doi: 10.59327/IPCC/AR6-9789291691647.001.

¹⁷⁶ Pobis, "Removing 'soft,'" Stanford Woods Institute for the Environment.

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Oversight and Investigations: Drug and Fentanyl Transportation

By: Hima Thota, Eli Sobel, Scarlet Sanders, Sofia Angileri and Nolan Murphy

The basis of our assessment in drug and fentanyl transportation in the U.S. is based upon these three following points of reform: border technology that is inadequate in stopping movement of illicit drugs, the lack of regulation and of pill presses in the U.S., and the lack of research into the impacts of fentanyl and potential preventative measures like fentanyl test strips.

Introduction

Background on the issue

Every day in the United States, over 150 Americans lose their lives to synthetic drug overdose, specifically from illicit fentanyl.¹⁷⁷ This problem is continuing to worsen, with roughly 55,000 Americans dying from overdose each year.¹⁷⁸ In a recent study, it was shown that if the influx of illicit drugs into the United States continues, then the number of deaths due to overdose will reach 1.22 million people.¹⁷⁹ Illegal fentanyl is entering the U.S. through two major methods: first, being smuggled past border control without detection, and second, the materials needed to create the drugs are being shipped into the U.S. without suspicion, and afterwards, the drugs are made on U.S. soil.¹⁸⁰ While there are current precautions put in place to stop the flow of fentanyl and other illegal drugs, such as new border control vehicle scanning devices, many of these new devices lack the adequate funding to be installed, or are ineffective.¹⁸¹ In addition, there is a lack of adequate research going into the negative impacts of fentanyl and measures that can be made to prevent it such as test strips. Lastly, there is a lack of regulation of pill presses in the U.S. Therefore, the precautions that have been put in place currently are not as effective as they need to be to help with this problem, and therefore, there is a need for greater oversight and regulation. Despite the numerous hearings that have been held in recent years in Congress to address this issue, it is clear that further action needs to be taken. For example, a hearing titled “Oversight and Reauthorization of the Office of National Drug Control Policy” took place in 2023 that addressed the need to reduce the amount of illicit drug income.¹⁸² This hearing also reviewed policies that were currently in place, such as President Biden’s border policies, which were shown to not be helping the current problem at hand. However, since this hearing and many others, despite the issue somewhat being addressed, change still needs to be made. Therefore we are calling for additional hearings in Congress which build upon the policies

¹⁷⁷ “CBP: America’s Front Line against Fentanyl,” U.S. Customs and Border Protection, accessed May 1, 2024, <https://www.cbp.gov/frontline/cbp-america-s-front-line-against-fentanyl>.

¹⁷⁸ “CBP: America’s Front Line,” U.S. Customs and Border Protection

¹⁷⁹ Keith Humphreys, Chelsea L Shover, and Christina M Andrews, et. al “The Lancet Commissions Responding to the Opioid Crisis in North America and beyond: recommendations of the Stanford-Lancet Commission,” The Lancet, February 22, 2022, <https://thewellnews.com/wp-content/uploads/2022/02/OpioidCommission.pdf>.

¹⁸⁰ “CBP: America’s Front Line,” U.S. Customs and Border Protection

¹⁸¹ Julia Ainsley, “ICE Will Face a \$500 Million Budget Shortfall for Border Security Unless Congress Takes Action, Officials Say,” NBCNews.com, February 13, 2024, <https://www.nbcnews.com/politics/immigration/border-deal-dead-money-border-security-might-run-out-rcna138499>

¹⁸² “Hearing Wrap up: Congress Must Address the Deadly Flow of Illicit Drugs and Historically High Overdose Deaths,” United States House Committee on Oversight and Accountability, July 27, 2023, <https://oversight.house.gov/release/hearing-wrap-up-congress-must-address-the-deadly-flow-of-illicit-drugs-and-historically-high-overdose-deaths%EF%BF%BC/>

proposed in previous hearings, and call for further oversight regarding border control funding and border control policies, pill press regulation and further adequate research into the important issue.

What we hope to achieve

In this call for further oversight, we hope to demonstrate why further hearings must be held, further regulation must be enforced for border policy regulation and technology, further research must be conducted, and more policies should be drafted regarding pill presses to address this growing problem which is putting the lives of many innocent people in danger. We hope that we can work to create better regulation and oversight of this current issue to continue working on reducing the amounts of illicit drugs which are entering the country and harming the community.

Status Quo and Shortcomings

Shortcomings of Current Border Technology

As a response to the border crisis, U.S. Customs and Border Protection is asking for an upgrade and expansion of a network of surveillance towers along the southern border.¹⁸³ This upgrade is expected to cost almost \$200 Million. Plans are to increase the number of towers on the border from 135 to 442, and upgrade technology in some of the towers.¹⁸⁴ Now, as of April 1, 2024, this number has grown to 467 towers. These towers, which form a trail from San Diego, California, to near Port Isabel, Florida, have cost over \$1 billion since 2005. Each tower is equipped with long-range cameras (some of which can detect a person 7.5 miles away), radar, and laser illuminators. Recently, a study by the EFF found that these towers are inefficient and not very successful. Despite this, the long term goal remains to expand the program to 723 towers.¹⁸⁵

A few problems arise from this tower project aside from its possible inefficiency. Not only are the towers covering roads and loosely populated rural areas, but they are also surveilling towns on both sides of the border. This could pose a potential threat to the privacy of American civilians. In addition, every photo of a license plate (with timestamp and coordinates attached) is stored as data for 15 years, longer than any retention policy for state or local agencies. This has left this data open to leaks and hacks, most notable of which occurring in 2019: CBP estimated that more than 184,000 traveler facial image files, as well as 105,000 license plate images were stored on a subcontractor's network at the time of the ransomware attack. The CBP has also believed that 19 of these pictures have been sold on the dark web, and has said that other sensitive information from this database has been sold on the dark web as well. However,

¹⁸³ Cal Biesecker, "CBP Releases RFP for Consolidated Border Surveillance Tower Program," Defense Daily, December 9, 2022, <https://www.defensedaily.com/cbp-releases-rfp-for-consolidated-border-surveillance-tower-program/homeland-security/>.

¹⁸⁴ "The US Is Pouring Money into Surveillance Tech at the Southern Border." n.d. MIT Technology Review.

<https://www.technologyreview.com/2023/04/17/1071682/us-pouring-money-surveillance-towers-southern-border/>.

¹⁸⁵ Dave Maass, "CBP Is Expanding Its Surveillance Tower Program at the U.S.-Mexico Border—and We're Mapping It," Electronic Frontier Foundation, April 2, 2024, <https://www.eff.org/deeplinks/2023/03/cbp-expanding-its-surveillance-tower-program-us-mexico-border-and-were-mapping-it>.

according to information from Border Patrol sectors, as of September 2021, only about half of checkpoints with license plate reader technology (21 of 43 checkpoints) were using this system as intended.¹⁸⁶ This not only poses a threat to the rights of American citizens but also may severely reduce trust amongst Americans.

National Committee-Driven Initiatives

Several committees and legislatures exist to promote awareness on the dangers surrounding opioid and fentanyl use as well as to propel medical research into drugs. Examples of such executive agencies include the White House Office of National Drug Control Policy (ONDCP), the Department of Health and Human Services, and the National Institutes of Health. Currently, most fentanyl substances are controlled under the CSA Schedule 1, which is the most restrictive of drug groups.¹⁸⁷ This means that such drugs are currently not accepted for medical use due to a high risk of addiction and abuse. Other substances in this category include heroin and cannabis.

The ONDCP facilitates the commencement of research on fentanyl-related substances as such research can help scientists and policy-makers better understand the risks and impact of fentanyl on public health, and what solutions can be developed.¹⁸⁸ Developing solutions include fentanyl test strips, which allow drug users to test whether or not a drug contains any traces of fentanyl.¹⁸⁹ The NIH and the FDA have also made it a priority to highlight the urgent need for research that can address the gaps in development and implementation of technology like fentanyl test strips and other such measures that can prevent overdoses due to fentanyl.¹⁹⁰ Currently, fentanyl strips are the most well-known and distributed preventative tool, but they are also extremely understudied. And in addition, they are not bulletproof in producing accurate results.¹⁹¹ Research not only ensures an assessment of fentanyl and preventative measures on public health, but also allows for the evaluation of quality of preventative measurements that specifically allow for immediate fentanyl testing and overdose prevention.

Specific Research Into Fentanyl Strips

Fentanyl test strips are a low-cost device that allow users to test products they have for any presence of fentanyl, and these results can be obtained in a little over five minutes. These strips are growing more and more important in the world today, as more fentanyl is being laced into drugs such as cocaine, or medications such as Adderall that appear to be harmless, but have in

¹⁸⁶ <https://www.documentcloud.org/documents/23702372-gao-22-104568#document/p46>

¹⁸⁷ Matthew S. Borman, “The Federal Register,” Federal Register: The Daily Journal of the United States Government, December 7, 2023, <https://www.federalregister.gov/documents/2023/12/07/2023-26694/schedules-of-controlled-substances-placement-of-nine-specific-fentanyl-related-substances-in#:~:text=It%20was%20suggested%20that%20the,uses%20in%20the%20United%20States.>

¹⁸⁸ “National Drug Control Strategy,” The White House Executive Office of the President, 2022, <https://www.whitehouse.gov/wp-content/uploads/2022/04/National-Drug-Control-2022Strategy.pdf>.

¹⁸⁹ “Fentanyl Test Strips: A Harm Reduction Strategy,” Centers for Disease Control and Prevention, September 30, 2022, <https://www.cdc.gov/stopoverdose/fentanyl/fentanyl-test-strips.html>.

¹⁹⁰ <https://www.nih.gov/news-events/news-releases/nih-fda-leaders-call-more-research-lower-barriers-improve-implement-drug-checking-tools-amid-overdose-epidemic>

¹⁹¹ <https://www.cdc.gov/stopoverdose/fentanyl/fentanyl-test-strips.html#:~:text=Negative%20results%3A%20Two%20pink%20lines,you%20receive%20a%20negative%20result.>

fact been laced with fentanyl. These strips allow users to check these products before they consume them to ensure that they have not been laced. The development of these strips is an example of a harm-reduction strategy. Researchers are calling for further research into strips to expand testing abilities to urine and hair. This would be invaluable for on-site workers in medical settings in assessing their plan of action. Additionally, developing solutions are still not completely foolproof—for example, fentanyl test strips are not always accurate.¹⁹² It is critical for the Biden-Harris administration to allocate sufficient funding and grants to medical research institutions such as the NIH and FDA to further research into such an essential aspect of preventative measures to avoid death from fentanyl ingestion.

Pill Presses Awareness and Regulation throughout the U.S.

Pill press laws are another crucial but understudied aspect of fentanyl and drug regulation in the U.S. The NIH suggested that pill press laws should receive more attention at the legislative level.¹⁹³ Pill presses and molds are used to produce counterfeit fentanyl-laced pills in countries like Mexico. The Biden-Harris administration also recently imposed sanctions on Chinese pill presses, which has raised awareness on pill press regulation.¹⁹⁴ However, throughout the U.S., pill press laws are federally limited, and are essentially non-existent at the state level, with very few individuals knowing about what they are.¹⁹⁵

Earlier this year, the DEA issued a letter to E-commerce companies after finding that pill presses and stamps which can be used to make counterfeit pills were being sold on e-commerce platforms.¹⁹⁶ Part of their letter suggested that such platforms should track the buyers and sellers of such products. And preceding this event, in 2019, the DEA also established the Industry Liaison Project, which works with dozens of E-commerce companies to regulate and track the sale of pill presses. Following the establishment of the Liaison Project, companies like Amazon and Etsy banned such products from their sites.¹⁹⁷ It would most likely benefit public health for other E-commerce companies to follow suit. Ultimately, it would be beneficial for U.S. public health if more awareness is raised about pill presses and sanctions, legislation is introduced, and awareness campaigns concerning pill presses are passed federally - especially in states like California where the opioid epidemic is out of control.¹⁹⁸

Current counter arguments and perspectives

In February of 2024, President Joe Biden signed a bill into law concerning border policy and asylum, however, this bill also included methods to help reduce the amount of illegal drugs that

¹⁹² <https://nida.nih.gov/news-events/news-releases/2023/06/nih-and-fda-leaders-call-for-more-research-lower-barriers-to-improve-and-implement-drug--checking-tools-amid-overdose-epidemic>

¹⁹³ <https://pubmed.ncbi.nlm.nih.gov/37694167/>

¹⁹⁴ <https://pubmed.ncbi.nlm.nih.gov/37694167/>

¹⁹⁵ <https://pubmed.ncbi.nlm.nih.gov/37694167/>

¹⁹⁶ <https://www.dea.gov/press-releases/2024/02/26/dea-issues-letter-e-commerce-companies-sale-pill-presses-used-make#:~:text=DEA%20has%20found%20that%20pill,Controlled%20Substances%20Act%2C%2021%20U.S.C.>

¹⁹⁷ <https://www.dea.gov/press-releases/2024/02/26/dea-issues-letter-e-commerce-companies-sale-pill-presses-used-make>

¹⁹⁸ <https://www.dea.gov/press-releases/2024/02/26/dea-issues-letter-e-commerce-companies-sale-pill-presses-used-make>

pass through U.S. borders. These methods all came in the form of providing funding that would be used to hire more workers at the border and ensure the amount of illegal substances entering the U.S. is reduced. Specifically, \$405 million has been spent on hiring 1,300 additional Border Patrol Agents in order to help secure the border, and \$239 million has been spent on hiring 1,000 additional CBP to stop dangerous drugs entering the country.¹⁹⁹ From this, it is stated that these changes have helped to seize record level mounts of illicit fentanyl at port entries, and that these policies are countering the problem of fentanyl in the U.S.²⁰⁰ While these policies have definitely made a significant impact, the belief that the problem of illegal drugs entering the U.S. is being fully managed is not true, and further action is necessary for the problem to be truly solved. This requires further oversight from the Office of National Drug Control, as current methods for ensuring that the income of illegal drugs is prevented are not working effectively.

Our Proposed Solution

What is our proposed solution?

Currently, there are many different agencies pushing for further research into the impacts of fentanyl, as well as efficient strategies to develop preventative measures, and there are concerns surrounding the lack of regulation and awareness surrounding pill presses, which can be used to manufacture counterfeit medications with fentanyl. Pill presses are common in countries like China and Mexico, but they are also for sale online in the U.S.

We call for a push upon agencies to require research that looks further into the efficacy of fentanyl testing and strategies of distribution. We also request a hearing about the efficacy of the Industry Liaison Project so that any shortcomings can be addressed in future efforts to prevent the circulation of counterfeit drugs. We lastly push for a hearing into how money is being efficiently distributed in different forms of border technology, as currently the border technology that is in place such as surveillance towers have millions of dollars invested into their production, but do not produce adequate results. We would like to propose a hearing that delves into new forms of technologies that could be produced that would be more cost effective, and produce overall better results for monitoring the income of illegal drugs at the border.

Conclusion

The significance and long-term impact of our proposed oversight

We call for an audit to assess the current budget allocations in the realm of current border technologies. Reforming border technology through the reallocation of budgets and assessing the

¹⁹⁹ <https://www.whitehouse.gov/briefing-room/statements-releases/2024/03/11/fact-sheet-the-presidents-budget-secures-our-border-combats-fentanyl-trafficking-and-calls-on-congress-to-enact-critical-immigration-reform/>

²⁰⁰ <https://www.whitehouse.gov/briefing-room/statements-releases/2024/03/11/fact-sheet-the-presidents-budget-secures-our-border-combats-fentanyl-trafficking-and-calls-on-congress-to-enact-critical-immigration-reform/>

²⁰¹ <https://energycommerce.house.gov/posts/e-and-c-republican-leaders-demand-big-tech-does-more-to-stop-illegal-fentanyl-sales-on-their-platforms>

²⁰² <https://energycommerce.house.gov/posts/the-fentanyl-crisis-in-america-is-getting-worse>

²⁰³ <https://energycommerce.house.gov/combating-the-opioid-crisis-spotlight>

technology used directly in relation to drug transportation allow for reformation of the broader issue of drug, specifically fentanyl, transportation in the U.S.. The current border technology used is ineffective in detecting and preventing the movement of opioids across the border. By reallocating our budgets to enhance border technology we can improve our ability to locate and intercept the shipments of drugs at the border, in turn reducing the rate of illicit drugs immigrating into our country. The lack of regulation and awareness surrounding the pill press lead to an influx in opioids in America. Pill presses are used to produce counterfeit prescription pills laced with fentanyl, contributing to the large number of illegal opioid imports into our country. Strengthening our border technology can help interfere with these illegal imports of not only the pill presses but also the chemicals used in the drug production.²² Increasing awareness regarding these items can aid in reducing the use of these presses and preventing the production of these killer counterfeit drugs filled with opioids such as fentanyl. The introduction of new more efficient technologies can improve border security and enhance the detention and prevention of drug and fentanyl transportation.²³ Technologies such as drones, ground-based radar systems, remote sensor networks, and blockchain technology would ultimately combat the opioid crisis and protect the health, safety, and lives of the American people. The lack of research into the impact of fentanyl and further preventative measures lead to a lack of general knowledge surrounding the harm of these drugs.²⁴ Fentanyl test strips are a harm reduction tool to help individuals detect the presence of fentanyl in drugs, but because of how fentanyl laced drugs are created, the tests are not fully accurate. Since limited research has been done regarding other efficient preventative methods there is still a large number of drug related risks. By reallocating funds towards research we can better understand the impacts of fentanyl and develop effective preventative methods and expand on already accessible faulty measures such as test strips. Overall, reforming border technology and addressing the deficiencies in detecting and preventing the transportation of is essential for combating the fight against the opioid crisis. Investing in advanced border surveillance technology, regulating pill presses, and research on fentanyl, are necessary steps in weakening spread and effects of opioid abuse.

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O&I Leaded Fuel Summary Memo

Audrey Sullivan, Daisy Casildo Abad, Jaya Urrutia, Jacob Kessler, Megan Hastings

Introduction

For most of the 20th century, lead use was everywhere, leading to increasingly high blood levels in the US population. When scientists discovered the extremely harmful effects of this lead exposure for children, from increased blood pressure to lowered IQ, many new regulations were implemented to lower lead concentrations in the environment. Lead use was banned in paint, automotive gasoline, and many other areas. But today, it is still used in the aviation industry, a key component of the avgas fuel mixture utilized in small piston-engine aircraft. According to the Environmental Protection Agency (EPA), there are approximately 20,000 small airports in the US where the use of leaded avgas is allowed²⁰⁴.

Health Risks & Lack of Regulation

The use of leaded fuel raises many issues, including lead pollution, health risks for children and climate inequality, conflicting with National Ambient Air Quality Standards (NAAQS). According to Representative Ro Khanna, “Children living near the Reid-Hillview Airport...have been found to have lead in their blood” and he declares this to be “a public health and environmental crisis.”²⁰⁵ A recent study of the airport found children's blood lead levels (BLLs) “increase significantly with proximity to [the airport]” and also “increase significantly with the volume of measured PEA [piston-engine aircraft] traffic at RHV from the date of blood draw.”²⁰⁶ These findings were comparable to the BLL levels of children during the Flint Water Crisis (FWC), making up roughly 50% of the increase seen in BLLs at the height of FWC. The scope of exposure is not small, as “four million persons reside, and about six hundred K-12th grade schools are located within 500 meters of PEA-servicing airports,” and even more live within a further radius, which still puts them at risk for lead poisoning.²⁰⁷ Any BLL above zero can cause significant health problems and damage developing organs, even affecting IQ scores: an estimated 824,097,690 IQ points were lost in Americans born from 1951 to 1980 due to lead exposure.²⁰⁸ The EPA lists 188 hazardous air pollutants (HAPs) that they are required to

²⁰⁴ Environmental Protection Agency. (2023, October 20). Regulations for Lead Emissions from Aircraft. US EPA. <https://www.epa.gov/regulations-emissions-vehicles-and-engines/regulations-lead-emissions-aircraft>

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²⁰⁶ Zahran, S., Keyes, C., & Lanphear, B. (2023). Leaded aviation gasoline exposure risk and child blood lead levels. *PNAS Nexus*, 2(1). <https://doi.org/10.1093/pnasnexus/pgac285>

²⁰⁷ Miranda, M. L., Anthopolos, R., & Hastings, D. (2011). A geospatial analysis of the effects of aviation gasoline on childhood blood lead levels. *Environmental Health Perspectives*, 119(10), 1513-1516. <https://doi.org/10.1289/ehp.1003231>

²⁰⁸ McFarland, M. J., Hauer, M. E., & Reuben, A. (2022). Half of US population exposed to adverse lead levels in early childhood. *Proceedings of the National Academy of Sciences*, 119(11). <https://doi.org/10.1073/pnas.2118631119>

regulate, including lead compounds. However, leaded fuel is still the primary avgas type used by PEAs in the US and has not been regulated at a national level. Additionally, small airports serve a more affluent community and yet tend to be located in lower-income communities of color. This is an environmental justice issue, as minorities and lower-income American children may be more affected by lead poisoning from aircraft emissions.

Policy Proposal

In October of 2023, the Environmental Protection Agency (EPA) declared leaded emissions from aircraft to be a serious threat to public health due to its effects on the environment and exposure to children. According to the EPA, this determination obligates the FAA and EPA to “develop standards that address the composition, chemical, or physical properties of an aircraft fuel or fuel additive to control or eliminate aircraft lead emissions.”²⁰⁹ However, as of now, no further steps have been taken to eliminate the usage of lead in avgas. This is due to a major issue, until October of last year, the EPA did not declare its intent to control, eliminate, or address the use of leaded avgas. We propose an EPA hearing on this issue where the EPA can explain the reasoning behind their lack of regulatory efforts on leaded fuel. With this hearing, we can establish any gaps that need to be addressed in the regulation of leaded fuel in PEAs and how we can prevent future oversights. While it may be difficult to instate a wide-scale ban on leaded fuel, it is necessary to at least further regulate the sale of it and create a distinct plan for a lead-free future. The alarming public health risks that lead exposure poses for children deem it necessary to increase regulation, despite possible pushback from the industry.

Legislative Analysis

The Clean Air Act (CAA) controls air pollution on a national level and sets standards for air quality and emissions. However, it lacks specific provisions regarding pollution and the health impacts of lead emissions from aircraft engines. For this reason, even as lead usage has been banned in many other contexts, it has remained common in aircraft fuel, and attempts to regulate it have been inconclusive. The recent determination by the EPA that lead emissions contribute to air pollution underscores the urgency of addressing this issue, highlighting how small piston-engine aircrafts affect public health not addressed by standards set in the original law.²¹⁰ The proposed hearing would improve existing regulations through the EPA, which was given authority to reinforce compliance with environmental requirements by the Clean Air Act,

²⁰⁹ EPA Press Office. (2023, October 18). EPA Determines that Lead Emissions from Aircraft Engines Cause or Contribute to Air Pollution [Press release]. <https://www.epa.gov/newsreleases/epa-determines-lead-emissions-aircraft-engines-cause-or-contribute-air-pollution#:~:text=WASHINGTON%20%E2%80%94%20Today%2C%20the%20U.S.%20Environmental%20Protection%20Agency,health%20and%20welfare%20under%20the%20Clean%20Air%20Act.>

²¹⁰ EPA Press Office. (2023, October 18). *EPA Determines that Lead Emissions from Aircraft Engines Cause or Contribute to Air Pollution* [Press release]. <https://www.epa.gov/newsreleases/epa-determines-lead-emissions-aircraft-engines-cause-or-contribute-air-pollution#:~:text=WASHINGTON%20%E2%80%94%20Today%2C%20the%20U.S.%20Environmental%20Protection%20Agency,health%20and%20welfare%20under%20the%20Clean%20Air%20Act>

and set the NAAQs and HAPs. It will also enhance the existing regulations and assess the effectiveness of EPA and CAA.

Furthermore, the California State Aeronautics Act (SAA), passed in 2023, regulates aviation within state jurisdiction but lacks provisions addressing public health concerns about leaded fuel. The provisions existing in SAA cover licensing and registration but make no mention of leaded fuel used in noncommercial airlines. While the commercial aircraft industry is highly regulated, privately owned aircraft face little of the same environmental policies and attention.

According to the CAA, the EPA is required to enforce regulations for lead emissions from certain aircrafts. The Federal Aviation Administration (FAA) must also formulate standards aimed at controlling or eradicating aircraft lead emissions by addressing the composition, chemical, or physical attributes of aircraft fuel or fuel additives. So far these regulations have not been implemented. The FAA has declared its goal of removing lead from aircraft entirely by the year 2030 but has yet to implement real regulations to reach this aim. With this hearing, we can redraw attention to the issue and encourage immediate action to eliminate lead usage.

Conclusion

Leaded fuel has been a controversial topic for a long time, and while advancements have been made in the creation of unleaded options, there is still a lack of regulation from the EPA and FAA to prevent its usage. Due to its incredibly harmful effects on IQ levels and children's health, and the inequities it raises for lower-income communities, it is necessary to increase regulation immediately. Through our proposed hearing, the EPA can come to understand what path to take as far as regulation and make the necessary plans for instating policy.

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Oversight & Investigations Subcommittee: Medication Distribution

Members: Myesha Phukan, Melinda Huang, Yuna Suh

Introduction:

In 2012, patients who used steroid injections from the New England Compounding Center became victims of a meningitis outbreak that gained national attention from organizations like the Food and Drug Administration, the Centers for Disease Control, and others.²¹¹ The outbreak impacted individuals from across the United States and caused the death of 64.²¹² During its investigation into the steroid injections, the FDA found that they had been contaminated with a fungal infection.²¹³ In order to prevent this from happening in the future, Congress passed the Drug Quality and Security Act in 2013, sponsored by Representative Fred Upton (R-MI-6), and co-sponsored by 10 other representatives.²¹⁴ The bill, which became law in November of that year, imposed stricter regulations on both compounding and supply chain-related actions. The act, consisting of two sections, has benefitted individuals greatly. It ensures uniform standards for sterile compounding, subsequently creating safer products, and places stricter rules on compounding pharmacies and outsourcing facilities.²¹⁵ However, the DQSA has gaps that must be filled in to ensure better patient safety, accessibility, and livelihood. The law places limits on interstate drug distribution, which can prevent patients from receiving the necessary drugs in an efficient manner. Some must ship the medication to their home, which costs even more money.²¹⁶ In its current form, the DQSA creates inaccessibility for patients and their medications. In addition, the DSQA limits the use of office-use, impacting the amount of medication available at physician's offices. This can not only deteriorate the relationships between pharmacists and physicians, but prevent offices from keeping up with patient demand. In order to remedy these oversights, we propose writing a letter to the FDA that requests them to prompt investigations into different sections of the DQSA.

The Proposal:

We propose that the Government Accountability Office (GAO) spearheads an investigation into the parts of the Drug Quality and Security Act that allow pharmacies and manufacturers to register their compounding status voluntarily. We propose that the

²¹¹ Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp>.

²¹² Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Protects Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp>.

²¹³ Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Protects Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp>.

²¹⁴ H.R.3204 - 113th Congress (2013-2014): Drug Quality and Security Act. (2013, November 27). <https://www.congress.gov/bill/113th-congress/house-bill/3204>

²¹⁵ Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Protects Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp>.

²¹⁶ Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Protects Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp>.

investigation scrutinizes the number of facilities that are currently registered. It would be beneficial to consumers and patients if there were large numbers of facilities that are participating in this sort of registration. A GAO investigation into these data points and numerical values would allow the organization to determine if the current policy of voluntary registration is fulfilling its intended purpose. Should the investigation yield that smaller than desired amounts of facilities are involved in the registration, the agency should look toward proposing a new rule that requires manufacturers and pharmacies to be registered under different categories to identify the types of compounding being practiced.

Additionally, we propose that the GAO open an investigation into how the limitation of office-use drug distribution impacts medication shortages in physician's offices and its influence on deteriorating pharmacist-physician relations. The investigation should look into whether the current limits placed on the amount of office-use drugs available for outsourcing facilities to distribute, affects the ability to fulfill the medication needs of a physician's office. Should the investigation yield that current restrictions are not fulfilling patient medication needs, we propose that the GAO look into adding a new rule that allows for more drugs to be distributed when needed.

Finally, we ask that the GAO investigate the results of section 503A of the DQSA and the memorandum of understanding (MOU) of 2015, which states that only 30% of a pharmacy's total distribution may be interstate. This investigation should detail the average number of pharmacies available to patients, and whether the 30% limit inhibits easy access to patient-specific drugs unavailable elsewhere. If this investigation reveals that the current system prevents patients from accessing needed drugs, limits on interstate distribution should not solely be determined by the FDA and states that accept memoranda of understanding, as compounding facilities are currently completely reliant on the FDA and the state. Hospitals should be excluded from interstate distribution limits as hospitals may need to transport drugs to their own facilities located in different states.

Legislative Analysis:

The Drug Quality and Security Act (DQSA), which serves as the primary framework for drug oversight and investigation, was passed by Congress in 2012. While the DQSA does impose necessary regulations among compounding pharmacies, there are two provisions that significantly hamper drug accessibility and patient equity. The first prohibited compounding pharmacies from distributing office-use prescriptions, and the second severely limits the number of prescriptions that compounding pharmacies can distribute out of state (5% of their total distribution).²¹⁷ A 30% limit can be achieved if pharmacies meet certain guidelines.²¹⁸

²¹⁷ Logan J. Eliassen. (2018). *The Drug Quality and Security Act: Providing quality, but not security, for patients*. Iowa Law Review - The University of Iowa. <https://ilr.law.uiowa.edu/print/volume-103-issue-3/the-drug-quality-and-security-act-providing-quality-but-not-security-for-patients>

²¹⁸ The Federal Register. Federal Register :Memorandum of Understanding Addressing Certain Distributions of Compounded Human Drug Products Between the State Board of Pharmacy or Other Appropriate State Agency and the Food and Drug Administration; Availability. (2020). <https://www.federalregister.gov/documents/2020/10/27/2020-23687/memorandum-of-understanding-addressing-certain-distributions-of-compounded-human-drug-products>

Exceptions to the DQSA do exist (“memorandum of understanding”), however, the DQSA prevents patients from accessing necessary drugs and incentivizes states to reach the memorandum of understanding.²¹⁹ The first provision has weakened the important and interpersonal relationship between pharmacists and physicians, as physicians now have less flexibility with which compound pharmacies they could choose to fulfill their prescriptions. As a result, physicians are forced to work with larger pharmacies rather than small ones, which reduces accessibility and personalized services, leaving potential for conflicts of interest.²²⁰

The second provision greatly reduces accessibility, as patients who aren’t able to travel across state lines are prevented from accessing potentially life-saving medication.²²¹ While Section 503A does acknowledge the 30% limit outlined in the memorandum, states are disincentivized from reaching it as they are forced to assume responsibility for over-regulation, greatly burdening their resources.²²²

Furthermore, the lack of nationwide identification of sterile compounders hampers FDA and state board inspections, with only 59 registered outsourcing facilities.²²³ High registration costs and the lack of uniform state requirements leave voluntary registration to be inefficient, as companies exhibit a willingness to evade regulatory scrutiny.²²⁴ The DQSA also fails to address the risks posed by non-sterile compounds, which prove to be just as dangerous as sterile compounds.²²⁵ Thus, it is crucial we address those oversight gaps to adequately promote public safety in drug supply chains.

Conclusion:

In conclusion, although the Drug Quality Security Act provides consumers and patients with increased protections in the security of facilities and pharmacies, the bill does not account for other integral protections that are vital to patient safety. Forcing companies to register under specific categories that dictate what they are compounding would allow for more transparency and permit the FDA to conduct more thorough and deep investigations. Furthermore, increasing the amount of medication that outsourcing facilities or pharmacies are able to distribute interstate would be extremely beneficial to patient care. The current limitations prevent many patients from accessing their medications without paying exorbitant fees to ship the medicine to their homes. Our proposal would ensure that there is sufficient investigation into existing policies and parts of the DQSA that may be detrimental to patients and physicians alike. These investigations are necessary to secure remedies for possible oversights present in the current

²¹⁹ Logan J. Eliassen. (2018). *The Drug Quality and Security Act: Providing quality, but not security, for patients*. Iowa Law Review - The University of Iowa. <https://ilr.law.uiowa.edu/print/volume-103-issue-3/the-drug-quality-and-security-act-providing-quality-but-not-security-for-patients>

²²⁰ Logan J. Eliassen. (2018). *The Drug Quality and Security Act: Providing quality, but not security, for patients*. Iowa Law Review - The University of Iowa. <https://ilr.law.uiowa.edu/print/volume-103-issue-3/the-drug-quality-and-security-act-providing-quality-but-not-security-for-patients>

²²¹ Logan J. Eliassen. (2018). *The Drug Quality and Security Act: Providing quality, but not security, for patients*. Iowa Law Review - The University of Iowa. <https://ilr.law.uiowa.edu/print/volume-103-issue-3/the-drug-quality-and-security-act-providing-quality-but-not-security-for-patients>

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²²³ Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Protects Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp> .

²²⁴ Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Protects Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp> .

²²⁵ Dinkelaker, T. (2016). (rep.). *A False Sense of Safety: How the Drug Quality and Security Act Fails to Protect Protects Patients from from Harm*. Saint Louis University. Retrieved 2024, from <https://scholarship.law.slu.edu/cgi/viewcontent.cgi?article=1067&context=jhlp> .

form of the act, remedies that are extremely important in ensuring the future of safe, efficient, and substantive medication distribution.

LETTER:

May 18, 2024

The Honorable Xavier Becerra
Secretary of Health and Human Services
U.S. Department of Health and Human Services
200 Independence Avenue, S.W.
Washington, D.C. 20201

The Honorable Gene L. Dodaro
Comptroller General of the United States
Government Accountability Office
441 G Street, N.W.
Washington, D.C. 20548

Dear Secretary Becerra and Comptroller General Dodaro,

We write to request that the GAO open an investigation into multiple stipulations of the Drug Quality and Security Act. The act, consisting of two sections, has benefitted individuals greatly. It ensures uniform standards for sterile compounding, subsequently creating safer products, and places stricter rules on compounding pharmacies and outsourcing facilities. However, the DQSA has gaps that must be filled in to ensure better patient safety, accessibility, and livelihood. The law places limits on interstate drug distribution, which can prevent patients from receiving the necessary drugs in an efficient manner. Some must ship the medication to their home, which costs even more money. In its current form, the DQSA creates inaccessibility for patients and their medications. In addition, the DSQA limits the use of office-use, impacting the amount of medication available at physician's offices. This can not only deteriorate the relationships between pharmacists and physicians, but prevent offices from keeping up with patient demand. In order to remedy these oversights, we believe an investigation is necessary.

Firstly, we propose that the GAO spearheads an investigation into the parts of the Drug Quality and Security Act that allow pharmacies and manufacturers to register their compounding status voluntarily. We propose that the investigation scrutinizes the number of facilities that are currently registered. It would be beneficial to consumers and patients if there were large numbers of facilities that are participating in this sort of registration. A GAO investigation into these data points and numerical values would allow the organization to determine if the current policy of voluntary registration is fulfilling its intended purpose. Should the investigation yield that smaller than desired amounts of facilities are involved in the registration, the agency should look toward proposing a new rule that requires manufacturers and pharmacies to be registered under different categories to identify the types of compounding being practiced.

Subsequently, we propose that the GAO open an investigation into how the limitation of office-use drug distribution impacts medication shortages in physician's offices and its influence on deteriorating pharmacist-physician relations. The investigation should look into whether the

current limits placed on the amount of office-use drugs available for outsourcing facilities to distribute, affects the ability to fulfill the medication needs of a physician's office. Should the investigation yield that current restrictions are not fulfilling patient medication needs, we propose that the GAO look into adding a new rule that allows for more drugs to be distributed when needed.

Finally, we ask that the GAO investigate the results of section 503A of the DQSA and the memorandum of understanding (MOU) of 2015, which states that only 30% of a pharmacy's total distribution may be interstate. This investigation should detail the average number of pharmacies available to patients, and whether the 30% limit inhibits easy access to patient-specific drugs unavailable elsewhere. If this investigation reveals that the current system prevents patients from accessing needed drugs, limits on interstate distribution should not solely be determined by the FDA and states that accept memoranda of understanding, as compounding facilities are currently completely reliant on the FDA and the state. Hospitals should be excluded from interstate distribution limits as hospitals may need to transport drugs to their own facilities located in different states.

Although the Drug Quality Security Act provides consumers and patients with increased protections in the security of facilities and pharmacies, the bill does not account for other integral protections that are vital to patient safety. Forcing companies to register under specific categories that dictate what they are compounding would allow for more transparency and permit the GAO to conduct more thorough and deep investigations. Furthermore, increasing the amount of medication that outsourcing facilities or pharmacies are able to distribute interstate would be extremely beneficial to patient care. The current limitations prevent many patients from accessing their medications without paying exorbitant fees to ship the medicine to their homes. The proposed inquiries would ensure that there is sufficient investigation into existing policies and parts of the DQSA that may be detrimental to patients and physicians alike. These investigations are necessary to secure remedies for possible oversights present in the current form of the act, remedies that are extremely important in ensuring the future of safe, efficient, and substantive medication distribution.

We look forward to your prompt response. Thank you in advance for your cooperation and time.

Sincerely,
Myesha Phukan, Melinda Huang, and Yuna Suh

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